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Financial Markets and Products 目录

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1. Consider an exposure with a value of \$30.0 million. In L(+)/P(-) format, its expected loss is 4.0% with volatility of 15.0%; i.e., $\mu(L/P) = +4.0\%$, $\sigma = 15.0\%$. Conveniently, the exposure happens to have a normal distribution. Which is **nearest** to the single-period 97.5% expected shortfall (ES)?

考虑一笔名义金额为 \$30.0 百万美元的风险敞口。在 L(+)/P(-) 格式下，其预期损失为 4.0%，波动率为 15.0%；即 $\mu(L/P) = +4.0\%$, $\sigma = 15.0\%$ 。为方便起见，假定该敞口的分布为正态分布。单期 97.5% 期望损失（ES）中，哪一个数值与其**最接近**？

- A. 6.60 million
六百六十万
- B. 9.38 million
九百三十八万
- C. 11.72 million
一千一百七十二万
- D. 14.50 million
一千四百五十万

答案:C

解析:

C 选项正确。若分布为正态分布，则 97.5% ES 可由 99.0% VaR 近似。在这种情况下，我们可以使用 $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$ 。精确的 97.5% ES 为 \$11.7210，因此该数值被作为正确选项。

考察: 市场风险 重要程度: [■■□]

科目: Quantitative Analysis

2. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

在 Zapex Financial Corporation 的金融头寸中，它写了看涨期权、进入了利率互换并写了信用违约互换。在这些（以下）四个头寸中，哪一个是最典型的反向风险？

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
Zapex 向同一行业的对手方出售了看涨期权
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
Zapex 是利率互换中的固定利率支付方，在经济复苏期间
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
Zapex 向信用质量与参考实体正相关的对手方出售了信用违约互换
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability
Zapex 在远期货币合约中持有正相关于对手方违约概率的标记市值收益

答案:D

解析:

D 选项正确。 Zapex 在远期货币合约中持有正相关于对手方违约概率的标记市值收益。

反向风险（信用风险）是敞口与信用质量之间的不利相关性，因此反向风险是敞口与违约概率（信用质量）之间的正（负）相关性。这是反向风险，因为更大的敞口与较低支付可能性相关。代数上，反向风险（或正向风险）是敞口/EAD 和 PD 之间的一些依赖性。我们通常写 $EL\% = PD \times EAD$ ，但假设 PD 和 EAD 之间是独立的，因此考虑 WWR 的一种简单方法是当 $EL > PD \times EAD$ 时，因为 $EL \neq PD \times EAD$ 也是非独立性的条件。

关于（A）、（B）和（C），每项都是正向风险的例子。

考察: Credit Risk 重要程度: [■■□]

科目: Financial Markets and Products

3. In December 2017, the Basel Committee (BCBS) made several revisions to the Basel III framework. This included revisions to both the standardized approach to credit risk (“Credit risk accounts for the bulk of most banks’ risk-taking activities and hence their regulatory capital requirements. The standardized approach is used by the majority of banks around the world, including in non-Basel Committee jurisdictions,” wrote the Committee) and the internal ratings-based (IRB) approaches to credit risk (“The financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes,” wrote the Committee).¹

In regard to these December 2017 revisions to credit risk approaches in Basel III, which of the following statements is TRUE?

¹ “High-level summary of Basel III reforms,” (Basel Committee on Banking Supervision Publication, December 2017)

2017 年 12 月，巴塞尔委员会（BCBS）对巴塞尔协议 III 框架进行了多项修订。其中包括对信用风险标准法的修订（委员会写道：“信用风险占据了大多数银行风险承担活动以及相应监管资本要求的主要部分。标准法被世界上大多数银行采用，包括那些不在巴塞尔委员会成员辖区内的银行”），以及对信用风险内部评级法（IRB）方法的修订（委员会写道：“金融危机凸显了与监管资本内部模型方法使用相关的一些缺陷，其中包括信用风险内部评级法（IRB）方法。这些缺陷包括 IRB 方法的过度复杂性、各银行内部模型化的 IRB 资本要求缺乏可比性，以及在对某些资产类别建模时缺乏稳健性”）。¹ 关于 2017 年 12 月对巴塞尔 III 中信用风险计量方法所作的这些修订，以下哪一项表述是正确的？¹ “High-level summary of Basel III reforms”（《巴塞尔协议 III 改革的高层次概要》），巴塞尔银行监管委员会出版物，2017 年 12 月

- A. In the standardized approach (SA) for residential real estate exposures, the risk weights are lower if the loan-to-value (LTV) falls into a lower band
在住宅房地产敞口中，标准法（SA）下，如果贷款价值比（LTV）落入较低的区间，则风险权重较低
- B. The standardized approach (SA) eliminated the treatment of unrated exposures because all exposures are required to have an external credit rating
标准法（SA）消除了对无评级敞口的治疗，因为所有敞口都必须具有外部信用评级
- C. The internal ratings-based (IRB) approach removed the option to use the foundation IRB (F-IRB) because the advanced IRB (A-IRB) must be used for all asset classes
内部评级法（IRB）消除了使用基础 IRB（F-IRB）的选项，因为所有资产类别都必须使用高级 IRB（A-IRB）
- D. The internal ratings-based (IRB) approach removed the parameter input floors (in the case of PD, LGD, and EAD) to increase RWA variability across different banks
内部评级法（IRB）消除了参数输入地板（在 PD、LGD 和 EAD 的情况下），以增加不同银行之间的 RWA 可变性

答案:A

解析:

A 选项正确。在住宅房地产敞口中，标准法（SA）下，如果贷款价值比（LTV）落入较低的区间，则风险权重较低。见下文。巴塞尔委员会（High-level summary of Basel III reforms, December 2017）写道：“对于住宅房地产敞口，已经开发了更多风险敏感的方法，其中风险权重根据抵押贷款的贷款价值比（而不是现有的单一风险权重）以及更好地反映市场结构差异的方式而变化。”

Residential real estate exposures								
LTV bands	Below 50%	50% to 60%	60% to 70%	70% to 80%	80% to 90%	90% to 100%	above 100%	Criteria not met
General RRE								
Whole loan approach RW	20%	25%	30%	40%	50%	70%	RW of counterparty	
Loan-splitting approach ² RW	20%		RW of counterparty					RW of counterparty
Income-producing residential real estate (IPRRE)								
Whole loan approach RW	30%	35%	45%	60%	75%	105%	150%	

关于（B）、（C）和（D），每项都是错误的。相反，以下是正确的：

- 巴塞尔 III 改革包括对无评级敞口的更细粒度方法，并减少了“机械依赖信用评级”
- 关于内部评级法，GARP 写道，“金融危机凸显了与监管资本内部模型方法使用相关的一些缺陷，其中包括信用风险内部评级法（IRB）方法。这些缺陷包括 IRB 方法的过度复杂性、各银行内部模型化的 IRB 资本要求缺乏可比性，以及在对某些资产类别建模时缺乏稳健性。为了解决这些缺陷，委员会对 IRB 方法进行了以下修订：(i) 消除了对某些资产类别使

- 用高级 IRB (A-IRB) 方法的选项；(ii) 采用了“输入”地板 (对于诸如违约概率 (PD) 和违约损失率 (LGD) 等指标), 以确保在 IRB 方法仍然可用的资产类别中模型参数的最小水平保守性；(iii) 提供了更详细的参数估计实践规范, 以减少 RWA 可变性。” — 来源: 2020 FRM 二级:《运营风险和复原力》, 第 10 版, Pearson Learning Solutions.
- 关于输入地板的规范:“修订后的 IRB 框架还引入了银行估计的 IRB 参数的最小地板值, 这些参数用作 RWA 计算的输入。这些包括 F-IRB 和 A-IRB 方法的 PD 地板, 以及 A-IRB 方法的 LGD 和 EAD 地板。在某些情况下, 这些地板由现有巴塞尔 II 地板的重新校准值组成。在其他情况下, 地板代表银行 IRB 模型的新的约束条件。” — 来源: 2020 FRM 二级:《操作风险和复原力》, 第 10 版, Pearson Learning Solutions.

考察: Operational Risk and Resiliency 重要程度: [■■□]

科目: Quantitative Analysis

4. An asset that is not very liquid is quoted bid \$89.00, offer \$96.00. Which is **nearest** to the proportional bid-offer spread?
一笔不太流动的资产的报价为买入价 \$89.00, 卖出价 \$96.00。哪一个数值与其**最接近**的相对买卖价差?

- A. 0.0378
- B. 0.0757
- C. 0.1538
- D. 7.000

答案:B

解析:

B 选项正确。 0.0757。因为 $(96.00 - 89.00)/92.50 = 0.075676$

根据 Hull (24.1.2. Measuring Market Liquidity), 两种衡量买卖价差的方法是:

- 美元买卖价差 $= p = \text{卖出价} - \text{买入价}$
- 相对买卖价差 $= s = (\text{卖出价} - \text{买入价}) / \text{中间价}$

(来源: John C. Hull, Risk Management and Financial Institutions, 5th Edition (Hoboken, NJ: John Wiley & Sons, 2018))

考察: Market Liquidity 重要程度: [■■□]

科目: Quantitative Analysis

5. Richard conducted a factor regression for his firm’s Flagship Portfolio. He used the three-factor Fama-French specification where the factors are MKT (the market), SMB (small stocks minus big stocks), and HML (value stocks minus growth stocks). His factor regression results are displayed below.

<i>Flagship excess returns, $R(p) - R_f$, Fama-French benchmark (monthly)</i>		
	Coeffi- cient	Standard Error (SE)
Alpha, α	0.300%	0.033%
MKT Loading	1.250	0.096
SMB Loading	0.640	0.043
HML Loading	-0.920	0.066
Adjusted R^2	0.290	

Please note the regression is based on monthly returns. Each of the following statements is TRUE about these factor regression results EXCEPT which is false?

Richard 对其公司旗舰组合进行了因子回归。他采用了三因子 Fama-French 模型设定，其中因子包括 MKT（市场）、SMB（小盘股减大盘股）以及 HML（价值股减成长股）。其因子回归结果如下所示。

<i>Flagship excess returns, $R(p) - R_f$, Fama-French benchmark (monthly)</i>		
	Coeffi- cient	Standard Error (SE)
Alpha, α	0.300%	0.033%
MKT Loading	1.250	0.096
SMB Loading	0.640	0.043
HML Loading	-0.920	0.066
Adjusted R^2	0.290	

请注意，该回归是基于月度收益率进行的。以下关于这些因子回归结果的陈述中，除了哪一项为假，其余均为真

- A. The portfolio’s style appears to growth-oriented, small-cap
该组合的风格似乎是成长型、小盘股
- B. This portfolio manager(s) appear to be skillful; that is, the alpha is significant
该组合经理似乎很擅长；即，alpha 是显著的
- C. The low adjusted R^2 and low standard errors imply these factors are not appropriate to this portfolio
低调整后的 R^2 和低标准误差表明这些因子不适合该组合
- D. The implied CAPM benchmark is leveraged as it borrows cash (i.e., T-bills) to invest more than 100% (i.e., > \$1.0) in the market portfolio
隐含的 CAPM 基准被杠杆化，因为它借入现金（即，T-bills）以投资超过 100%（即，> \$1.0）的市场组合

答案:C

解析:

C 选项正确。 C. 错误的：这些因子每个都是显著的，调整后的 R^2 对于实证金融来说相对较高。关于（A）、（B）和（D），每项都是正确的。

考察: Finance: Portfolio Management **重要程度:** [■■■]

科目: Quantitative Analysis

6. The riskfree rate is 5.0% while the current stock price is \$80.00. Two most liquid options both have a strike price of \$70.00 such that the ratio $K/S(0) = 0.90$. In addition to the same strike price, both options have the same maturity of three months, $T = 0.25$ years. The implied volatility of the call is 30.0%, and the implied volatility of the put is 38.0%. What is the trade that exploits this scenario, if any?

无风险利率为 5.0%，当前股票价格为 \$80.00。两种最流动的期权都以 \$70.00 的执行价格，使得比率 $K/S(0) = 0.90$ 。此外，两种期权都具有相同的到期时间，即三个月， $T = 0.25$ 年。看涨期权的隐含波动率为 30.0%，看跌期权的隐含波动率为 38.0%。如果存在这样的交易机会，请指出哪种交易策略最合适？

- A. Buy call, sell put, and short the stock
买入看涨期权，卖出看跌期权，并卖空股票

- B. Buy put, sell call, and buy the stock
买入看跌期权，卖出看涨期权，并买入股票
- C. Buy the call, buy the put, and short a straddle
买入看涨期权，买入看跌期权，并卖空跨式期权
- D. Sell the call, sell the put, and buy a straddle
卖出看涨期权，卖出看跌期权，并买入跨式期权

答案:A

解析:

A 选项正确。正确的：买入看涨期权，卖出看跌期权，并卖空股票

考察: Derivatives: Options **重要程度:** [■■■]

科目: Quantitative Analysis

7. Consider a netting set with two derivative positions: an interest rate swap (IRS) with a marginal CVA of -1.450 and a cross-currency swap (CCS) with a marginal CVA of -1.550 . If the first transaction is the cross-currency swap (CCS) and its incremental CVA is -2.100 , then what is the incremental CVA of the IRS?

考虑一个净额结算集，其中包含两个衍生品头寸：一个利率互换（IRS），其边际 CVA 为 -1.450 ，一个交叉货币互换（CCS），其边际 CVA 为 -1.550 。如果第一个交易是交叉货币互换（CCS），其增量 CVA 为 -2.100 ，那么利率互换（IRS）的增量 CVA 是多少？

- A. -0.900
- B. -1.100
- C. -1.550
- D. -2.220

答案:A

解析:

A 选项正确。总 CVA 是边际 CVA 的总和，因此必须为 $-1.450 + (-1.550) = -3.000$ 。增量 CVA 必须等于总（净额）CVA，因此，如果 IRS 是第二个，其增量 CVA 等于 $-3.000 - (-2.100) = -0.900$ 。

考察: Credit Risk **重要程度:** [■□□□□]

科目: Foundations of Risk Management

8. The central feature of Basel III's first pillar is the regulatory capital (i.e., CET1, Tier 1, and total capital) requirements which require minimum fractions of risk-weighted assets (RWA). In addition to these minimums, as of early 2019 (when the buffers were fully phased in), Basel specifies three buffers: the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB). This is all very confusing, so you ask your colleague Mary to summarize these additional buffers and their motivations. She makes the following five points:

- I. For all three, the additional buffer must be CET1
- II. For all three, breach of the buffer requirements implies the bank's ability to pay dividends will be restricted
- III. The CCB requires an additional 2.5% of CET1 capital and is meant to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred.
- IV. The additional G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} https://www.bis.org/fsi/fsisummary_sib_framework.htm and is meant to reduce the likelihood and severity of the failure of a global systemically important financial institution
- V. The CCyB requirement varies between zero and 2.5% and is meant to protect the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risks

Is Mary correct in her summary?

巴塞尔 III 第一支柱的核心特征是监管资本（即 CET1、一级资本和总资本）要求，这些要求规定了相对于风险加权资产（RWA）的最低资本比例。除这些最低要求之外，截至 2019 年初（当各类缓冲全部过渡到位时），巴塞尔还规定了三种缓冲：资本留存缓冲（CCB）、全球系统重要性银行缓冲（G-SIB 缓冲）以及逆周期资本缓冲（CCyB）。这些内容非常容易让人困惑，因此你请同事 Mary 总结这些额外缓冲及其设立动机。她给出了以下五点：

- I. 对这三类缓冲而言，额外缓冲部分都必须由 CET1 构成
- II. 对这三类缓冲而言，一旦未满足缓冲要求，银行支付股利的能力将受到限制
- III. CCB 要求额外 2.5% 的 CET1 资本，旨在确保银行拥有一层在发生损失时可以被动用的额外可用资本。
- IV. 额外的 G-SIB 要求包含五个档次 {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} [https://www.bis.org/fsi/fsisummaries/g-sibframework.h](https://www.bis.org/fsi/fsisummaries/g-sibframework.htm)，旨在降低全球系统重要性金融机构倒闭的可能性及其严重程度
- V. CCyB 要求在零和 2.5% 之间变动，旨在保护银行体系免受总量信贷过度增长时期的影响，而这种过度增长往往伴随着系统性风险的积聚

Mary 在她的总结中是否正确？

- A. No, unfortunately, none of her statements is correct
不幸的是，她的陈述没有一项是正确的
- B. I. and II. are correct, but the others (III., IV., and V.) are wrong
I. 和 II. 是正确的，但其他（III.、IV. 和 V.）是错误的
- C. III., IV., and V. are correct, but the others (I. and II.) are wrong
III.、IV. 和 V. 是正确的，但其他（I. 和 II.）是错误的
- D. Yes, all five statements are correct
是的，所有五项陈述都是正确的

答案:D

解析:

D 选项正确。是的，所有五项陈述都是正确的

- 关于正确的陈述（I.）和（II.），资本留存缓冲（CCB）、G-SIB 缓冲和逆周期缓冲（CCyB）每个都需要普通股一级资本（CET1），这是最高质量的缓冲。此外，任何一项的违约都意味着对股息支付的限制（除了其他监管工具，如对高管奖金的限制）
- 关于正确的陈述（III.），BCBS 写道，“资本留存缓冲（CCB）被引入以确保银行在发生损失时拥有额外的可用资本层，可以被提取”（见 https://www.bis.org/fsi/fsisummaries/b3_capital.htm）和 GARP 写道，“在 CCB 的情况下，大致遵循了美国资本监管中内置的促动纠正行动（PCA）系统的逻辑；即，比例开始接近最低值的银行应受到越来越严格的监管干预，以促使回到资本充足的状态。”—2020 FRM 二级：《操作风险和复原力》，第 10 版，Pearson Learning Solutions.
- 关于正确的陈述（IV.），G-SIB 要求包括五个档次 {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%}，GARP 解释说，“在 G-SIB 缓冲的情况下，大致遵循了 CCB 的逻辑，但也认识到 G-SIBs 在社会中造成巨大成本的困境（以及它们中一些损失的更高波动性）。因此，较大的缓冲被指定为进一步降低失败的可能性。”
- 关于正确的陈述（V.），“逆周期资本缓冲（CCyB）有两个理性。一个是提供一个宏观审慎的工具来抑制过热；另一个是关注资本成本。” GARP 说道。进一步，BCBS 解释说，“逆周期资本缓冲（CCyB）旨在确保银行资本要求考虑银行运营的宏观金融环境。其主要目标是使用资本缓冲来实现更广泛的宏观审慎目标，即保护银行体系免受过量信贷增长时期的影响，而这种过量信贷增长往往与系统性风险的积累有关。由于其逆周期性质，逆周期资本缓冲制度也可能首先有助于对抗信贷周期积累阶段。在衰退期间，该制度应有助于减少信贷供应可能受到监管资本要求制约的风险，这可能削弱实体经济的表现，并在银行系统中导致额外的信贷损失。” 见 <https://www.bis.org/bcb/ccyb/>

记住这三个额外缓冲的基本要求：CCB 需要 +2.5% CET1; G-SIB 需要 {+1.0%, +1.5%, +2.0%, +2.5%, or + 3.5%} 取决于银行的评分（见 <https://trtl.bz/fsb-g-sibs>）; 和 CCyB 需要在零和 2.5% 之间。

考察: Finance: Regulatory Capital 重要程度: ■■□

9. A portfolio holds 100,000 shares of a stock and this single position has a value of \$3.0 million. The stock is quoted bid \$29.00, offer \$31.00. The stock’s daily volatility is 1.43% or 143 basis points. For purposes of value at risk (VaR), we will assume the stock’s arithmetic returns are normally distributed (aka, normal VaR) and the expected daily return rounds to zero (under these assumptions absolute VaR is identical to relative VaR). Which is **NEAREST** to the position’s one-day 99.0% confident liquidity-adjusted value at risk (LVaR)?

一个投资组合持有 100,000 股股票，该单一头寸的价值为 300 万美元。股票的买卖报价为 29 美元/股和 31 美元/股。股票的日波动率为 1.43% 或 143 个基点。为了计算价值风险（VaR），我们假设股票的算术回报率是正态分布的（即，正态 VaR），并且预期日回报率近似为零（在这些假设下，绝对 VaR 与相对 VaR 相同）。哪一个选项最接近该头寸的单日 99.0% 置信水平的流动性调整价值风险（LVaR）？

- A. \$15,000
- B. \$80,000
- C. \$200,000
- D. \$1.0 million

答案:C

解析:

C 选项正确。 单日 99.0% VaR 由 $-0 + 1.43\% * 2.326 * \$3.0 \text{ million} = \$99,800$ 给出。

流动性成本（LC）由 $(31.00 - 29.00)/30.00 * 0.5 * \$3.0 \text{ million} = \$100,000$ 给出。

因此，单日 99.0% VaR 为 $99,800 + 100,000 = 199,800$

考察: Market Risk: Liquidity Risk **重要程度:** [■■■□]

科目: Quantitative Analysis

10. According to Grinold, typical or illustrative information ratios (IR) include: 0.500 is good, 0.750 is very good and 1.00 is exceptional. In regard to risk aversion, λ , 0.150 is restrained, 0.100 is moderate, and 0.050 is aggressive. Patricia is a manager with a “very good” information ratio of 0.750 and a “moderate” risk aversion of 0.10. If she shifts from this moderate risk aversion to an aggressive risk aversion of 0.050, what is the impact on the optimal level of active risk?

根据 Grinold 的说法，典型的或说明性的信息比率（IR）包括：0.500 是好的，0.750 是非常好的，1.00 是异常的。在风险厌恶程度上， λ , 0.150 是抑制的，0.100 是温和的，0.050 是激进的。帕特丽夏是一位经理，她有一个“非常好”的信息比率 0.750 和一个“温和”的风险厌恶 0.10。如果她从这种温和的风险厌恶转变为激进的风险厌恶 0.050，对最优主动风险水平有什么影响？

- A. No effect
没有影响
- B. Optimal active risk reduces by one-half
最优主动风险减少一半
- C. Optimal active risk Increases by about +41%
最优主动风险增加约 +41%
- D. Optimal active risk doubles
最优主动风险翻倍

答案:D

解析:

D 选项正确。 正确的：最优主动风险翻倍因为最优主动风险由 $IR/(2 * \lambda)$ 给出，帕特丽莎在从 $0.750/(2 * 0.10) = 3.750$ 到 $0.750/(2 * 0.050) = 7.500$ 时，最优主动风险翻倍。

考察: Portfolio Management: Fundamental Law of Active Management **重要程度:** [■■■□]

科目: Quantitative Analysis

11. For his term structure model, Peter considered the Vasicek model, but on reflection, he prefers the Cox-Ingersoll-Ross (CIR). In regard to the CIR, which of the following statements is TRUE?
- 彼得考虑使用 Vasicek 模型来构建他的期限结构模型，但他后来更喜欢 Cox-Ingersoll-Ross (CIR) 模型。在 CIR 方面，以下哪个陈述是正确的？
- A. A drawback of the CIR is that it does not mean revert to a long-term rate
一个缺点是 CIR 不能恢复到长期利率
 - B. In the CIR, the drift is time-dependent such that CIR is a no-arbitrage model
CIR 的漂移是时间依赖的，因此 CIR 是一个无套利模型
 - C. An advantage of the CIR is that the terminal distribution of the short rate is eventually (e.g., > 10 years) normally distributed
CIR 的一个优点是，短期利率的最终分布（例如，>10 年）最终是正态分布的
 - D. Unlike the Vasicek model's constant volatility, the CIR's basis point volatility is not constant, and this ensures the short rate cannot become negative
与 Vasicek 模型的恒定波动率不同，CIR 的基点波动率不是恒定的，这确保了短期利率不会变为负数

答案:D

解析:

D 选项正确。正确的：与 Vasicek 模型的恒定波动率不同，CIR 的基点波动率不是恒定的，这确保了短期利率不会变为负数。关于 (A)、(B) 和 (C)，每项都是错误的。

考察: Term Structure Models **重要程度:** [■■■□]

科目: Quantitative Analysis

12. Jason is a small business owner specializing in the wholesale of steel. Jason worked hard on his business, so he purchased insurance to help safeguard his company on January 4th, 2023. Later that month, on January 31st, he made a significant payment to secure a shipment of steel that will arrive on February 2nd, 2023. Jason delivered the steel to two clients on February 8th and 9th; however, he did not receive payment from the clients until February 13th. After receiving the proceeds Jason deposited the money at ABC bank.
- Among these transaction dates, on which did Jason increase his credit risk?
- 贾森是一家钢铁批发商。贾森努力经营他的业务，所以他在 2023 年 1 月 4 日购买了保险，以帮助保护他的公司在 1 月 31 日，他支付了一大笔钱以确保在 2 月 2 日到货的钢铁。贾森在 2 月 8 日和 2 月 9 日向两个客户交付了钢铁；然而，他直到 2 月 13 日才收到客户的付款。收到款项后，贾森将钱存入 ABC 银行。在这些交易日期中，哪一天贾森增加了他的信用风险？
- A. None of the transactions increase his credit risk
这些交易没有增加他的信用风险
 - B. His credit risk increased on the Jan dates (Jan 4th and 31st) but was unchanged on Feb dates (Feb 8th, 9th and 13th)
他的信用风险在 1 月日期（1 月 4 日和 31 日）增加，但在 2 月日期（2 月 8 日、9 日和 13 日）保持不变
 - C. His credit risk increased on the Feb dates (Feb 8th, 9th, and 13th) but was unchanged on Jan dates (Jan 4th and 31st)
他的信用风险在 2 月日期（2 月 8 日、9 日和 13 日）增加，但在 1 月日期（1 月 4 日和 31 日）保持不变
 - D. His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)
他的信用风险在所有引用的交易日期增加；即，1 月日期（4 日和 31 日）和所有 2 月日期（8 日、9 日和 13 日）

答案:D

解析:

D 选项正确。正确的：他的信用风险在所有引用的交易日期增加；即，1 月日期（4 日和 31 日）和所有 2 月日期（8 日、9 日和 13 日）

- 1 月 4 日：购买保险增加了信用风险，因为贾森购买了一份保险，如果发生事件，保险将不会交付。*contingent claims* 是一种信用风险。根据文本，“第六种类型的事务是资产索赔的特殊情况——contingent claim. 索赔取决于某些事件的发生，例如保险政策覆盖的损失。在政策开始时，政策持有者对保险公司没有索赔。然而，一旦被保险人遭受 covered loss, 被保险人有索赔。如果保险公司未能支付索赔，这将构成信用损失。” (†)
- 1 月 31 日：他预付了订单，因此有他的对手方不履行义务的可能性。根据文本，“*Prepayment of goods and services* 是一种涉及信用风险的交易类型。交付预计在某个时间以某种质量和/或性能进行，对手方的失败可能导致预付款的损失，并产生业务中断成本。” (†)
- 2 月 8 日和 9 日：贾森向他的客户交付了钢铁订单。他的任何一个客户都可能不履行向他付款的义务，因此，贾森增加了他的信用风险。根据文本，“第三种类型是产品或服务的销售，没有立即现金支付。卖方在产品发货或服务完成后向买方发送发票，买方有几周时间付款。这被称为 *account receivable*.” (†)
- 2 月 13 日：他将销售所得和上一季度的收益存入 ABC 银行。贾森将他的资产委托给第三方保管，因此，他面临银行无法偿还其资产的风险。根据文本，“第五种类型的事务涉及一方对另一方保管或管理的资产的索赔，例如银行存款。” (†)

信用风险可以通过多种方式产生。文本 (†) 列出了七种：

1. 贷款：借款人向贷款人偿还的风险。
2. 租赁：承租人无法履行未来对资产使用费的支付的风险。
3. 应收账款（产品/服务销售，没有立即现金支付）：买方在收到产品或服务后不支付的风险。
4. 预付款：对手方在收到付款后不交付或不符合质量/性能的风险。
5. 存款和银行余额：持有存款或管理资产的金融机构违约的风险。
6. contingent claims (例如，保险，养老金基金索赔)：保险公司或赞助商无法支付索赔或养老金负债的风险。
7. 衍生敞口：衍生交易（如互换或期货）未来支付义务的风险，这些义务依赖于底层资产的市场价值。

(†) Sylvain Bouteille 和 Diane Coogan-Pushner, 《信贷风险管理手册：发起、评估和管理信贷敞口》(第 2 版, 美国新泽西州霍博肯：约翰威利与子公司, 2022 年)。

考察: Finance: Credit Risk 重要程度: ■■□

科目: Quantitative Analysis

13. Peter is studying for the Financial Risk Manager (FRM) exam, and he thinks the historical evolution of the Basel regulations is confusing. In particular, he wants to better understand the difference between the original Basel I accord (aka, the 1988 Basel Accord that was implemented by 1992) and Basel II, which was finally published over ten years later in 2004. His colleague Mary explains that, in comparison to the original accord, Basel II contained four significant innovations. Specifically, in comparison to the original Basel I accord (which do include the 1995/1996 Amendments) that came before, Mary says that significant innovations in Basel II included the following:

- I. In addition to credit risk and market risk, Basel II required capital for operational risk
- II. Basel II eliminated risk weights and risk-weighted assets (RWA) and replaced them with direct calculation of risk charges
- III. Basel II contained specific requirements for supervision related to capital and risk management (Pillar 2) and required public disclosures (Pillar 3)
- IV. To fine-tune the accord’ s design, Basel II made repeated use of Quantitative Impact Studies (QIS) to which banks contributed data (i.e., feedback) that was analyzed by supervisors

Is Mary correct?

Peter 正在备考金融风险管理师 (FRM) 考试，他觉得巴塞尔法规的历史演进非常令人困惑。具体来说，他想更好地理解最初的巴塞尔 I 协议（又称 1988 年巴塞尔协议，并在 1992 年前后实施）与巴塞尔 II（于十多年后的 2004 年发布）之间的差异。他的同事 Mary 解释说，与最初的协议相比，巴塞尔 II 包含了四项重要创新。特别是，相较之前的巴塞尔 I 原始协议（其中包括 1995/1996 年的修订），Mary 认为巴塞尔 II 的重大创新包括以下内容：

- I. 除了信用风险和市场风险之外，巴塞尔 II 还要求对操作风险计提资本
- II. 巴塞尔 II 取消了风险权重和风险加权资产（RWA），改为直接计算风险资本占用
- III. 巴塞尔 II 包含与资本和风险管理相关的监管要求（第二支柱），并要求公开披露（第三支柱）
- IV. 为了对协议设计进行微调，巴塞尔 II 多次采用量化影响研究（QIS），银行提供数据（反馈），由监管方进行分析

Mary 的说法正确吗？

- A. Yes, all four statements are correct
是的，所有四个陈述都是正确的
- B. I. and II. are correct but III and IV are inaccurate
I. 和 II. 是正确的，但 III 和 IV 是不准确的
- C. Only II. is inaccurate but I, III, and IV are correct
只有 II. 是不准确的，但 I, III, 和 IV 是正确的
- D. Unfortunately, none of her statements are correct
不幸的是，她的陈述都不正确

答案:C

解析:

C 选项正确。巴塞尔 II 并没有取消风险加权资产（或风险权重）；风险加权资产（RWA）在巴塞尔 III/IV 中仍然存在。根据 GARP 的 Mark Carey，作为对原始巴塞尔 I 协议的改进，巴塞尔 II 的四项重大创新包括：更为复杂的风险权重公式；新增了操作风险资本要求；引入了第二支柱和第三支柱；以及多次采用量化影响研究（QIS）。许多专家认为新增的支柱是十分重大的进展。Carey 解释道：“三大支柱代表了各国监管实践趋同的努力，特别是在‘规则与原则’分歧的背景下。具体来说，第二支柱要求监管者让银行持有高于最低要求的资本，并建立与其风险状况相符的内部资本充足评估流程（ICAAP）……协议条款执行方面的国家自主权被削弱，国家监管机构须对其执行协议的努力保持透明，包括那些高于最低要求的措施。第三支柱则要求进行更多定性和定量的信息披露，希望市场参与者的压力能帮助提升银行的实践水平（即市场约束）。”（来源：2020 FRM Part II: Operational Risk and Resiliency，第 10 版，Pearson Learning Solutions）

考察: Finance: Basel Accords 重要程度: [■ ■ □]

科目: Quantitative Analysis

14. Assume the spot foreign exchange (FX) rate between the US dollar and the Japanese Yen is 110.00 which can be represented as USDJPY 110.00 or JPY 110.00 where USD is the base currency and JPY is the quote currency; aka, quote 110.00 yen per one dollar of the base or “per unit” currency. The USD interest rate is 2.50% and the JPY interest rate is 0.450%. The period is one year, and the compound frequency is annual. If the covered interest rate (CIP) arbitrage framework enforces a perfectly accurate one-year forward exchange rate, then what is the implied one-year “swap rate” which is here simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference, $F(\text{USDJPY}) - S(\text{USDJPY})$, or $F(\text{USDJPY}) - 110.00$?

假设美元/日元即期汇率是 110.00，可以表示为 USDJPY 110.00 或 JPY 110.00，其中 USD 是基础货币，JPY 是报价货币；即，报价 110.00 日元每单位基础货币美元。美元利率为 2.50%，日元利率为 0.450%。期限为一年，复利频率为年复利。如果套利框架中的覆盖利率（CIP）强制执行一个完美准确的远期汇率，那么什么是隐含的一年“掉期率”，这里 simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference, $F(\text{USDJPY}) - S(\text{USDJPY})$, or $F(\text{USDJPY}) - 110.00$?

- A. -2.20 JPY (or -2.20 USDJPY)
- B. Zero
- C. + 5.80 JPY
- D. + 13.70 JPY

答案:A

解析:

A 选项正确。正确的： $F - S = -2.20$ USDJPY

如果 CIP 被强制执行，那么 $F - S = S \times [(1 + Rf_JPY)/(1 + Rf_USD) - 1] = 110.00 \times [(1 + 0.45\%)/(1 + 2.50\%) - 1] = -2.20$ 。

考察: Financial Markets and Products **重要程度:** [■■■]

科目: Quantitative Analysis

15. Volatility is an important risk factor according to Andrew Ang. The strong negative relationship between VIX changes and stock returns is explained by (at least) two dynamics. The first is called the leverage effect. The “second channel is a time-varying risk premium story and is the one the basic CAPM [capital asset pricing model] advocates,” explains Ang.(†)

Assume the riskfree rate is 3.0% while the equity risk premium (EPR; aka, the market’s expected excess return) is 6.0% because the market’s expected return is 9.0%. According to the CAPM, and consistent with Ang’s theory on the transmission channel between volatility and stock returns, which of the following should cause a DECLINE in a stock’s price?

根据 Andrew Ang 的说法，波动性是一个重要的风险因素。VIX 变化和股票回报之间的强负相关关系至少由两个动态解释。第一个被称为杠杆效应。第二个通道是一个时间变化的风险溢价故事，是基本 CAPM（资本资产定价模型）倡导的。Ang 解释说：假设无风险利率为 3.0%，而股权风险溢价（EPR；即，市场预期超额回报）为 6.0%，因为市场预期回报为 9.0%。根据 CAPM，并且与 Ang 关于波动率和股票回报之间的传导渠道的理论一致，以下哪个应该导致股票价格下降？

- A. The VIX Index declines from 21.00 to 19.00
VIX 指数从 21.00 下降到 19.00
- B. The market’s volatility decreases and the stock’s volatility decreases along with it
市场波动率下降，股票波动率也随之下降
- C. The stock’s volatility decreases, which corresponds to an increase in its leverage via the leverage effect
股票波动率下降，对应杠杆效应增加
- D. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged
股票波动率增加（伴随着与市场的相关性增加），而市场波动率保持不变

答案:D

解析:

D 选项正确。股票的波动率增加（伴随着与市场的相关性增加），而市场波动率保持不变。

股票的波动率增加（伴随着与市场的相关性增加），而市场波动率保持不变（aka, ceteris paribus），意味着股票的 beta 增加。这增加了股票的预期回报。因此，股票的价格应该下降。

Ang（强调 ours）：“VIX 变化和股票回报之间的相关性是-39%，所以当波动率上升时，股票表现不好。波动率和回报之间的负关系被称为杠杆效应。当股票回报下降时，公司的财务杠杆增加，因为债务大致保持不变，而股权的市场价值下降。这使得股权风险更大，增加了它们的波动性。另一个通道是高波动率导致低股票回报：**波动率上升提高了投资者要求的股权回报，也导致股票价格下降。**这个第二个通道是一个时间变化的风险溢价故事，是基本 CAPM 倡导的：随着市场波动率增加，折现率增加，股票价格必须今天下降，以便未来股票回报可以高。” (†)

关于（A）、（B）和（C），每项都是错误的。

(†) **Andrew Ang**, 《资产管理：因子投资系统方法》（纽约：牛津大学出版社，2014 年）。

考察: Finance: CAPM **重要程度:** [■■■]

科目: Quantitative Analysis

16. An analyst fits a Vasicek model when the current short-term rate is 4.60%. If the theta, θ , parameter is 13.50%, then each of the following statements is true EXCEPT which is false?

分析师在当前短期利率为 4.60% 时拟合 Vasicek 模型。如果 theta 参数为 13.50%，那么以下哪个陈述是正确的，除了哪个是错误的？

- A. A key advantage of the Vasicek model is that it is a relatively simple model
Vasicek 模型的一个关键优势是它是一个相对简单的模型

- B. If the half-life is nine years, then we might expect the short rate to reach $\sim 9.0\%$ in nine years or so
如果半衰期为九年，那么我们可能期望短期利率在九年内达到 $\sim 9.0\%$
- C. The standard deviation of the short rate in ten years is less than it would be without mean reversion
十年期短期利率的标准差小于没有均值回归的情况
- D. Because it is a multi-factor model, the Vasicek is popularly applied to hedge portfolios that contain short-, medium- and long-term bonds
因为它是一个多因素模型，Vasicek 被广泛应用于对包含短期、中期和长期债券的对冲组合

答案:D

解析:

D 选项正确。错误的：Vasicek 是一个单因素模型；为了对冲短期、中期和长期债券，需要例如三因素 Gauss+ 模型。

关于 (A)、(B) 和 (C)，每项都是正确的。

考察: Fixed Income: Vasicek Model **重要程度:** [■■□]

科目: Quantitative Analysis

17. Firm A, Firm B, and Firm C are three companies in the bottled water industry. All companies have a mix of debt and equity in their capital structure. In 2022 all three companies had credit losses that lowered EBIT. Below is a summary of their financial information for the year 2022.

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for Loan Loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%

Given the above information, which firm has the higher return on equity (ROE)?

公司 A、公司 B 和公司 C 是瓶装水行业的三家。所有公司都有债务和股权的混合资本结构。在 2022 年，三家公司的信用损失都降低了 EBIT。以下是 2022 年他们的财务信息总结。

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for Loan Loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%

给出以上信息，哪家公司拥有更高的净资产收益率 (ROE)?

- A. Firm C has the highest ROE due to its lower equity position.
公司 C 拥有最高的 ROE，因为它有较低的股权地位。
- B. Firm A had the higher ROE because of its large debt position.
公司 A 拥有更高的 ROE，因为它有较大的债务地位。

C. Without credit losses, firm A would have had the highest ROE.

如果没有信用损失，公司 A 将拥有最高的 ROE。

D. Without credit losses, firm B would have had the highest ROE.

如果没有信用损失，公司 B 将拥有最高的 ROE。

答案:A

解析:

A 选项正确。正确的：公司 C 拥有最高的 ROE，因为它有较低的股权地位。

- 公司 A 拥有负的 ROE (−0.33%)，因为它有较大的债务地位（如下图所示）。即使遭受信用损失，公司仍必须偿还其债务持有者。公司 A 遭受了严重的信用损失，降低了 EBIT。然而，即使调整了其信用损失，公司 A 仍然会比公司 B 和 C 拥有更小的 ROE。
- 公司 B 拥有正的 ROE，但 ROE 非常小，因为它与公司 C 和 A 相比有较大的股权。公司 A 遭受了较大的信用损失，降低了 EBIT。然而，即使调整了其信用损失，公司仍然会比公司 C 拥有更小的 ROE。
- 公司 C 拥有最高的 ROE，因为它有较小的股权和适度的信用损失。

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for loan loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%
Solution			
EBIT	\$75,000	\$75,000	\$75,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	0	\$8,610.00	\$8,610.00
NI	(\$10,000.00)	\$32,390.00	\$32,390.00
ROE	-0.33%	0.81%	1.12%
Without Credit Loss			
EBIT	\$125,000	\$90,000	\$76,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	\$8,400.00	\$11,760.00	\$8,820.00
NI	\$31,600.00	\$44,240.00	\$33,180.00
ROE	1.05%	1.11%	1.14%

考察: Finance: Financial Statement Analysis 重要程度: [■ ■ □]

科目: Quantitative Analysis

18. The three lines of defense model (3LoD) has existed in some variation for over 30 years, although it achieved more visibility in 2013 when the Institute of Internal Auditors (IIA) published its initial position paper on the 3LoD, which it updated in 2021 (The IIA’s Three Lines Model, an update of the Three Lines of Defense). GARP recommends firms refer to this update because the IIA “has refined the model to reflect the evolution of risk management within organizations and aim to foster closer collaboration between business functions including internal audit.” In regard to the 3LoD model, each of the following is true EXCEPT which is false?

三道防线模型 (3LoD) 已经存在了 30 多年，尽管它在 2013 年获得了更多的关注，当时 Institute of Internal Auditors (IIA) 发布了其最初的立场文件，该文件在 2021 年进行了更新 (The IIA’s Three Lines Model, an update of the Three Lines of Defense)。GARP 建议公司参考这个更新，因为 IIA “已经完善了模型，以反映组织内部风险管理的发展，并旨在促进业务职能之间的更紧密合作，包括内部审计”。关于三道防线 (3LoD) 模型，下列各项陈述均为真，除了哪一项是假的

- A. The first line defines the organization’s risk appetite but should not attempt to directly influence its risk culture
第一道防线定义了组织的风险偏好，但不应试图直接影响其风险文化
- B. The first line is responsible for implementing and maintaining effective controls to manage material operational risks
第一道防线负责实施和维护有效的控制，以管理重要的运营风险
- C. The second line is part of management (and owns the ORM methodology) but provides support to the first line in order to help the firm create and protect value
第二道防线是管理的一部分（并拥有 ORM 方法论），但为第一道防线提供支持，以帮助公司创造和保护价值
- D. The third line has the clearest boundaries due to its independence
第三道防线由于其独立性，边界最清晰

答案:A

解析:

A 选项正确。错误的：两项。相反，以下是正确的：风险偏好（即风险偏好声明）由高级管理层开发并由董事会批准。风险文化不是唯一责任，而是公司实际 ORMF 的基础，因此所有员工都对文化承担一定责任。

关于（B）、（C）和（D），每项都是正确的。

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

19. Liquidity Early Warning Indicators (EWI) can be compared to an automobile’s dashboard signal system where the appearance of a red light points our attention to something that may become a problem if not addressed. Each of the following is a credible red indicator (aka, red flag) in a Liquidity Early Warning Indicator (aka, Liquidity EWI) dashboard EXCEPT which is not a red (red flag) liquidity indicator?

流动性早期预警指标 (EWI) 可以与汽车仪表盘信号系统进行比较，其中红色灯光的出现提醒我们注意可能成为问题的事情，如果不及早解决。以下各项都是流动性早期预警指标（流动性 EWI）仪表盘中的可信红色指示器（红色标志），除了哪一项不是红色（红色标志）流动性指示器？

- A. Narrowing debt/CDS spreads
债务/CDS 利差缩小
- B. Sudden increase in debt costs
债务成本突然增加
- C. Sudden growth in assets accompanied by volatile liabilities
资产突然增长伴随着波动性负债
- D. Rapid decline in the weighted average maturity of liabilities
负债加权平均期限突然下降

答案:A

解析:

A 选项正确。错误的：流动性 EWI 是：债务/CDS 利差扩大（与债务成本突然增加一致）

关于（B）、（C）和（D），每项都是流动性 EWI 的候选，根据研究笔记：“巴塞尔银行监管委员会（BCBS）建议将以下措施作为流动性早期预警指标（EWI）。这些措施包括 [重点摘要如下]：

- 资产的异常增长，尤其是伴随着负债的波动性增强时；
- 债务（信用）利差扩大，和/或信用违约掉期（CDS）利差扩大；
- 资产和负债结构多样性的下降；货币错配情况加剧；
- 负债加权平均到期期限下降时；
- 持仓超出或接近监管限额时；
- 某些产品线出现不利趋势；银行的财务状况恶化；负面新闻报道；

- 信用评级被下调；股价下跌；
- 债务成本上升；零售和/或批发融资成本上升；
- 交易对手对银行的财务状况产生紧张情绪；
- 信贷额度被下调；
- 零售存款外流加速；定期存单（CDs）被大量提前赎回；
- 长期融资渠道变得更困难；短期负债发行变得更困难”

考察: Liquidity Risk Management **重要程度:** [■ ■ □]

科目: Quantitative Analysis

20. In factor theory, an asset’s risk premium (aka, expected excess return) is given by $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where the asset’s beta is with respect to the stochastic discount factor (SDF; aka, pricing kernel) and (m) in an index of bad time. Let’s make the following assumptions about an asset:

- The riskfree rate is 3.0% such that $E(m) = 1/1.03 = 0.9709$
- The variance (m) equals $30.0\%^2 = 0.090$
- The covariance (r_i, m) equals -0.10680

Which is **nearest** to the asset’s risk premium which refers to the asset’s expected excess return, $E(r_i) - R_f$?

在因子理论中，资产的风险溢价（aka，预期超额回报）由 $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ 给出，其中资产的 beta 是相对于随机贴现因子（SDF；aka，定价核）和 (m) 中的一个指数的坏时间。让我们对资产做出以下假设：

- 无风险利率为 3.0%，使得 $E(m) = 1/1.03 = 0.9709$
- 方差 (m) 等于 $30.0\%^2 = 0.090$
- 协方差 (r_i, m) 等于 -0.10680

哪一个选项最接近资产的风险溢价，即资产的预期超额回报， $E(r_i) - R_f$?

- A. -5.0%
- B. +4.0%
- C. +8.0%
- D. +11.0%

答案:D

解析:

D 选项正确。 The expected excess return, $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where $\beta(i, m) = \text{cov}(r_i, m)/\text{var}(m)$ and $\lambda(m) = -\text{var}(m)/E(m)$.

In this case, $\beta(i, m) = \text{cov}(r_i, m)/\text{var}(m) = -0.10680/0.090 = -1.186667$, and $\lambda(m) = -\text{var}(m)/E(m) = -0.090/0.9709 = -0.0926975$.

Therefore, $E(r_i) - R_f = \beta(i, m) \times \lambda(m) = -1.186667 \times -0.0926975 = 0.1100011$ or about 11.0%.

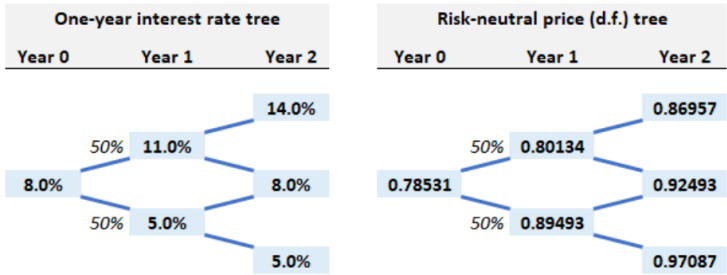
考察: Finance: Factor Theory **重要程度:** [■ ■ □]

科目: Quantitative Analysis

21. Below (in the left-hand panel) is an interest rate tree for the one-year rate where the initial rate of 8.0% jumps 300 basis points to either 11.0% or 5.0% next year. In two years, the one-year rate will be either 2.0%, 8.0%, or 14.0%.

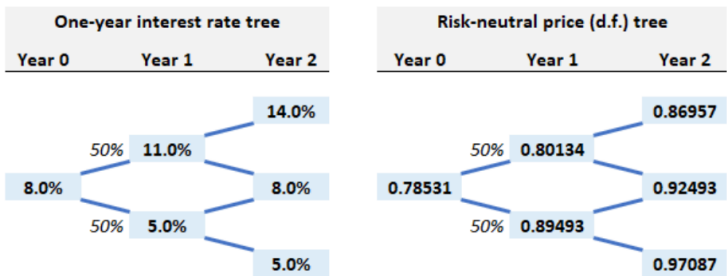
Next, we will assume a three-year zero-coupon bond. Without a risk premium, for example, the date 0, state 2 price would be $0.877193 = 1/(1 + 14\%)$. However, we will instead assume a risk premium. Specifically, investors require compensation of 50

basis points for each year of duration risk. The right-hand panel displays the associated risk-neutral (i.e., , assumes the risk premium) price tree for this three-year zero-coupon bond



In regard to the three-year zero-coupon bond, which of the following is true?

下面（左侧面板）给出了一个一年期利率树，初始利率为 8.0%，在下一年将跳升 300 个基点到 11.0%，或下降至 5.0%。在两年后，一年期利率将变为 2.0%、8.0% 或 14.0%。接下来，假设一只三年期零息债券。在无风险溢价的情况下，例如，时点 0、状态 2 的价格为 $0.877193 = 1/(1 + 14\%)$ 。但是，我们在此假设存在风险溢价：具体来说，投资者要求每一年久期风险获得 50 个基点的补偿。右侧面板展示了这只三年期零息债券的对应风险中性（即考虑风险溢价）的价格树。



关于这只三年期零息债券，下列哪项说法是正确的？

- A. The expected return over the next year is 7.50%
接下来一年的预期回报是 7.50%
- B. The expected return over the next year is 9.00%
接下来一年的预期回报是 9.00%
- C. The implied 3-year spot rate term structure is downward sloping (aka, inverted)
隐含的三年期即期利率期限结构是向下倾斜的（aka，倒置）
- D. Jensen’s inequality implies that expected excess return over two years is less than 50 basis points
詹森不等式意味着两年期预期超额回报小于 50 个基点

答案:B

解析:

B 选项正确。正确：接下来一年的预期回报是 9.00%。

我们不需要套用公式。每一年久期风险的风险溢价为 50 个基点，而这只三年期债券具有两年的久期风险。因此预期回报比 8.0% 高出 100 个基点。

考察: Fixed Income 重要程度: [■][■][□]

科目: Quantitative Analysis

22. Old Crow Bank had a credit committee meeting yesterday, and the following is a copy of the notes taken during the meeting:

- All members are present Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), and Ashton (Chief Financial Officer).
- Jan initiated the meeting by confirming if everyone had received an email containing approval documents from credit officer Doug, who had originated the loan. Everyone confirmed receiving the email with the documents yesterday.

- Jan provided an overview of the loan for approximately five minutes, after which she proceeded to inquire about the opinions of those present. It was the consensus of all individuals, except for Alyssa, that the loan be approved. Alyssa expressed concern regarding the trustworthiness of the counterparty.
- Jan respects Alyssa’s opinion but disagrees and initiates a vote. Two other committee members agree with Jan, resulting in loan approval.

What is the most probable issue with the credit committee’s meeting?

老鹰银行昨天召开了信用委员会会议，以下是会议记录的副本：

- 所有成员都出席了：Jan（首席风险官）、Eddie（合规主管）、Alyssa（法律主管）、Don（税务主管）和 Ashton（首席财务官）。
- Jan 首先确认每个人是否收到了一封包含信用官员 Doug 批准文件的电子邮件。每个人都确认收到了带有这些文件的电子邮件。
- Jan 用大约五分钟的时间介绍了贷款情况，然后询问了与会人员的意见。除了 Alyssa 之外，所有人都同意批准贷款。Alyssa 对交易对手的可靠性表示担忧。
- Jan 尊重 Alyssa 的意见，但不同意，并发起投票。另外两名委员会成员也同意 Jan，导致贷款获得批准。

信用委员会会议最可能存在的问题是什么？

- A. The approval documents should have been distributed earlier.
批准文件应该更早分发。
- B. Jan monopolized the conversation and did not properly facilitate discussion.
Jan 垄断了整个对话，没有起到恰当的引导讨论作用
- C. The meeting was not properly documented.
会议没有恰当的证明文件。
- D. The vote was not unanimous, so the loan should have been rejected.
投票结果并非一致通过，因此这笔贷款本应被否决

答案:C

解析:

C 选项正确。在这个场景中，所有批准文件都在会议前恰当地分发给委员会。Jan 确实花了时间向委员会解释投资，但她还帮助促进了委员会成员之间的讨论，并听取了每个人的意见。投票不必一致通过，但除非决定一致通过，否则应该举行投票。然而，会议应该准确记录，以防投资出现问题。在不良交易的情况下，审计师希望知道会议中讨论了什么，以及投资决策背后的推理。

考察: Credit Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

23. A bank is using the VaR and stressed VaR (SVaR) market risk framework according to the Basel 2.5 guidelines. The bank’s internal models for market risk have generated the following risk measures for the current trading book positions (VaR and SVaR are in USD millions):

Multiplicative factors, m and m(S)				3.0
Confidence Level	Previous day		60-day average	
	VaR	SVaR	VaR	SVaR
	(mm)	(mm)	(mm)	(mm)
95.00%	\$110.0	\$190.0	\$127.0	\$238.0
99.00%	185.0	360.0	140.0	270.0
99.90%	283.0	740.0	228.0	490.0

The supervisor has set both multiplication factors, m and m(S), to 3.0. What is the correct capital requirement for general market risk for the bank under Basel 2.5? (note: this question inspired by Question 42 of GARP’s 2020 Part 2 Practice Exam). 一家银行正在按照巴塞尔 2.5 指南使用 VaR 和压力 VaR (SVaR) 市场风险框架。该银行的内部市场风险模型为当前交易头寸 (VaR 和 SVaR 以百万美元为单位) 生成了以下风险衡量：

Multiplicative factors, m and m(S)				3.0
Confidence Level	Previous day		60-day average	
	VaR	SVaR	VaR	SVaR
	(mm)	(mm)	(mm)	(mm)
95.00%	\$110.0	\$190.0	\$127.0	\$238.0
99.00%	185.0	360.0	140.0	270.0
99.90%	283.0	740.0	228.0	490.0

监管机构将乘数 m 和 m(S) 都设置为 3.0。在巴塞尔 2.5 下，该银行的一般市场风险资本要求是多少？（注意：这个问题受到了 GARP 2020 Part 2 模考题第 42 题的启发）

- A. \$630.0 million
- B. \$810.0 million
- C. \$1,230.0 million
- D. \$2,154.0 million

答案:C

解析:

C 选项正确。正确的：\$1,230.0 million

巴塞尔 2.5 市场风险资本要求需要 99.0% 的置信水平，由以下公式给出：

$$MRC = \max[VaR(t - 1), m \times VaR(60\text{-day avg})] + \max[SVaR(t - 1), m(S) \times SVaR(60\text{-day avg})]$$

在这种情况下，

$$MRC = \max[185.0, 3.0 \times 140.0] + \max[360.0, 3.0 \times 270.0] = \$1,230.0 \text{ million.}$$

考察: Market Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

24. If a corporation’s marginal tax rate is 30.0% and it purchases an A-rated municipal bond that offers a 4.20% gross yield (aka, yield to maturity), what is the tax equivalent yield (TEY)? 如果一家公司的边际税率是 30.0%，它购买了一只 A 级市政债券，该债券提供 4.20% 的毛收益率 (aka，到期收益率)，那么税等效收益率 (TEY) 是多少？

- A. 2.45%
- B. 3.50%
- C. 6.00%
- D. 16.67%

答案:C

解析:

C 选项正确。正确的：6.00%。TEY = after-tax return on tax-exempt investment/(1 – marginal tax rate); in this case, TEY = 4.20%/(1 – 0.30) = 6.00%

考察: Finance: Fixed Income **重要程度:** [■□□□□]

科目: Foundations of Risk Management

25. Andrew Ang introduces factors and divides them primarily into two types: macro, fundamental-based factors versus investment-style factors. He writes, “Factors drive risk premiums. One set of factors describes fundamental, economywide variables like growth, inflation, volatility, productivity, and demographic risk. Another set consists of tradeable investment styles like the market portfolio, value-growth investing, and momentum investing.” (†)

Ang also claims that the three most important macro factors are growth, inflation, and volatility. His evaluation of these macro factors is based on a long-term historical sample (specifically, 1952:Q1 to 2011:Q4). It is important to qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window.

In regard to his historical analysis, each of the following statements is true EXCEPT which is false?

安德鲁·安（Andrew Ang）介绍了因素并将其主要分为两类：宏观、基本面因素 versus 投资风格因素。他写道：“因素驱动风险溢价。一组因素描述了基本面的、经济范围的变量，如增长、通货膨胀、波动性、生产率和人口风险。另一组包括可交易的、投资风格，如市场投资组合、价值增长投资和动量投资。” (†)

安还声称，三个最重要的宏观因素是增长、通货膨胀和波动性。他对这些宏观因素的评估是基于一个长期的历史样本（具体来说，1952 年第一季度至 2011 年第四季度）。qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window. 重要的是要限定样本窗口，因为我们不能确定过去是序言；例如，利率在这段时间内经历了两个长期的大趋势。

关于他的历史分析，下列各项陈述均为真，除了哪一项是假的？

- A. Government and investment-grade bonds performed BETTER during economic recessions than during expansions
政府和投资级债券在经济衰退期间表现优于在扩张期间
- B. Both large and small (cap) stocks perform significantly BETTER during economic expansions than during recessions
无论是大型还是小型（cap）股票，在经济扩张期间表现显著优于在经济衰退期间
- C. During periods of high inflation, all five asset classes perform significantly BETTER than during periods of low inflation
在高通货膨胀期间，所有五个资产类别都比在低通货膨胀期间表现显著更好
- D. All five assets classes are much MORE VOLATILE during recessions (or periods of low GDP growth) than during expansions
在衰退期间（或低 GDP 增长期间），所有五个资产类别都比在扩张期间更不稳定

答案:C

解析:

C 选项正确。错误的：为了正确，这应该改为：在高通货膨胀期间，所有五个资产类别都比在低通货膨胀期间表现显著更差。关于（A）、（B）和（D），每项都是正确的。

考察: Factor Investing **重要程度:** [■■□]

科目: Financial Markets and Products

26. Below is the price tree for a stylized constant-maturity Treasury (CMT) swap struck at 2.0% with a notional of \$1.0 million. The risk-neutral probabilities are displayed, specifically: 80.09% (d = 19.91%) from date 0 to date 1; and 65.20% (d = 34.80%) from date 1 to date 2. Using these risk-neutral probabilities returns an expected discounted value (EDV) of \$3,702.11 for the value of the CMT swap. Please notice this exactly matches Tuckman’s Figure 7.8 in his 4th edition value. This is the so-called model value.

Notional	\$1,000,000	
Strike rate	2.00%	
<u>i-rate evolution</u>	<u>T= 0.5</u>	<u>T= 1.0</u>
		3.00%
	2.50%	2.00%
2.0%	1.50%	1.00%
Risk-neutral	80.09%	65.20%
Probabilities:	19.91%	34.80%
<u>Price tree: 2 dates</u>	<u>T= 1.0</u>	
		\$5,000.00
	\$5,719.52	\$0.00
\$3,702.11	-\$4,227.29	-\$5,000.00
\$3,694.81	← Observed Market Price	
0.25%	← OAS	

Assume the observed market price of this CMT swap is \$3,694.81 (as highlighted in purple), which is \$7.30 less than the market price. This implies an option-adjusted spread (OAS) of 25 basis points. Which of the following is the **best** interpretation of this OAS?

下面给出了一个简化的恒定到期国债（CMT）掉期的价格树，执行利率为 2.0%，名义本金 \$1.0 百万。给出了风险中性概率：从时点 0 到时点 1 为 80.09%（d = 19.91%）；从时点 1 到时点 2 为 65.20%（d = 34.80%）。使用这些风险中性概率，CMT 掉期的预期贴现价值（EDV）为 \$3,702.11。请注意，这与 Tuckman 第四版图 7.8 的数值完全一致。这就是所谓的模型价值。

Notional	\$1,000,000	
Strike rate	2.00%	
i-rate evolution	T= 0.5	T= 1.0
		3.00%
	2.50%	2.00%
2.0%	1.50%	1.00%
Risk-neutral	80.09%	65.20%
Probabilities:	19.91%	34.80%
Price tree: 2 dates	T= 1.0	
		\$5,000.00
	\$5,719.52	\$0.00
\$3,702.11	-\$4,227.29	-\$5,000.00
\$3,694.81	← Observed Market Price	
0.25%	← OAS	

假设这只 CMT 互换的市场价格为 \$3,694.81（紫色突出显示），比市场价格低 \$7.30。这意味着一个选项调整的利差（OAS）为 25 个基点。下列哪一项是对这个 OAS 的最佳解释？

- A. The model returns the market price if we adjust the risk-neutral probabilities to 55.09% and 44.91%
模型返回市场价格，如果我们调整风险中性概率为 55.09% 和 44.91%
- B. The model returns the market price if we assume adjusted cash flows; e.g., \$3,750 instead of \$5,000 on (state 2, date 2)
模型返回市场价格，如果我们假设调整后的现金流；例如，(state 2, date 2) 上的 \$3,750 而不是 \$5,000
- C. The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)
模型返回市场价格，如果我们使用贴现率为 2.25%（时点 0，状态 0）、1.75%（时点 1，状态 0）和 2.75%（时点 1，状态 1）
- D. We cannot interpret the OAS here because this is a CMT swap, and the OAS requires an embedded option; e.g., a bond with an embedded call option
我们在这里无法解释 OAS，因为这个是 CMT 互换，而 OAS 需要一个嵌入式选项；例如，一个嵌入式看涨期权的债券

答案:C

解析:

C 选项正确。正确的：模型返回市场价格，如果我们使用贴现率为 2.25%（时点 0，状态 0）、1.75%（时点 1，状态 0）和 2.75%（时点 1，状态 1）

考察: Fixed Income Analysis 重要程度: [■ ■ □]

科目: Quantitative Analysis

27. Strongcore Financial Corporation has purchased a long-term at-the-money (ATM) call option from a risky counterparty. The option's stock and strike price are \$25.00 million. The underlying asset's volatility, σ , is estimated to be 20.0% while the

risk-free rate is 3.0% per annum. The European-style option matures in 3.0 years. In regard to the standard normal cumulative distribution function, in this scenario $N(d1)$ equals 0.66750 and $N(d2)$ equals 0.53450.

With respect to its counterparty risk, Strongcore assigns a hazard rate estimate of 6.0% and believes its counterparty could default at any time during the three-year option period; i.e., the cumulative default probability is relevant for this approximation. Further, if its counterparty defaults, Strongcore assigns a conservative recovery rate estimate of 30.0%.

In summary, the assumptions are: $S = K = \$25.0$ million, $\sigma = 20.0\%$, $R_f = 3.0\%$, $T = 3.0$ years; λ (i.e., hazard rate) = 6.0%, and $LGD = 70.0\%$.

In this situation, Strongcore does NOT want to compute the standard credit value adjustment (CVA) formula which is the summation of discounted expected exposures, $EE(t)$, over multiple future periods. Instead, Strongcore wants to rely on the simpler “semi-analytical method” that produces an estimate as a function of the position’s current value, the cumulative default probability, $PD(0,T)$, and the loss given default, LGD . Under this simpler approximation, which of the following is **nearest** to the CVA charge for the long option position?

Strongcore Financial Corporation 从一家存在信用风险的交易对手处购买了一份长期平值（ATM）看涨期权。该期权的标的股票价格和执行价均为 \$25.00 百万。标的资产波动率 σ 估计为 20.0%，无风险利率为年化 3.0%。该欧式期权到期时间为 3.0 年。关于标准正态分布的累积分布函数，在本情境下 $N(d1)$ 等于 0.66750, $N(d2)$ 等于 0.53450。就交易对手风险而言，Strongcore 给出的违约强度 (hazard rate) 估计为 6.0%，并认为其交易对手可能在三年期权期内的任意时点发生违约；即，对本近似而言需要使用累积违约概率。进一步地，如果交易对手违约，Strongcore 采用 30.0% 的保守回收率估计。总结假设如下： $S = K = \$25.0$ 百万， $\sigma = 20.0\%$ ， $R_f = 3.0\%$ ， $T = 3.0$ 年； λ （即违约强度）= 6.0%，且 $LGD = 70.0\%$ 。在此情况下，Strongcore 不想计算标准的信用估值调整（CVA）公式，即对多个未来时点的贴现预期敞口 $EE(t)$ 求和。相反，Strongcore 希望使用更简化的“半解析方法”，该方法给出一个依赖于头寸当前价值、累积违约概率 $PD(0,T)$ 以及违约损失率 LGD 的估计。在这种更简单的近似下，下列哪一个数值与该多头期权头寸的 CVA 费用 **最接近**？

- A. \$15,600
- B. \$90,000
- C. \$516,000
- D. \$1.1 million

答案:C
解析:
C 选项正确。\$516,000

$CVA = -LGD \times PD(0,T) \times V(t)$ 。本题中， $V(t) = S \times N(d1) - K \times \exp(-rt) \times N(d2) = 25 \times 0.66750 - 25 \times \exp(-0.030 \times 3) \times 0.53450 = \4.475 百万美元。累积 $PD = 1 - \exp(-0.060 \times 3) = 16.473\%$ 。因此，CVA 的近似计算为 $70\% \times 16.473\% \times 4.475$ 百万美元 = \$0.5160 百万美元，约为 \$516,002。

考察: Credit Risk 重要程度:
科目: Quantitative Analysis

28. Andrew is preparing to make a presentation to the board. The presentation’s goal is to introduce a new, proposed operational risk management (ORM) framework and get the board’s feedback on the way to eventual approval. It happens to be the case that his company is publicly traded with a high cost of capital due to its above-average beta. The board members are generally more familiar with market and credit risk but less familiar with operational risk. Therefore, his presentation will review several of the features (and nature) of operational risk, including its dynamic evolution. His presentation will briefly explore the skewed, heavy-tailed feature of operational risk. For candidate frequency distributions, he will illustrate the Poisson and negative binomial. For candidate severity distributions, he will illustrate the log-normal and generalized Pareto Distribution (GPD).

In addition to these features (i.e., dynamic, heavy-tailed), which of the following is TRUE about operational risk?
安德鲁正在准备向董事会做一次汇报。这次汇报的目的是介绍一个新的、拟议的运营风险管理（ORM）框架，并获得董事会的反馈，以便最终获得批准。事实证明，他的公司是一家公开交易的公司，由于其高于平均水平的贝塔系数，其资本成本较高。董事会成员对市场和信用风险比较熟悉，但对运营风险不太熟悉。因此，他的汇报将回顾运营风险的一些特征（及其性质），包括其动态演变。他的汇报将简要探索运营风险的偏斜、重尾特征。对于候选频率分布，他将说明泊松分布和负二项分布。对于候选严重分布，他将说明对数正态分布和广义帕累托分布（GPD）。

除了这些特征（即动态、重尾）外，下列哪一项关于运营风险的说法是正确的？

- A. The easiest way for a high-beta firm to lower its beta is to centralize the operational risk management (ORM) function
高贝塔公司降低其贝塔系数的最简单方法是集中运营风险管理（ORM）功能
- B. Operational risk is highly homogeneous because so-called boundary events define clusters that contain loss events and their shared (aka, common) features
运营风险高度同质化，因为所谓的边界事件定义了包含损失事件和它们的共享（aka，共同）特征的集群
- C. ORM is valuable because it reduces regulatory capital, and regulatory capital is a good indicator of the firm’s quality of risk management
ORM 是有价值的，因为它减少了监管资本，而监管资本是衡量公司风险管理质量的良好指标
- D. Most causes of operational risk trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes
大多数运营风险的原因归因于控制弱点；人类偏见和行为失败；以及/或 operating environment changes

答案:D

解析:

D 选项正确。大多数运营风险的原因归因于控制弱点；人类偏见和行为失败；以及/或 operating environment changes 关于（A）、（B）和（C），每项都是错误的。

- 运营风险主要是异质性的，并在公司内部扩散。Beta 是一种系统性风险。
- 运营风险高度异质但相互关联。此外，边界事件是“风险事件，它们表现为不同于它们导致的事件”–Source: GARP Introduction to Operational Risk and Resilience, Section 1.5)
- 根据 GARP, “监管运营风险资本对公司的风险状况不敏感，因此没有真正的激励来优化运营风险的管理。事实上，监管资本通常是风险管理质量的糟糕指标。”–Source: Chapter (GARP Introduction to Operational Risk and Resilience, Section 1.7)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

29. Over the next 24 hours, Greenlux State Bank estimates that the following cash inflows and outflows (all figures in millions) will occur:

Deposit withdrawals	\$70.0
Deposit inflows	\$100.0
Scheduled loan repayments	\$60.0
Acceptable loan requests	\$90.0
Borrowings from the money market	\$80.0
Sales of bank assets	\$30.0
Stockholder dividend payments	\$20.0
Revenues from sale of nondeposit services	\$10.0
Repayment of bank borrowings	\$50.0
Operating expenses	\$40.0

What is the bank’s projected net liquidity position?

通过以下现金流入和流出（所有金额以百万为单位），格林卢克斯州银行预计将在接下来的 24 小时内发生：

Deposit withdrawals	\$70.0
Deposit inflows	\$100.0
Scheduled loan repayments	\$60.0
Acceptable loan requests	\$90.0
Borrowings from the money market	\$80.0
Sales of bank assets	\$30.0
Stockholder dividend payments	\$20.0
Revenues from sale of nondeposit services	\$10.0
Repayment of bank borrowings	\$50.0
Operating expenses	\$40.0

银行预计的净流动性头寸是多少？

- A. -30.0 million
- B. +10.0 million
- C. +40.0 million
- D. +90.0 million

答案:B

解析:

B. TRUE: +10.0 million

流入包括：存款流入 = 100+

计划贷款偿还 = 60+

从货币市场借款 = 80+

银行资产的出售 = 30+

非存款服务销售收入 = 10;

总流入为 $\$100 + \$60 + \$80 + \$30 + \$10 = 280.0$ 百万。

流出包括：

存款提取 = 70+

可接受的贷款请求 = 90+X

股东分红支付 = 20+

银行借款的偿还 = 50+

运营支出 = 40;

总流出为 $\$70 + \$90 + \$20 + \$50 + \$40 = \270.0 百万

因此，预计的净流动性头寸为 $\$280 - 270 = +10.0$ 百万。

考察: Finance 重要程度: [■ ■ □]

科目: Quantitative Analysis

30. Emily manages a \$20.0 million portfolio allocated into two positions (aka, two components): a Technology ETF, and a Real Estate Investment Trust (REIT) ETF. She compares four different allocations, where the weight assigned to the Tech ETF component is either 35.0%, 50.0%, 65%, or 80.0%.

	Mix (Allocation) Scenarios			
	#1	#2	#3	#4
Weight in Tech ETF (%)	35.0%	50.0%	65.0%	80.0%
Position (000s)	\$7,000	\$10,000	\$13,000	\$16,000
Individual VaR (000s)	\$2,303	\$3,290	\$4,277	\$5,264
Marginal VaR	0.020	0.118	0.231	0.305
	#1	#2	#3	#4
Weight in REIT ETF (%)	65.0%	50.0%	35.0%	20.0%
Position, USD (000s)	\$13,000	\$10,000	\$7,000	\$4,000
Individual VaR (000s)	\$6,415	\$4,935	\$3,454	\$1,974
Marginal VaR	0.461	0.386	0.232	0.039
Undiversified Portfolio VaR	\$8,718	\$8,224	\$7,731	\$7,237

If her goal is to minimize the portfolio’s diversified value at risk (VaR), which of the above allocations is **best**?
 艾米莉管理着一个 \$20.0 百万的投资组合，分配到两个头寸 (aka, 两个组件): 一个科技 ETF, 和一个房地产投资信托 (REIT) ETF。她比较了四种不同的分配，其中分配给科技 ETF 组件的权重分别是 35.0%、50.0%、65.0% 或 80.0%。

	Mix (Allocation) Scenarios			
	#1	#2	#3	#4
Weight in Tech ETF (%)	35.0%	50.0%	65.0%	80.0%
Position (000s)	\$7,000	\$10,000	\$13,000	\$16,000
Individual VaR (000s)	\$2,303	\$3,290	\$4,277	\$5,264
Marginal VaR	0.020	0.118	0.231	0.305
	#1	#2	#3	#4
Weight in REIT ETF (%)	65.0%	50.0%	35.0%	20.0%
Position, USD (000s)	\$13,000	\$10,000	\$7,000	\$4,000
Individual VaR (000s)	\$6,415	\$4,935	\$3,454	\$1,974
Marginal VaR	0.461	0.386	0.232	0.039
Undiversified Portfolio VaR	\$8,718	\$8,224	\$7,731	\$7,237

如果她的目标是最大限度地降低投资组合的分散价值风险 (VaR)，那么哪一种分配是 **最佳的**？

- A. Mix #1 with 35.0% allocated to the Tech ETF
混合 #1 分配给科技 ETF 35.0%
- B. Mix #2 with 50.0% allocated to the Tech ETF
混合 #2 分配给科技 ETF 50.0%
- C. Mix #3 with 65.0% allocated to the Tech ETF
混合 #3 分配给科技 ETF 65.0%
- D. Mix #4 with 80.0% allocated to the Tech ETF
混合 #4 分配给科技 ETF 80.0%

答案:C

解析:

C 选项正确。混合 #3 分配给科技 ETF 65.0%

当边际 VaR 保持不变时，风险被最小化（即，投资组合的分散 VaR 被最小化）。在四种混合中，混合 #3 显然是边际 VaR 最接近的选择（即，0.231 与 0.232 几乎完全匹配）。混合 #1 具有最大的分散 VaR。从混合 #1 开始，风险通过添加边际 VaR 较低的组件（在本混合中，科技 ETF 的边际 VaR 仅为 0.020，而 REIT ETF 的边际 VaR 为 0.461）并减少边际 VaR 较高的组件来减少。如果我们从混合 #1 开始，并添加 15.0% 到科技 ETF，那么我们“到达”混合 #2。然而，在混合 #2 中，REIT ETF 的边际 VaR 仍然比科技 ETF 的边际 VaR 高得多（即，0.386 与 0.118），因此我们可以继续向边际 VaR 较低的组件（仍然是科技 ETF）转移。如果我们再添加 15% 到科技 ETF，那么我们到达混合 #3：这里边际 VaR 几乎相等，因此我们非常接近于最小化投资组合的分散 VaR 的分配。记录在案，四种混合的分散投资组合 VaR 分别为 \$6,131.14（混合 #1）、\$5,043.03（混合 #2）、\$4,621.42（混合 #3）和 \$5,036.58（混合 #4）。

考察: Finance: Risk Management **重要程度:** [■ ■ □]

科目: Quantitative Analysis

31. In comparison to a yield-based DV01 hedge, which of the following is the **best** argument in favor of a regression hedge?
与基于收益率的 DV01 对冲相比，下列哪一项是对回归对冲的最佳论据？

- A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position
回归对冲最小化了对冲头寸的利润和损失（P&L）方差
- B. The regression hedge is based on economic theory and future developments rather than empirical analysis
回归对冲基于经济理论和未来发展，而不是经验分析
- C. The regression hedge does not require any out-of-sample test(s), which are often unreliable if the sample is small
回归对冲不需要任何样本外测试，如果样本很小，这些测试通常不可靠
- D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset (RWA) under Basel regulations
回归对冲求解一个风险权重，该权重符合巴塞尔法规下的有效风险加权资产（RWA）

答案:A

解析:

A 选项正确。正确的：回归对冲最小化了对冲头寸的利润和损失（P&L）方差。

如 Tuckman 所解释（加粗为我们所强调）：“**回归对冲的最佳论据**实际上并不是先前关于债券收益率完全依赖回归模型变化的假设。将头寸的盈亏（P&L）写为……公式 6.11……其中负号表示，名义本金为正且收益率上升时，P&L 会减少。可以证明，公式 (6.8) 中的回归对冲能够最小化方差。换句话说，**只要将 P&L 方差作为适当的风险度量，回归对冲就能最小化被对冲头寸的风险。**”——引自 Tuckman, Bruce; Serrat, Angel. 《固定收益证券》（Wiley Finance）（第 159 页），Wiley 出版社，Kindle 版。关于（B）、（C）和（D），每项都是错误的。相反，以下是正确的陈述：

- 回归对冲基于经验分析，而不是经济理论和未来发展
- 回归对冲通常需要（或至少偏好）样本外测试（s）
- 风险权重是 beta 系数，与巴塞尔法规下的风险加权资产（RWA）无关

考察: Fixed Income: Hedging **重要程度:** [■ ■ □]

科目: Quantitative Analysis

32. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

在 Zapex Financial Corporation 的金融头寸中，它写了一个看涨期权，进入了一个利率互换，并写了一个信用违约互换。在这四个头寸中（以下），哪一个是最坏的例子？

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
Zapex 向同一行业的交易对手出售（aka，卖出）一个看涨期权
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
Zapex 在经济复苏期间是固定利率支付者
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
Zapex 向信用质量与参考实体正相关的交易对手出售（aka，卖出）一个信用违约互换（CDS）
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability
Zapex 有一个远期货币合同头寸，其公允价值收益与交易对手的违约概率正相关

答案:D

解析:

D 选项正确。 Zapex 有一个远期货币合同头寸，其公允价值收益与交易对手的违约概率正相关
反向风险（信用风险）是敞口与信用质量之间的不利相关性，因此反向风险是敞口与违约概率（信用质量）之间的正（负）相关性。这是反向风险，因为更大的敞口与较低支付可能性相关。代数上，反向风险（或正向风险）是敞口/EAD 和 PD 之间的一些依赖性。我们通常写 $EL\% = PD \times EAD$ ，但假设 PD 和 EAD 之间是独立的，因此考虑 WWR 的一种简单方法是当 $EL > PD \times EAD$ 时，因为 $EL \neq PD \times EAD$ 也是非独立性的条件。
关于 (A)、(B) 和 (C)，每项都是正向风险的例子。

考察: Credit Risk: Wrong-Way Risk **重要程度:** [■ ■ □]

科目: Quantitative Analysis

33. According to the FDIC, model risk management starts with robust model development, implementation, and use (aka, the first element. The second element is a sound validation process. The third element is governance). Which of the following is a TRUE statement about this *first element* (development, implementation and use) of model risk management?
根据 FDIC，模型风险管理始于强大的模型开发、实施和使用（aka，第一个元素。第二个元素是一个健全的验证过程。第三个元素是治理）。下列哪一项是关于模型风险管理 第一个元素（开发、实施和使用）的 TRUE 陈述？

- A. Model development should be a straightforward and routine technical process
模型开发应该是一个简单且常规的技术过程
- B. To exercise conservatism, pick an extreme point on a given model
为了体现审慎性，应在给定的模型分布上选择一个极端点
- C. Banks should never model output with judgmental or qualitative adjustments
行在对模型输出进行调整时绝不应使用主观判断或定性调整
- D. Although the nature of testing varies by the context and type of model, testing is an integral part of model development
尽管测试的性质因上下文和模型类型而异，但测试是模型开发不可或缺的一部分

答案:D

解析:

D. TRUE: 尽管测试的性质因上下文和模型类型而异，但测试是模型开发不可或缺的一部分
关于 (A)、(B) 和 (C)，每项都是错误的。

- 模型开发既不简单也不是一个常规的技术过程！
- 在给定的模型分布上选择一个极端点并不一定是审慎的，如果分布首先被错误指定或错误估计。
- 主观判断或定性调整是模型开发/实施/使用的关键部分：“此外，在模型开发、实施、使用和验证的各个阶段都进行了相当数量的主观判断。决策者应认识到这种主观性提高了对健全和综合模型风险管理过程重要性的认识。” ... “银行应确保其模型的主观和定性方面的发展也是健全的。在某些情况下，银行可能从模型中获取统计输出，并将其与主观或定性调整一起作为模型开发的一部分进行修改。”

考察: Operational Risk: Model Risk Management **重要程度:** [■ ■ □]

科目: Quantitative Analysis

34. Suppose that Acme Bank reports interest-sensitive assets (ISA) of \$490.0 million and interest-sensitive liabilities (ISL) of \$610.0 million. Recall that the relative interest-sensitive gap (aka, Relative IS GAP) is equal to the Dollar IS GAP divided (aka, scaled) by the size of the bank (where IS assets is a valid measure of size). Respectively, what is the bank’s relative IS GAP; its interest-sensitivity ratio (ISR); and the impact of rising interest rates on its net interest margin (NIM)?
假设 Acme Bank 报告的利息敏感资产（ISA）为 \$490.0 百万，利息敏感负债（ISL）为 \$610.0 百万。请记住，相对利息敏感缺口（aka，相对 IS 缺口）等于利息敏感缺口（Dollar IS GAP）除以银行规模（其中利息敏感资产是一个有效的规模衡量标准）。分别是什么？

- A. The Relative IS GAP is -120.0; the ISR is 1.245; rising interest rates will lower the NIM
相对 IS 缺口为 -120.0; ISR 为 1.245; 上升的利率将降低 NIM
- B. The Relative IS GAP is -0.245; the ISR is 0.803; rising interest rates will lower the NIM
相对 IS 缺口为 -0.245; ISR 为 0.803; 上升的利率将降低 NIM
- C. The Relative IS GAP is -0.245; the ISR is 1.245; rising interest rates will increase the NIM
相对 IS 缺口为 -0.245; ISR 为 1.245; 上升的利率将增加 NIM
- D. The Relative IS GAP is +4.08; the ISR is 0.803; rising interest rates will increase the NIM
相对 IS 缺口为 +4.08; ISR 为 0.803; 上升的利率将增加 NIM

答案:B

解析:

B 选项正确。利息敏感缺口 (Dollar IS GAP) = 利息敏感资产 (ISA) - 利息敏感负债 (ISL) = \$490.0 – \$610.0 = –120.0。相对利息敏感缺口 (Relative IS GAP) = –120.0/490.0 = –0.245。利息敏感比率 (ISR) = 490.0/610.0 = 0.803。这个银行是负债敏感的，因此上升的利率将降低其净利息收入 (NIM)。

考察: Market Risk: Interest Rate Risk **重要程度:** [■■■□]

科目: Quantitative Analysis

35. You work for the Private Client division of an international bank. The bank’s Private Client Advisors work with high net worth (NNW) individuals and families. Your boss has tasked you to perform due diligence on a roster of hedge funds; the funds that survive your due diligence will become candidates for presentation (and recommendation) to suitable customers of the bank who will become investors in the fund(s).

Your boss says that your due diligence should include a careful screen for fraud risk on behalf of the investors. In the case of your due diligence, which of the following is the most important ingredient?

你在一个国际银行的私人客户部门工作。该银行的私人客户顾问与高净值 (NNW) 个人和家庭合作。你的老板让你对一组对冲基金进行尽职调查；通过尽职调查后幸存的基金将成为候选基金，将推荐给合适的银行客户，成为基金的投资者。你的老板说，你的尽职调查应该包括对投资者的欺诈风险进行仔细筛查。在尽职调查中，哪一项是最重要的成分？

- A. Diligent application of a comprehensive checklist including a list of specific questions
仔细应用包括特定问题列表的综合检查清单
- B. Strong pre-existing personal relationships among the fund’s managers and all related parties
基金经理和所有相关方之间强有力的现有个人关系
- C. Guarantees by the fund of exceptional performance (e.g., > 30% per annum) with little or no risk
基金对卓越绩效（例如，年化 > 30%）的保证，风险很小或没有风险
- D. The investors’ membership in an affinity group, preferably a religious group due to high correlation with ethics
投资者在同质群体中的成员资格，最好是宗教群体，因为与伦理的高相关性

答案:A

解析:

A. TRUE: 仔细应用包括特定问题列表的综合检查清单

作者建议审查 SEC 网站，包括其问题列表；例如，投资欺诈红旗检查清单 at <https://www.investor.gov/>.

关于 (B)、(C) 和 (D)，每项都是错误的：

- 与相关方之间强有力的现有个人关系是投资者应该要求的**独立性**的相反。
- 对卓越绩效的保证，风险很小或没有风险是一个经典的**红旗**。
- 关于同质群体：这些是欺诈的著名目标，因为参与者倾向于信任他们的同伴。

考察: Finance: Due Diligence **重要程度:**

科目: Valuation and Risk Models

36. Which of the following is the BEST summary of the purpose of a copula function?

哪一项是对 copula 函数目的的最佳总结？

- A. To utilize the Gaussian correlation matrix such that tractable pairwise correlation parameters replace intractable dependence values
利用高斯相关矩阵，使得可处理的成对相关参数取代不可处理的依赖值
- B. To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure
将各种（甚至不寻常）单变量边际分布映射到具有可访问或熟悉依赖结构的分布，指定一个多元分布作为这些单变量边际分布的函数
- C. To translate a vector of outcomes into quantiles and map the quantiles to probabilities such that virtually any combination of multiple outcomes can be expressed as a single vector
将结果向量转换为分位数，并将分位数映射到概率，使得几乎任何多个结果的组合都可以表示为一个单一向量
- D. To decompose a posterior conditional distribution into a combination of the prior probability, the likelihood, and the marginal probability, typically by identifying Cholesky’s decomposition
将后验条件分布分解为先验概率、似然和边际概率的组合，通常通过识别 Cholesky 分解

答案:B

解析:

B 选项正确。将各种（甚至不寻常）单变量边际分布映射到具有可访问或熟悉依赖结构的分布，指定一个多元分布作为这些单变量边际分布的函数

考察: Quantitative Analysis: Copulas 重要程度: [■□□□□]

科目: Foundations of Risk Management

37. Retail exposures include the following: credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit), and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

零售风险敞口包括以下内容：信用卡、分期付款（例如，个人财务、教育贷款、汽车贷款、租赁）、循环信贷（例如，透支、房屋净值信贷额度）和住宅抵押贷款。下列哪一项通常是零售信用风险而不是企业信用风险的一个更强特征？

- A. Portfolio tends to behave like a well-diversified portfolio in normal markets
投资组合倾向于在正常市场中表现得像一个分散良好的投资组合
- B. Default by a single customer is never expensive enough to threaten a bank
单个客户违约永远不会昂贵到足以威胁银行
- C. A rise in defaults is often signaled in advance by a change in customer behavior
默认率的上升通常通过客户行为的变化提前发出信号
- D. Portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures
投资组合风险由由于暴露集中而导致信用损失上升到意外水平的风险主导

答案:D

解析:

D 选项正确。这通常是企业信用风险的一个特征：投资组合风险由由于暴露集中而导致信用损失上升到意外水平的风险主导。

- 关于（A）、（B）和（C），每项都是零售信用风险的一个倾向。

考察: Credit Risk: Retail vs Corporate 重要程度: [■■□□]

科目: Quantitative Analysis

38. According to the Basel Committee on Banking Supervision (BCBS, 2018), in regard to cyber-security and cyber-resilience metrics, “Some jurisdictions have methodologies to assess or benchmark regulated institutions’ cyber-security and resilience ...

None of these methodologies produce quantitative metrics or risk indicators comparable to those available for financial risks and resilience; e.g., standardized quantitative metrics where established data are available. Instead, indicators provide information on regulated institutions’ approach to building and ensuring cyber-security and resilience more broadly. Supervisory authorities also rely on entities’ own management information, although this differs across entities and is not yet mature.” (†)

With respect to current cybersecurity and resilience metrics, each of the following statements is true EXCEPT which is inaccurate?

根据巴塞尔委员会（BCBS, 2018），关于网络安全和弹性指标，“一些司法管辖区有方法来评估或基准化受监管机构的网络安全和弹性... 这些方法没有一个产生可量化的指标或风险指标，与金融风险 and 弹性指标相比，例如，可获得的已建立数据的标准化可量化指标。相反，指标提供有关受监管机构建立和确保网络安全和弹性的一般方法的信息。监管机构还依赖于实体自己的管理信息，尽管这因实体而异，尚未成熟。” (†)

关于当前的网络安全和弹性指标，下列陈述中哪一项是正确的，除了哪一项是不准确的？

- A. No single cybersecurity/resilience metric in isolation is sufficient
没有一个单独的网络安全/弹性指标是足够的
- B. Patch aging (aka, days to patch) is a widespread and comparable metric
补丁老化（aka，补丁到天数）是一个广泛且可比较的指标
- C. Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt
向后看指标是有帮助的，但不足；向前看指标是必要的，因为对手动态适应
- D. The penetration ratio (i.e., percentage of banks in a jurisdiction that can be penetrated) is a useful metric that compares a supervisor’s effectiveness to other jurisdictions
渗透率（即，一个司法管辖区中可以渗透的银行百分比）是一个有用的指标，用于比较监管机构的有效性与其他司法管辖区

答案:D

解析:

D 选项正确。 错误。这里所说的“渗透率”这一指标并不合理（用它来比较监管机构的有效性也同样如此），不过巴塞尔委员会多次讨论了渗透测试（penetration testing）作为网络风险有效的控制方法。不过需要注意，渗透测试（按类型：测试次数和发现等级）确实是一种常见且有效的网络弹性度量指标。详见 4.2.2 渗透测试部分。

关于（A）、（B）和（C），每一项都是正确的。具体来说：

- 没有任何单一的网络安全/弹性指标可以独立满足需求：“这些指标（即由受监管机构收集的网络指标）可以整体反映机构的网络和操作弹性水平对其业务需求和风险偏好的适应性。然而，单独任何一项指标都不足以作为充分的度量，而且到目前为止还没有确定出可以提供有意义基准的标准化指标集。” (†)
- 补丁老化（即打补丁所需的天数）是一项广泛且可比较的指标：“附录 C 包含了一组受监管机构为决策机构（董事会及其下属委员会）整理的以网络为中心的指标。值得注意的是，这些数据通常能体现出趋势信息，让审核者能够评估情况是在好转还是恶化。有些指标用于跟踪内部政策的合规性，其他指标则衡量固有风险。特别是补丁老化是一项广泛且具有可比性的指标。” (†)
- 向后看的指标有帮助但不充分；由于对手会动态适应，因此还需要向前看的指标：“向后看的指标是用以往的表现作为未来表现提供参考，这在机构的运营和风险环境长期相对稳定且较少受外部影响时是合理的。但在网络风险管理中，这是不够的，因为攻击者是动态的，会根据机构的应对和防护措施调整自己的策略，有时甚至在一次网络事件过程中就会改变他们的战术和策略。分布式拒绝服务（DDoS）事件就是一个典型例子，相关网络流量在最近两年内显著增长，而攻击者会针对机构防御进行技巧上的调整。尽管向后看的指标依然重要，但各司法管辖区日益认识到需要引入向前看的指标，作为直接或间接衡量弹性的指标，以判断受监管机构在风险事件发生时是否具备更强或更弱的应对能力。” (†)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

39. Venkat explains that “all risk management frameworks start with a governance structure that defines the roles and responsibilities of various bank employees and committees in overseeing risk-related activities,” and effective governance includes the oversight of intraday liquidity risk. In regard to the governance structure of intraday liquidity risk management which of

the following statements is TRUE?

Venkat 解释说, “所有风险管理框架都以一个治理结构开始, 该治理结构定义了各种银行员工和委员会在监督风险相关活动中的角色和责任,” 并且有效的治理包括对日内流动性风险的管理。关于日内流动性风险管理的治理结构, 下列哪一项陈述是正确的?

- A. Intraday liquidity risk should be accepted as a cost of doing business
日内流动性风险应该被接受为经营成本
- B. Roles and responsibilities should be defined within the eight lines (i.e., two times four) of defense model
角色和责任应该在八条防线（即, 两条四条）的防御模型中定义
- C. PCS Systems should only be a source of funds; if PCS becomes a use of funds, then a yellow flag should be triggered
PCS 系统应该只是资金的来源; 如果 PCS 成为资金的用途, 那么应该触发一个黄色标志
- D. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.
日内流动性风险应该纳入风险分类, 是风险自评估的一部分, 包括结算风险。

答案:D

解析:

D 选项正确。日内流动性风险应该纳入风险分类, 是风险自评估的一部分, 包括结算风险。
关于 (A)、(B) 和 (C), 每项都是错误的。

- Venkat(†) 解释说, 主动风险管理是日内流动性风险管理治理的一个特征: “主动风险管理。在许多机构中, 日内流动性风险被接受为经营成本, 并且不如其他类型的企业风险或甚至其他流动性风险那样积极管理。拥有大量 PCS 活动的领先银行认识到减少日内流动性风险的重要性。这些机构将结算和系统性风险分类为风险分类的组成部分, 并且至关重要, 将它们纳入公司的风险偏好框架” (†)
- 与风险治理的整合包括三条防线的模型: Treasure, Corporate Risk Management, 和 Internal Audit。Venkat(†) 说, “日内流动性风险管理框架遵循行业三条防线的模型, 特别强调第二条防线的专业知识, 以协调机构之间的活动... Corporate Risk Management 是第二条防线, 负责监督与资金相关的政策和程序: 在风险管理计划的发展方面提供建议, 监控银行所有资金部门的持续风险承担活动, 汇总银行的报告, 并提供银行整体日内流动性风险管理计划有效性的独立观点。” (†)
- 支付、清算和结算 (PCS) 系统可以是资金的来源或用途: “PCS 系统的结算。大多数 PCS 系统每天有一个结算, 许多发生在下午早些时候。这可能作为资金的来源或用途, 取决于参与者在任何给定日的净头寸。” (†)

考察: Liquidity Risk Management 重要程度: [■ ■ ■ □]

科目: Valuation and Risk Models

40. Hedge funds are less regulated than mutual funds. According to the authors (Fung and Hsieh), this lighter regulation of hedge funds offers genuine benefits but also creates unintended consequences. Which of the following is both a negative consequence and a TRUE statement about this relatively light regulation?

对冲基金的监管比共同基金要轻。根据作者 (Fung 和 Hsieh), 这种较轻的对冲基金监管确实带来了真正的利益, 但也产生了意想不到的后果。下列哪一项是相对较轻的监管的负面后果和 TRUE 陈述?

- A. Lack of reliable, standardized, and long-term performance reporting
缺乏可靠、标准化和长期绩效报告
- B. Lack of agreement on the specific definitions of absolute return and alpha
缺乏对绝对回报和 alpha 的具体定义的一致意见
- C. Ability of global macro funds to secretly drift into various strategies and styles with reckless impunity
全球宏观基金能够秘密地进入各种策略和风格, 毫无顾忌
- D. Ability of hedge funds to advertise (and specifically solicit) to individual investors without any restrictions
对冲基金能够秘密地进入各种策略和风格, 毫无顾忌

答案:A

解析:

A 选项正确。缺乏可靠、长期绩效报告

根据作者（加粗为我们所强调）：“然而，轻监管确实带来了意想不到的后果。首先，**对冲基金的绩效记录通常不是标准化的，可用的报告容易出现测量误差**。直到 1990 年代早期，对冲基金的历史绩效往往与他们的投资者一样私密，对冲基金绩效的评估既像艺术又像科学... **1994 年事件**促使投资者重新审视他们对对冲基金持有的看法的多样化。反过来，这有助于开发电子可用的对冲基金数据库，其中可以进行更正式的对冲基金绩效分析及其伴随风险。电子数据库的到来使得对对冲基金的学术研究成为可行的提议。” —来源：Handbook of the Economics of Finance, Volume 2B, edited by G. Constantinides, M. Harris, and R. Stulz.

- 关于（B）和（D），这些是错误的陈述。
- 关于（C），它是全球宏观基金的一个特征（aka，优势），它们可以利用各种风格的机会主义。

考察: Finance: Hedge Funds **重要程度:** [■ ■ □]

科目: Quantitative Analysis

41. Meissner conducted a detailed study of equity correlations of the 30 stocks in the Dow (DJIA) over 44.5 years (i.e., the 534 months from January 1972 to July 2017) that generated $534 * 30^2 = 480,600$ correlations values. In regard to his empirical findings with respect to equity correlations, each of the following statements is TRUE, except which is false?
Meissner 对道琼斯工业平均指数（DJIA）30 只股票的股权相关性进行了详细研究，持续了 44.5 年（即，1972 年 1 月至 2017 年 7 月 534 个月），产生了 $534 * 30^2 = 480,600$ 个相关值。在关于他对股权相关性的经验发现中，下列陈述中哪一项是 TRUE，除了哪一项是 FALSE?

- A. Mean reversion should be included in equity correlation models
均值回归应该包含在股权相关模型中
- B. For equity correlations, the Johnson SB distribution is better than the normal or lognormal
对于股权相关性，Johnson SB 分布比正态或对数正态更好
- C. A stressed scenario of bad economic times should incorporate higher correlation levels and higher correlation volatility
一个压力情景下的糟糕经济时期应该包含更高的相关水平和更高的相关波动
- D. Autocorrelation is a monotonically increasing function of the monthly lag; i.e., higher lags imply higher autocorrelations
自相关是月滞后的单调递增函数；即，更高的滞后意味着更高的自相关

答案:D

解析:

D 选项正确。错误。相反，自相关倾向于随着更大的滞后而衰减（即，减少）。关于（A）、（B）和（C），每项都是正确的。

考察: Quantitative Analysis: Correlation **重要程度:** [■ ■ □]

科目: Quantitative Analysis

42. Your colleague has drafted guidance for your firm’s computation of the credit value adjustment (CVA). The guidance begins with four principles: risk-neutrality, discounting, proportionality, and inseparability. The principles are summarized as follows:
I. Risk-neutrality: CVA can be computed either with risk-neutral (aka, market-implied) or real-world (aka, historical) parameters
II. Discounting: CVA should be discounted to present value, either via discounted expected exposure (EE) or via explicit discount factors
III. Proportionality: CVA will be proportional to the likelihood of default (PD), the exposure (EE) and the percentage amount lost in default (LGD).
IV. Inseparability: It should be impossible to separate the valuation of a transaction from its counterparty risk as measured by CVA

Each of the principles is valid, EXCEPT which of the principles is INCORRECT?

你的同事已经为你公司的信用价值调整（CVA）的计算起草了指导原则。指导原则以四个原则开始：风险中性、折现、比例性

和不可分离性。原则总结如下：I. 风险中性：CVA 可以使用风险中性（aka，市场隐含）或实际世界（aka，历史）参数计算 II. 折现：CVA 应该折现到现值，无论是通过折现预期暴露（EE）还是通过显式折现因子 III. 比例性：CVA 将与违约概率（PD）、暴露（EE）和违约损失率（LGD）成比例。IV. 不可分离性：应该不可能将交易的价值与其对手方风险（由 CVA 衡量）分开每个原则都是有效的，除了哪一项原则是不正确的？

- A. Risk-neutrality
风险中性
- B. Discounting
折现
- C. Proportionality
比例性
- D. Inseparability
不可分离性

答案:D

解析:

D 选项正确。错误（不可分离性）。相反，CVA 是可分离的，这一点具有实际重要性。最基本的、关键的关系是：风险价值 = 无风险价值 − CVA。关于（A）、（B）和（C），每项都是正确的原则。

考察: Counterparty Credit Risk **重要程度:** [■■■□]

科目: Quantitative Analysis

43. Acme Financial is a large bank whose Business Indicator (BI), for purposes of operational risk capital under the standardized measurement approach (SMA), is well in excess of EUR 1.0 billion. Therefore, Acme is required to use loss data as a direct input into its operational risk capital calculations. In regard to the general and/or specific criteria recommended by the Basel Committee, which of the following is TRUE about the identification, collection, or treatment of operational loss data?
Acme Financial 是一家大型银行，其业务指标（BI），在标准化测量方法（SMA）下，对于操作风险资本而言，远远超过 EUR 1.0 亿。因此，Acme 被要求将损失数据作为直接输入到其操作风险资本计算中。在巴塞尔委员会推荐的一般和/或特定标准中，哪一项关于识别、收集或处理操作损失数据是 TRUE？

- A. P&L provisions or reserves against potential loss impacts are included in the gross loss computation
损益（P&L）中针对潜在损失影响计提的准备金或储备应计入总损失计算
- B. All internal loss events must be included in the gross loss computation regardless of the size of the loss
所有内部损失事件必须包括在总损失计算中，无论损失的大小如何
- C. Legal expenses are excluded from the gross loss computation in the operational loss data because they represent the realization of legal risk
法律费用被排除在操作损失数据中的总损失计算之外，因为它们代表了法律风险的实现
- D. Receivables (a short-term asset) should be counted as recoveries unless the particular receivable causes the current ratio to exceed an internal limit
应收账款（一个短期资产）应该被计入 recoveries，除非特定的应收账款导致当前比率超过内部限制

答案:A

解析:

A 选项正确。正确：损益（P&L）中针对潜在损失影响计提的准备金或储备应计入总损失计算
关于选项（B）、（C）和（D），每一项都是错误的。实际上，正确的内容如下：

- 虽然损失数据必须涵盖“所有重大活动”，但低于最低阈值的损失不必被包括在内。
- 与事件直接相关的法律费用应包括在内（见下文）。请注意，操作风险包括法律风险。
- 应收账款不算作回收（recoveries）（见下文）。

摘自“总损失、净损失及回收的定义”(6. 损失数据识别、收集与处理的具体标准):

“21. 总损失是指在任何类型回收前的损失。净损失是指在考虑回收影响后的损失。回收是与原始损失事件相关、在时间上分开的独立事件, 由第三方支付资金或经济利益流入的情况。

22. 银行必须能够识别所有操作损失事件的总损失金额、非保险回收和保险回收。银行应在损失数据集中使用扣除回收后的净损失(包括保险回收)。但只有在银行实际收到回收款后, 回收才能用于减少损失。应收账款不计作回收。收到回收款以净额结算损失的验证材料须根据监管要求提供。

23. 损失数据集中总损失计算必须包括以下项目:

(a) 由于操作风险事件对银行损益账户的直接计提, 包括减值和和解, 以及相关的资产减记;

(b) 作为事件后果产生的费用, 包括与操作风险事件直接相关的外部支出(如与事件直接相关的法律费用及支付给顾问、律师或供应商的费用) 及为恢复事件发生前状态而产生的修理或替换费用;

(c) 损益中就潜在操作损失影响计提的准备金或储备;

(d) 因操作风险事件而产生具有明确财务影响的损失, 这些损失暂时计入过渡和/或暂记账户, 在损益尚未反映(“待决损失”)。重大待决损失应在符合其金额和时间要求的期限内计入数据集;

(e) 某一财务会计期间因操作风险事件影响以前期间现金流或财务报表而记入的负面经济影响(“跨期损失”)。当操作风险事件跨越多个财务会计期间并引发法律风险时, 重大“跨期损失”应计入损失数据集。”

(来源:《巴塞尔协议 III: 危机后改革最终方案》, 巴塞尔银行监管委员会出版物(2017 年 12 月), 第 132 页。)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

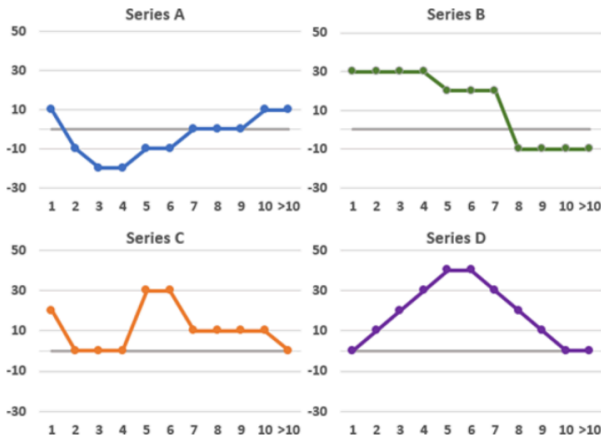
44. Geofinancial Bank currently has the following (very) simplified balance sheet:

Assets		Liabilities	
Bonds	\$30.0	Deposits	\$30.0
Loans	\$70.0	Bonds	\$50.0
		Equity	\$20.0

Further, the maturities of these accounts are as follows:

- **Assets:** The bonds (\$30.0 million) expire in one year. In regard to the loans (\$70.0), \$40.0 million expire in five (5) years, \$10.0 million expire in seven (7) years, and \$20.0 million expire in ten (10) years.
- **Liabilities:** In regard to the deposits (\$30.0 million), \$10.0 million expire in one (1) year, and \$20.0 million expire in two (2) years. In regard to the bonds (\$50.0 million), \$10.0 million expire in five (5) years, \$30.0 million expire in seven (7) years, and \$10.0 million expire in beyond ten (> 10) years.
- **Equity** (\$20.0 million) is presumed to expire in ten (10) years

Consider the following possible term structures of expected cash flows:



Which term structure of expected cash flows is accurate for Geofinancial Bank?

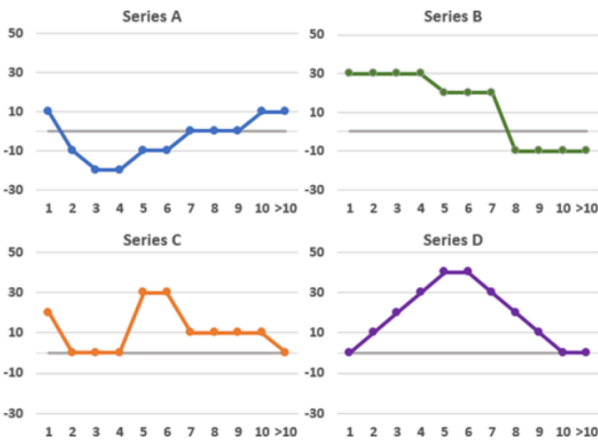
哪一种预期现金流量期限结构适用于 Geofinancial Bank? Geofinancial Bank 目前拥有如下(非常)简化的资产负债表:

Assets		Liabilities	
Bonds	\$30.0	Deposits	\$30.0
Loans	\$70.0	Bonds	\$50.0
		Equity	\$20.0

此外，这些科目的到期情况如下：

- 资产：**债券（\$30.0 百万）在一年后到期。就贷款（\$70.0 百万）而言，\$40.0 百万在五（5）年后到期，\$10.0 百万在七（7）年后到期，\$20.0 百万在十（10）年后到期。
- 负债：**就存款（\$30.0 百万）而言，\$10.0 百万在一（1）年后到期，\$20.0 百万在两（2）年后到期。就债券（\$50.0 百万）而言，\$10.0 百万在五（5）年后到期，\$30.0 百万在七（7）年后到期，\$10.0 百万在十年以上（> 10 年）到期。
- 权益**（\$20.0 百万）假定在十（10）年后到期

考虑以下可能的预期现金流期限结构：



对于 Geofinancial Bank，哪一种预期现金流期限结构是准确的？

- A. Series A
系列 A
- B. Series B
系列 B
- C. Series C
系列 C
- D. Series D
系列 D

答案:A

解析:

A 选项正确。正确：系列 A。见下文。

Liabil-				
Maturity	Assets	ities	Δ	Net
1	30	10	20	20
2		20	-20	0
5	40	10	30	30
7	10	30	-20	10
10	20	20	0	10
>10		10	-10	0
Sum	100	100		

考察: Finance: Asset Liability Management 重要程度: [■□□]

科目: Quantitative Analysis

45. Acme Industrial Corporation just finished establishing its new financial accounting controls which includes three components: planning, budgeting, and variance monitoring. The company’s next step is to establish a three-legged risk management stool that will analogously include the risk plan, the risk budget, and the risk monitoring process. In regard to the risk plan, each of the following will be an element EXCEPT which is the LEAST likely to be included in Acme’s risk plan?

Acme Industrial Corporation 刚刚完成了其新的财务会计控制体系的建立，其中包括三个组件：计划、预算和差异监控。公司的下一步是建立一个三足风险管理凳子，该凳子将类似地包括风险计划、风险预算和风险监控过程。在关于风险计划时，下列哪一项将是一个元素，除了哪一项是最不可能被包含在 Acme 的风险计划中？

- A. Diversification or risk decomposition policy
多样化或风险分解政策
- B. Strategic responses for events that are low-probability but franchise-threatening
针对低概率但具有威胁性的事件的战略响应
- C. Ample budget for losses that are material, predictable, and avoidable that cascade down
针对可预测且可避免的重大损失的充足预算
- D. Identification of critical dependencies that exist inside and outside the organization
识别存在于组织内部和外部的关键依赖关系

答案:C

解析:

C 选项正确。 错误。风险计划应该对“重大、可预测且可避免”的错误/损失零容忍。

关于 (A)、(B) 和 (D)，每项都是正确的。根据作者，风险计划中的五个指导原则是：

- 预期回报和波动率目标；以及里程碑，以识别失败。
- 成功或失败的定义
- 风险资本如何部署的愿景；以及多样化或风险分解政策
- “仅仅是令人失望的事件”和严重损害损失之间的明确界限。此外，“战略响应应该存在任何威胁 franchise 的事件——即使这些事件是低概率情况。风险计划应该识别那些如此严重以至于应寻求保险覆盖（例如，资产类别 put）来覆盖下行风险的损失类型。”（来源：2020 年财务风险管理第二部分：风险管理和投资管理，第 10 版。Pearson Learning Solutions, 2019。VitalBook 文件。）
- 识别存在于组织内部和外部的关键依赖关系

考察: Financial Risk Management **重要程度:** [■ ■ □]

科目: Quantitative Analysis

46. Bertha has a view about correlation, and she is evaluating several rainbow (aka, multi-asset) options. Her view is that correlation is going to decrease. Among the following trades, which **best** expresses her view?

Bertha 对相关性有一个看法，她正在评估几种彩虹（aka，多资产）选项。她的看法是相关性将会下降。在下列交易中，哪一种**最佳**表达了她的看法？

- A. Buy (aka, long position in) basket option
购买（aka，多头头寸）篮子选项
- B. Buy (aka, long position in) better-of-two where $\text{payoff} = \max(S_1, S_2)$
购买（aka，多头头寸）更好的两个选项，其中 $\text{payoff} = \max(S_1, S_2)$
- C. Buy correlation (aka, pay the fixed rate) in a correlation swap
购买相关性（aka，支付固定利率）在相关性交换中
- D. Short position in an exchange option where $\text{payoff} = \max(0, S_2 - S_1)$
空头头寸在交换选项中，其中 $\text{payoff} = \max(0, S_2 - S_1)$

答案:B

解析:

B 选项正确。 Bertha 想要“做空相关性”，并且随着相关性的下降，更好的两个选项变得更有价值。另一方面：

- 篮子选项的价值随着相关性的下降而减少，因此购买篮子选项是做多相关性
- 在相关性交换中购买相关性（aka，支付固定利率）是做多相关性
- 交换选项随着相关性的下降而变得更有价值，因此交换选项的多头头寸是做空相关性，交换选项的空头头寸是做多相关性

考察: Market Risk: Correlation **重要程度:** [■ ■ □]

科目: Quantitative Analysis

47. Julia the analyst is analyzing the credit risk of a contract to which her firm is considering entering into as a party. If true, which feature of her work (among the choices below) **most likely** signifies or indicates that this is a case of counterparty credit risk rather than (traditional) lending risk or some other type of risk?

Julia 分析师正在分析她的公司考虑作为一方进入的合同中的信用风险。如果为真，她的工作中哪一项特征（在下列选项中）**最有可能**表明或指示这是一个 counterparty 信用风险而不是（传统）贷款风险或其他类型的风险的情况？

- A. Interest rate (as a market variable) fluctuations have little impact on the amount owed
利率（作为市场变量）波动对欠款的影响很小
- B. The exposure at default (EAD) is 100% of the notional over most of the life of the contract
违约时的暴露（EAD）是合同生命周期的大部分时间内名义金额的 100%
- C. Her firm is the only party assuming credit risk: if her firm defaults, the other party faces no losses
她的公司是唯一承担信用风险的方：如果她的公司违约，其他方不会面临损失
- D. She is employing Monte Carlo simulation because the future value of the contract is highly uncertain
她正在使用蒙特卡洛模拟，因为合同的未来价值高度不确定

答案:D

解析:

D 选项正确。她正在使用蒙特卡洛模拟，因为合同的未来价值高度不确定。

根据 Jon Gregory 的说法，对手方信用风险与贷款风险之间的两个主要区别是：（1）贷款方知道其所承担的风险金额，但在对手方信用风险的情况下，风险金额高度不确定；（2）贷款通常只涉及单方风险（由贷款方承担），而对手方信用风险则是双边的，即双方都面临风险。

关于错误选项 (a), Herstatt 风险是结算风险的一种特殊情形 (跨币种), 而不是预结算风险。

考察: Counterparty Credit Risk **重要程度:** [■ ■ □]

科目: Quantitative Analysis

48. The Basel regulatory framework has undergone several major renovations but started in the late 1980s when the Basel Committee on Banking Supervision published the first Basel accord, which is now called Basel I; this original Basel I was implemented in 1992. According to Carey (GARP Chapter 19), which BEST summarizes the motivation(s) of the original Basel?

巴塞尔监管框架已经经历了几次重大改革, 但始于 1980 年代末, 当时巴塞尔银行监管委员会发布了第一份巴塞尔协定, 现在称为巴塞尔 I; 这个原始的巴塞尔 I 于 1992 年实施。根据 Carey (GARP 第 19 章), 哪一项 BEST 总结了原始巴塞尔的动机?

A. To formalize regulatory capital requirements with a set of regulations (i.e., Basel) that would become legal requirements and therefore carried the force of law; i.e., the motivation was legal

通过一套监管规定 (即巴塞尔协议) 形式化监管资本要求, 使其成为具有法律约束力的法律要求; 即其动机是法律层面的考虑

B. Regulators realized that specific measures of short- and long-term liquidity were necessary given that liquidity crunches had been the essential common denominator of recent financial disasters; i.e., the motivation was liquidity

监管者意识到, 有必要对短期和长期流动性进行具体度量, 因为流动性紧缩一直是近期金融灾难的共同关键因素; 即其动机是流动性管理

C. The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (aka, level playing field) between banks competing in each other’s home countries; i.e., the motivations were bank safety and competitive concerns

跨境金融的增长要求银行在压力时期保持偿付能力, 并要求在彼此本国市场竞争的银行之间实现公平竞争 (即 “公平竞争环境”); 因此其动机是银行安全性和竞争方面的考量

D. In the wake of a global recession, there was a motivation to incentivize banks to increase their lending (which had been historically low) in order to grow the developed and emerging economies; i.e., the motivation was economic recovery especially of emerging economies

在全球经济衰退之后, 有动机激励银行增加放贷 (此前历史上处于较低水平), 以推动发达经济体和新兴经济体的增长; 即其动机是促进经济复苏, 尤其是新兴经济体

答案:C

解析:

C 选项正确。正确: 跨境金融的增长要求银行在压力时期保持偿付能力, 并要求在彼此本国市场竞争的银行之间实现公平竞争 (即 “公平竞争环境”); 因此其动机是银行安全性和竞争方面的考量。

Carey 解释道: “有两个事件促成了巴塞尔 I 的诞生。

- 首先, Herstatt 银行倒闭后, 跨境金融的增长仍在继续, 这使得 G10 国家意识到, 他们有共同的利益, 确保银行拥有足够的资本来吸收大量损失。
- 其次, 国际银行在彼此的本国市场激烈竞争。然而, 各国所要求的最低资本标准差别显著, 这导致人们认为总部设在最低标准较低国家的银行具有竞争优势。为此, 巴塞尔银行监管委员会 (BCBS) 成员决定制定全球统一的最低标准, 以 “创造公平竞争环境”, 避免向下竞赛。也就是说, 虽然巴塞尔协议部分目的是保证银行安全与稳健, 但谈判本身也包含了对竞争优势的博弈成分。” (来源: 2020 FRM Part II: Operational Risk and Resiliency, 第 19 章, 第 10 版, Pearson Learning Solutions。)

考察: Finance: Basel Accords **重要程度:** [■ ■ □]

科目: Quantitative Analysis

49. Which of the following types of liquidity is available in the form of the institution’s liquidity asset buffer (LAB) and represents liquidity available to meet general financial obligations under a stress scenario?

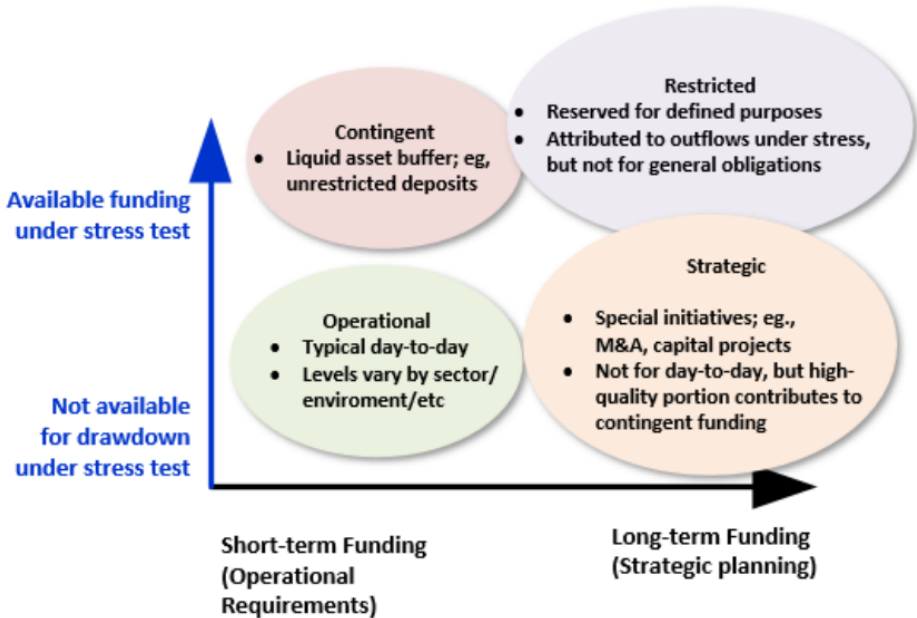
哪一种流动性形式存在于机构的流动性资产缓冲 (LAB) 中, 并代表在压力情景下可用于满足一般财务义务的流动性?

- A. Operational
运营性
- B. Strategic
战略性
- C. Contingent
偶然性
- D. Restricted
限制性

答案:C

解析:

C 选项正确。偶然性流动性捕捉现金或资产，当压力情景出现时可用。流动性取决于压力情景的发生。包括在这种情况下金融资产，可以在不损失价值的情况下转换为现金。捕捉偶然性流动性以覆盖压力现金流出是流动性压力测试的主要原因。Venkat 写道：“**流动性资产缓冲**：流动性资产缓冲代表当前存在的偶然性流动性... 流动性资产缓冲证券的基本特征应包括低信用和市场风险、估值的易性和确定性、在活跃且规模较大的市场上交易、以及买家和卖家集中度低。流动性资产缓冲还应满足运营要求，确保流动性在实体承受压力测试时处于中央 Treasury 部门的控制之下。” (†)



考察: Finance: Liquidity Risk 重要程度: [■ ■ □]
科目: Quantitative Analysis

50. Acme Investment Bank recently finished building the third leg of its three-legged risk management stool, after the risk plan and the risk budget: the bank installed its risk monitoring process. The bank shares a belief with the authors (Rosengarten and Zangari) that “risk is the shadow cost behind returns.” Each of the following is probably an element or plank in the bank’ s risk monitoring process EXCEPT which is the least likely to be found in the bank’ s risk monitoring process?

Acme Investment Bank 最近完成了其三足风险管理凳子的第三条腿的建设，在风险计划和风险预算之后：该银行安装了其风险监控过程。该银行与作者（Rosengarten 和 Zangari）分享了“风险是回报背后的影子成本”的信念。下列哪一项可能是银行风险监控过程中的元素或 plank，除了哪一项最不可能在银行的风险监控过程中找到？

- A. The bank’s monitoring is not dependent on any single tool or metric, but instead relies on an ensemble of various tools and metrics
银行的监控不依赖于任何单一工具或指标，而是依赖于各种工具和指标的集合
- B. The bank measures alpha as a regression intercept where alpha can help evaluate manager skill but unfortunately, its statistical significance often requires more data (i.e., longer history) than is available
银行测量 alpha 作为回归截距，其中 alpha 可以帮助评估经理技能，但不幸的是，其统计显著性通常需要更多数据（即更长的历史），而不是可用的数据

- C. The bank measures return on risk capital (RORC)) where value at risk (VaR) is a measure of risk capital; in addition to the RORC's level, the bank watches its variability
银行测量风险资本回报 (RORC)，其中价值 at risk (VaR) 是风险资本的度量；除了 RORC 的水平，银行还监控其可变性
- D. After a threshold level of data is collected, subjective articulations of each portfolio manager's philosophy becomes unnecessary, and should be discarded, in favor of quantitative performance evaluation
在收集到阈值水平的数据后，每个投资组合经理的主观哲学陈述变得不必要，应该被抛弃，取而代之的是定量绩效评估

答案:D

解析:

D 选项正确。错误。相反，每个投资组合经理的主观哲学陈述总是必要的，作为量化工具的补充，以澄清是否存在风格漂移。根据 GARP 的阅读，显式（和主观）的哲学陈述总是量化工具的必要补充，因为除了帮助识别 风格漂移之外，没有其他原因。摘自阅读：“在讨论这些 [即 commonly used performance tools] 之前，值得注意的是，尽管性能工具是必要的，但它们并不是在出现异常行为迹象时及时进行管理干预的替代品。当异常行为以 poor performance statistics 的形式显现时，损害可能已经无法挽回。因此，我们相信性能工具必须补充：

- 每个投资组合经理的清晰管理哲学陈述。该哲学陈述应确定经理如何期望从市场中提取回报。它应确定经理何时成功、何时失败。
- 设计用于识别哲学或流程偏离的例行位置和风格监控流程。这是一种早期预警系统。” —来源：2020 年财务风险管理第二部分：风险管理和投资管理，第 10 版。Pearson Learning Solutions，2019。VitalBook 文件。

关于 (A)、(B) 和 (C)，每项都是正确的。具体来说：

- 银行的监控不依赖于任何单一工具或指标，而是依赖于各种工具和指标的集合
- 银行测量 alpha 作为回归截距，其中 alpha 可以帮助评估经理技能，但不幸的是，其统计显著性通常需要更多数据（即更长的历史），而不是可用的数据
- 银行测量风险资本回报 (RORC)，其中价值 at risk (VaR) 是风险资本的度量；除了 RORC 的水平，银行还监控其可变性

考察: Financial Risk Management 重要程度: ☒☒☐

科目: Quantitative Analysis

51. Acme FinGlobal Bank has engaged consultants to redefine the organization's risk measures. The project includes value at risk (VaR) mapping across several portfolios (aka, asset classes) at the organization:

- A fund that invests in initial public offerings (IPOs)
- A fund that invests in emergent assets but where there is little historical data
- Small equity portfolio with less than 30 positions
- Large equity portfolio with 1,000 positions
- A portfolio of corporate bonds that carries six different ratings and has maturities of 3, 5, and 7 years

Each of the following statements is TRUE except which is false?

Acme FinGlobal Bank 聘请了顾问对机构的风险度量进行重新定义。该项目包括在机构内部对若干投资组合（即资产类别）进行价值-at-风险 (VaR) 映射：

- 一只投资于首次公开募股 (IPO) 的基金
- 一只投资于新兴资产、但历史数据很少的基金
- 持仓少于 30 个头寸的小型股票投资组合
- 持仓 1,000 个头寸的大型股票投资组合
- 一只公司债券投资组合，包含六种不同评级，期限为 3 年、5 年和 7 年

以下关于这些内容的陈述均为真，除了哪一项是假的？

- A. The corporate bond can be mapped to nine general risk factors
公司债券可以映射到九个一般风险因素
- B. The IPO and emergent asset funds cannot be mapped because these instruments lack historical data such that sensitivities cannot be estimated
首次公开募股和新兴资产基金无法映射，因为这些工具缺乏历史数据，因此无法估计敏感性
- C. The large equity portfolio is an ideal candidate for mapping because mapping to general and specific market primitives requires far fewer numerical estimates than specification of a covariance matrix
大型股票投资组合是映射的理想候选，因为映射到一般和特定市场原语需要比指定协方差矩阵更少的数值估计
- D. The equity portfolio VaR mapping process contains sources of model risk, including factor sensitivities (aka, betas) estimation error, subjective risk factor choices, and specific risk omission
股票投资组合价值-at-风险（VaR）映射过程包含模型风险，包括因子敏感性（aka，贝塔）估计误差、主观风险因子选择和特定风险遗漏

答案:B

解析:

B 选项正确。 错误。相反，映射是专门解决数据不足问题的解决方案。根据 Jorion 的说法，“风险经理必须用系统中已经存在的类似风险因素的敞口替换这些头寸。”^(†)

关于（A）、（C）和（D），每项都是正确的。

(†) Philippe Jorion, 《价值-at-风险：金融风险管理的新基准》，第 3 版。（纽约：麦格劳-希尔，2007 年）

考察: Market Risk Measurement and Management **重要程度:** [■ ■ □]

科目: Quantitative Analysis

52. Freecone Financial LLC entered into an interest rate swap some time ago where it agreed to make semi-annual fixed payments at a rate of 3.0% per annum, in exchange for receiving floating payments on a notional amount of \$50.0 million. The floating rate is the six-month secured overnight financing rate (SOFR) and this is also the discount rate used to value the swap. The swap has a remaining life of 1.25 years. The SOFR rate applicable to the exchange in three months was determined three months ago: it was 2.40%. Just as with a LIBOR-based swap, the floating rate is observed at the beginning of each six-month period but paid at the end of the period. Currently, the SOFR term structure is flat at 2.60% per annum with continuous compounding. With respect to its position in this swap, which of the following is nearest to Freecone Financial’s current exposure? Freecone Financial LLC 不久前进入了一个利率互换，其中它同意每半年支付一次固定利率，年利率为 3.0%，以换取收到名义金额为 \$50.0 百万的浮动支付。浮动利率是六个月期有担保隔夜融资利率（SOFR），这也是用于估值互换的贴现率。互换剩余期限为 1.25 年。三个月前确定的适用于交换的 SOFR 利率为 2.40%。与基于 LIBOR 的互换一样，浮动利率在每个六个月期初观察，但在每个六个月期末支付。目前，SOFR 期限结构是平坦的，年利率为 2.60%，连续复利。就其在该互换中的头寸而言，下列哪一项最接近 Freecone Financial 的当前暴露？

- A. Zero
- B. -\$335,900
- C. +\$335,900
- D. +1,335,900

答案:A

解析:

A 选项正确。 当前敞口 = $\max(\text{MtM value}, 0)$ 。在这种情况下， $\text{CE} = \max(-335,900, 0) = 0$ 。Freecone 在互换中处于亏损状态；其对手方由于对手方在违约事件中遭受损失，因此具有正的当前敞口。但 Freecone 在违约事件中不会遭受当前损失。

考察: Credit Risk: Current Exposure **重要程度:** [■ ■ □]

科目: Quantitative Analysis

53. According to the Guidance on Managing Outsourcing Risk (by the Board of Governors of the Federal Reserve System), with respect to a third-party service provider arrangement, the overall due diligence process by a financial institution should at least include a review of each of the following EXCEPT which is not essential?

根据联邦储备委员会董事会关于管理外包风险的《指导》，在第三方服务提供商安排方面，金融机构的总体尽职调查过程至少应包括对以下各项的审查，除了哪一项不是必要的？

- A. Ensure service provider has an appropriate background check program for its employees
- B. Financial review of service provider’s most recent annual report and financial statements
对服务提供商最近年度报告和财务报表的财务审查
- C. Evaluate the service provider’s performance along environmental, social, and governance (ESG) factors
评估服务提供商在环境、社会 and 治理（ESG）因素方面的表现
- D. Evaluate the adequacy of standards, policies and procedures including adherence to applicable laws, regulations and supervisory guidance
评估标准、政策和程序的充分性，包括遵守适用法律、法规和监管指导

答案:C

解析:

C 选项正确。并未提及对 ESG 因素的审查。根据美联储理事会的指导，尽职调查应涵盖以下三大类：

- 1. 业务背景、声誉和战略：“金融机构应当审查潜在服务提供商在行业内的地位及其公司历史与资质；审查服务提供商及其高级管理人员的背景与声誉；并确保服务提供商为其员工建立了合适的背景审查程序。”
- 2. 财务状况与表现：“金融机构应审查服务提供商及其密切相关关联公司的财务状况。财务审查可包括：服务提供商最近的财务报表和年度报告，涵盖未了承诺、资本实力、流动性及经营业绩；还应考虑服务提供商的可持续性，包括该服务提供商经营年限以及其在特定服务领域的市场份额增长等因素……”
- 3. 运营与内部控制。

考察: Operational Risk: Outsourcing Risk **重要程度:** [■ ■ □]

科目: Quantitative Analysis

54. The staff at Umbrella Street Bank, which is a commercial bank (it could also be a savings institution or credit union), produces several liquidity risk reports on a daily, weekly, monthly, and quarterly basis. Among these liquidity risk reports is a deposit tracker report. Among the following metrics, which is MOST LIKELY to appear in their deposit tracker report?

Umbrella Street Bank 的员工（这是一家商业银行，也可以是一家储蓄机构或信用合作社）会按日、周、月和季度频率编制多种流动性风险报告。在这些流动性风险报告中包含一份存款追踪报告。在下列指标中，哪一项最有可能出现在他们的存款追踪报告中？

- A. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved lower floor (limit) of 120.0%
贷款-存款（LTD）比率（当前和预测）与董事会批准的最低地板（限制）120.0%
- B. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved upper ceiling (limit) of 80.0%
贷款-存款（LTD）比率（当前和预测）与董事会批准的最高天花板（限制）80.0%
- C. Market-to-book ratio of common equity (current and forecast) versus investor-communicated target of 1.30
普通股的市场-账面比率（当前和预测）与投资者沟通的目标 1.30
- D. The leverage-adjusted duration gap (current and forecast) versus board-approved upper ceiling of 3.5 years
杠杆调整后的期限缺口（当前和预测）与董事会批准的最高天花板 3.5 年

答案:B

解析:

B 选项正确。正确：贷款-存款（LTD）比率（当前和预测）与董事会批准的最高天花板（限制）80.0% 关于（A）、（C）和（D），每项都是错误的。

55. You work as a pension fund administrator and the Investment Committee asks for your input. The pension fund seeks to increase its allocation to alternative investments and, in particular, will add a new hedge fund strategy; aka, style. You are limited to four strategies. You are perusing the brochures of four hedge funds (each named after Greek gods per a popular marketing convention!). The four strategies are global macro, merger arb, long/short equity, and equity market neutral. Below are the four funds. Each fund has a name, a strategy (aka, style), a tagline, and a very brief description.

- *Acheilus* is a Global Macro strategy. We are your effective asset (factor) allocator! We opportunistically make bets in different markets utilizing a range of strategies; you can think of us as betting on a range of factors such as commodities, interest rates, exchange rates.
- *Boreas* is a Merger Arb (aka, Risk Arb) strategy. We collect nickels (and dimes) in front of the proverbial steamroller! Like our sibling event-driven strategy (i.e., Distressed), we can offer you nonlinear returns but with tail exposure to extreme market conditions.
- *Chimera* is a Long/Short Equity strategy. We are friendly but will never hug a benchmark! Although we have some directional market exposure, we can be classified as relative value because we are stock pickers who seek alpha by way of a diversity of opinions and ability.
- *Daphne* is an Equity Market Neutral strategy. We bring you a basket of betas! As an event-driven strategy, our returns are highly linked to several common market risk factors including equity indices, bond spreads, and volatility factors.

Each of these descriptions is consistent with its strategy (aka, style) category EXCEPT for one. Select the fund that is NOT consistent with its description.

你在一家养老金基金担任管理人，投资委员会向你征求意见。该养老金基金计划提高对另类投资的配置，特别是将新增一种对冲基金策略（即风格）。你最多只能选择四种策略。你正在翻阅四只对冲基金的宣传册（每只基金都以希腊神命名，这是常见的营销惯例!）。这四种策略分别是全球宏观、并购套利、多空股票以及股票市场中性。下面是这四只基金。每只基金都有名称、策略（即风格）、一句宣传语以及非常简短的说明。

- *Acheilus* 是一只全球宏观（Global Macro）策略基金。我们是你高效的资产（因子）配置者！我们会在不同市场中以机会主义方式进行押注，并运用多种策略；你可以将我们视为在大宗商品、利率、汇率等一系列因子上进行押注。
- *Boreas* 是一只并购套利（Merger Arb，也称风险套利 Risk Arb）策略基金。我们是在俗称的“蒸汽压路机”前面捡镍币（和一角钱）的人！与我们的兄弟事件驱动策略（即困境投资 Distressed）类似，我们可以为你提供非线性的收益，但同时在极端市场环境下暴露于尾部风险。
- *Chimera* 是一只多空股票（Long/Short Equity）策略基金。我们很友好，但绝不会去“拥抱”基准！尽管我们有一定方向性的市场敞口，但可以被归类为相对价值策略，因为我们是通过择股来获取阿尔法，依靠的是多样化的观点与能力。
- *Daphne* 是一只股票市场中性（Equity Market Neutral）策略基金。我们为你带来一篮子贝塔！作为一只事件驱动策略基金，我们的收益与若干共同的市场风险因子高度相关，包括股票指数、债券利差以及波动率因子。

在上述描述中，除了其中一只基金以外，其余每只基金的描述都与其策略（即风格）类别相一致。请选择那只与其描述不一致的基金。

- A. Acheilus
阿基里斯
- B. Boreas
波瑞阿斯
- C. Chimera
奇美拉
- D. Daphne
达芙妮

答案:D

解析:

D 选项正确。标签/描述存在三重不一致：(1) 股票市场中性策略与可观察的市场风险因子无关（在引用的研究中，这是唯一没有与市场风险因子关联的策略）；因此，股票市场中性基金不可能“带来一篮子贝塔”，其目的正是中和贝塔，即对主流市场因子的贝塔为零；(2) 此外，股票市场中性策略并不属于事件驱动型策略，它本身是非方向性的，被作者归类为“小众策略”（niche strategy），有时也归于股票对冲（equity hedge）策略。(3) 描述将其归为事件驱动，也是错误的。

关于 (A)、(B) 和 (C)，其他三只基金（Acheilus、Boreas 和 Chimera）的策略、宣传语及描述彼此一致，具有可信度。

考察: Finance: Hedge Fund Strategies **重要程度:** [■■■□]

科目: Quantitative Analysis

56. Betty’s firm wants to allocate an economic capital cushion based on a one-day 99.5% value at risk (VaR) model. Further, she will conduct a basic backtest (aka, exception or frequency test) where the frequency of tail losses fit a binomial distribution. Her backtest window is limited to two years, $T = 500$ days. In regard to this backtest, which of the following statements is TRUE?

Betty 的公司想要基于一天的 99.5% 价值 at risk (VaR) 模型分配经济资本缓冲。此外，她将进行一个基本回测（aka，异常或频率测试），其中尾部损失的频率拟合二项分布。她的回测窗口限制为两年， $T = 500$ 天。就这个回测而言，下列哪一项陈述是正确的？

- A. To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier
为了增加回测的威力，她可以降低 VaR 置信水平并应用更高的 (k) 资本乘数
- B. At this high VaR confidence level and relatively short backtest window, the normal approximation is a better fit to test the model’s bias than the binomial distribution
在这个高 VaR 置信水平和相对较短的回测窗口下，正态近似比二项分布更适合测试模型的偏差
- C. She can achieve a more powerful tests of the same hypothesis, and without the need for any additional information, by utilizing any of several extensions approved by regulators
她可以通过利用任何由监管机构批准的扩展，在没有额外信息的情况下，实现更强大的同一假设测试
- D. Although VaR is always a one-tailed quantile (e.g., the 95.0% standard normal VaR is always 1.645 and never 1.960), the backtest is always a two-tailed (2T) backtest such that Betty must have a lower and an upper cutoff
尽管 VaR 总是单尾分位数（例如，95.0% 标准正态 VaR 总是 1.645 且从不 1.960），但回测总是双尾（2T）回测，因此 Betty 必须有一个下限和一个上限

答案:A

解析:

A 选项正确。为了增加回测的威力，她可以降低 VaR 置信水平并应用更高的 (k) 资本乘数。

一个关键的统计挑战是，高置信水平意味着异常样本量很小。解决方法是降低置信水平，但应用乘数（降低置信水平的 VaR）以近似高置信水平的 VaR。

根据 Jorion 的说法，“这个框架缺乏威力的原因是选择了高 VAR 置信水平（99 百分比），这导致异常样本量太少，无法进行可靠的测试。相比之下，考虑 95 百分比 VAR 置信水平的效果。（为了确保资本金额不受影响，我们可以使用更大的乘数 k。）” — Philippe Jorion. Value at Risk, 3rd Ed.: The New Benchmark for Managing Financial Risk (p. 150). Kindle Edition.

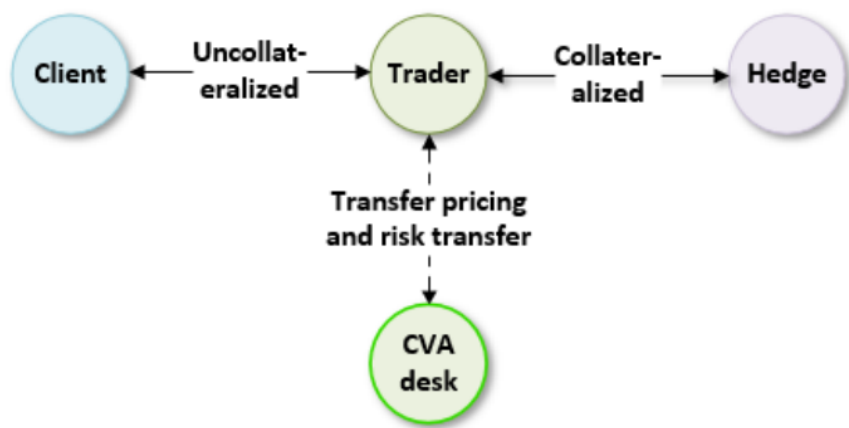
关于 (B)、(C) 和 (D)，每项都是错误的。具体来说：

- 一般准则表明，当 $n \times p$ 和 $n \times (1 - p)$ 大于 10 时，可以使用正态近似；有些人认为大于 5。在这种情况下， $0.5\% \times 500 = 2.5$ ，因此二项分布可能过于偏斜。
- 阅读（Jorion、Hull 或 Dowd）并未提及通过监管批准的扩展获得功率增益
- VaR 总是单尾分位数（例如，95.0% 标准正态 VaR 总是 1.645 且从不 1.960），但回测是统计检验，可以是单尾（1T）或双尾（2T）

考察: Value at Risk (VaR) **重要程度:** [■■■□]

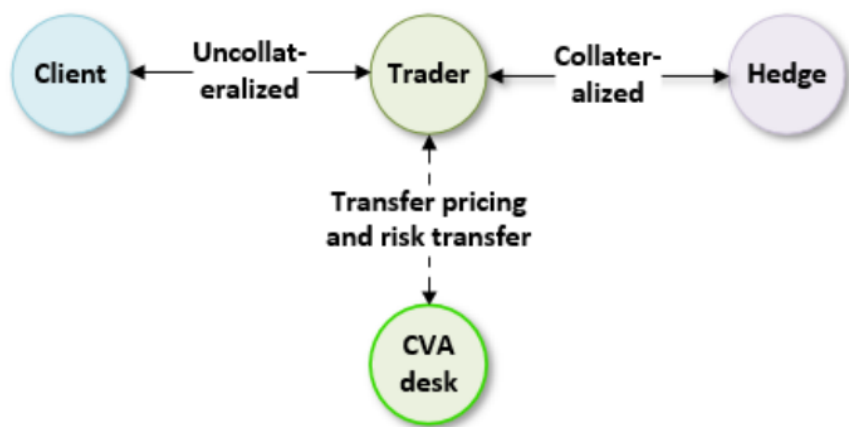
科目: Quantitative Analysis

57. The figure below illustrates the general role of the credit value adjustment (CVA) desk at Heroic Multifunction Bank (aka, the bank). Assume the trader is a market maker. The uncollateralized trade with the customer (on the left) is hedged with a collateralized trade (on the right):



Assume the uncollateralized transaction (with the client) has a positive mark-to-market (MTM) value such that, by definition, the collateralized hedge position, therefore, has a negative MTM value. If the market moves such that, with respect to the position with the client, the position's positive MTM value INCREASES and becomes even more positive, then which of the following implications is most likely to be TRUE?

下图展示了 Heroic Multifunction Bank（即该银行）中信用估值调整（CVA）交易台的一般作用。假设交易员是做市商。与客户之间的无抵押交易（左侧）通过一笔有抵押交易（右侧）进行对冲：



假设与客户之间的无抵押交易具有正的市值（MTM），则按定义，相应的有抵押对冲头寸必然具有负的市值。如果市场发生变动，使得客户头寸的正向市值进一步上升、变得更加为正，则下列哪一项推论最有可能为真？

- A. Funding cost (FVA) is positive
融资成本（FVA）为正
- B. From the trader's (market maker's) perspective, CVA decreases
从交易员（做市商）的角度来看，CVA 减少
- C. Because the position is hedged, there is no associated regulatory cost of capital to the bank
由于头寸得到对冲，银行没有与监管资本相关的成本
- D. The trader (market maker) writes a contingent credit default swap (CCDS) to transfer its counterparty risk
交易员（做市商）写一份或有信用违约互换（CCDS）来转移其对手方风险

答案:A

解析:

A 选项正确。正确：融资成本（FVA）为正。为了对冲，需要抵押品，这笔金额需要融资。关于（B）、（C）和（D），每项都是错误的。

- 关于错误的（B），CVA 可能会随着进一步增加正向敞口而增加

- 关于错误的 (C)，他们的对冲不会消除对监管资本的需求！Gregory 说：“资本：会有资本要求，这些要求会变化——这很重要，因为资本是一种成本。无抵押交易的资本会随着成为实值而增加。对冲的资本不会抵消这一点，因为它是抵押的——即使它是无抵押的，也可能会有凸性影响（一笔交易的资本会增加超过另一笔交易的资本减少）。”
- 关于错误的 (D)，这里的问题是交易员需要购买（购买）而不是出售（卖出）一个 CCDS。Gregory 说：“更定制化的信用衍生产品，如或有 CDS (CCDS)，已被设计为更直接地对冲对手方风险。CCDS 本质上是一种 CDS，但保护的名义金额与合同指定的衍生品（或甚至衍生品组合）的敞口挂钩。它们允许将特定交易和对手方的对手方风险合成转移给第三方。然而，CCDS 市场从未发展出任何重要的流动性。”

(来源：Jon Gregory, 《xVA 挑战：对手方信用风险、融资、抵押和资本》，第 3 版（西萨塞克斯，英国：约翰·威利父子有限公司，2015 年）)

考察: Finance: Counterparty Credit Risk **重要程度:** [■ ■ □]

科目: Quantitative Analysis

58. Andrew Ang makes an important, provocative statement when he writes “Reported illiquid asset returns are not returns.” He claims that people overstate the expected returns and understate the risk of illiquid assets, and he attributes this to three key biases. According to Ang, each of the following is a bias that overstates the expected returns (and/or understates the risk) of illiquid assets EXCEPT which is not accurate?

Andrew Ang 写了一句重要、有争议的话：“报告的流动性较差资产的回报不是回报。”他声称人们高估了流动性较差资产的预期回报，低估了其风险，并将此归因于三个关键偏差。根据 Ang，下列哪一项是高估了流动性较差资产的预期回报（和/或低估了风险）的偏差，除了哪一项不准确？

- A. Survivorship bias can inflate returns by 4.0% or more
生存偏差可以将回报提高 4.0% 或更多
- B. Infrequent sampling (aka, infrequent trading) artificially reduces risk and risk-related metrics such as volatility, correlation and beta
不频繁采样（aka，不频繁交易）人为地降低了风险和与风险相关的指标，如波动性、相关性和 beta
- C. Turnover bias decreases the typical time between transactions and tends to artificially increase the expected return by 5.0% or more
turnover bias 降低了交易之间的典型时间，并倾向于人为地增加预期回报 5.0% 或更多
- D. Selection bias (aka, reporting bias) is a distortion of the sample that artificially increases (ie, overestimates) alpha and artificially decreases (ie, underestimates) beta
选择偏差（aka，报告偏差）是样本的扭曲，人为地增加（即高估）alpha 和人为地减少（即低估）beta

答案:C

解析:

C 选项正确。 错误。turnover bias（换手率偏差）是杜撰出来的。关于 (A)、(B) 和 (D)，每一项都正确。根据 Ang 的说法，对于流动性较差资产，使投资者高估预期回报并低估风险的三种偏差分别是：生存偏差、不频繁采样和选择偏差。

考察: Alternative Investments: Illiquid Assets **重要程度:** [■ ■ □]

科目: Quantitative Analysis

59. Which of the following provides the BEST explicit link(s) between a bank’s contingency funding plan (CFP) and its liquidity stress testing framework?

哪一项提供了银行 contingency funding plan（CFP）与其流动性压力测试框架之间最佳的显式链接？

- A. Communication plan
沟通计划
- B. Limit structure and escalation levels
限制结构和升级级别

C. Funding sources concentration and non-core funding
资金来源集中度和非核心资金

D. Industry factors such as profitability trends in the financial sector
行业因素，如金融行业的盈利趋势

答案:B

解析:

B 选项正确。正确：限制结构和升级级别

解释 Lai 和 Tuosto 的观点（加重号为原文所有）：“II. 与更广泛的风险管理框架整合：CFP（应急融资计划）不是一个孤立的工具，而是机构流动性风险管理和集团范围风险管理框架的有机组成部分……CFP 应当与流动性风险度量框架和流动性压力测试框架明确地结合起来，尤其是通过其限额结构和升级级别。举例来说，机构在日常（BAU）风险管理活动中使用的流动性风险度量，为 CFP 确定其预警指标（EWI）提供了基础。此外，与业务连续性和危机管理框架的关联，会在危机时期加强关键的运营和沟通流程。”——摘自 Chi Lai 和 Richard Tuosto，《Venkat 的流动性风险管理》中的《应急融资计划》

考察: Liquidity Risk Management **重要程度:** [■■■□]

科目: Quantitative Analysis

60. Bodie, Kane and Marcus explain, based on an analogy originally developed by Robert C. Merton, that we can attempt to quantify the value of market timing by recognizing that foresight is similar to holding a call option on the owned equity portfolio (without paying the premium): the perfect market timer exercises her option, so to speak, by switching from the riskfree asset to the equity portfolio (or vice-versa) depending on which offers the better return. Consequently, the Black-Scholes (BSM) formula can be used, but when the strike price is given by $\$1.00 \cdot \exp(rT)$, the BSM call option formula conveniently reduces to $c = 2N[0.5\sigma(e)\sqrt{T}] - 1$.

If we assume annual forecasts ($T = 1.0$ year), and the volatility of the equity portfolio, $\sigma(e) = 36.0\%$ per annum, which of the following is **nearest** to the value of perfect market timing, as a percentage of the equity portfolio’s value?

Bodie, Kane 和 Marcus 基于 Robert C. Merton 最初发展的类比，解释了我们可以尝试量化市场时机价值，通过认识到远见卓识类似于持有 owned equity portfolio（即股票投资组合）的看涨期权（无需支付溢价）：完美的市场时机者行使她的期权，换句话说，根据哪一方提供更好的回报，从无风险资产切换到股票投资组合（或反之亦然）。因此，Black-Scholes（BSM）公式可以被使用，但当敲定价格为 $\$1.00 \cdot \exp(rT)$ 时，BSM 看涨期权公式方便地简化为 $c = 2N[0.5\sigma(e)\sqrt{T}] - 1$ 。

如果我们假设年度预测（ $T = 1.0$ 年），以及股票投资组合的波动率， $\sigma(e) = 36.0\%$ 每年，哪一项最接近完美市场时机的价值，作为股票投资组合价值的百分比？

- A. 7.39%
- B. 14.28%
- C. 46.50%
- D. 83.77%

答案:B

解析:

B 选项正确。在这种情况下， $0.5\sigma(e)\sqrt{T} = 0.5 \times 36.0\% \times \sqrt{1} = 18.0\% = 0.180$ 和 $N(0.18) = 0.57140$ 如果按照查找表四舍五入。然后 $c = 2 \times 0.57140 - 1 = 14.28\%$ 。

考察: Portfolio Management **重要程度:** [■■■□]

科目: Quantitative Analysis

61. Patricia has been asked by the Chief Risk Officer (CRO) to estimate value at risk (VaR) at high confidence levels. Initially, she fits a generalized extreme value (GEV) distribution to the loss data. However, some members of the ultimate audience (the board) do not understand the math. She observes that Dowd(†) explains, ”[t]here are also short-cut ways to estimate VaR (or ES) using EV theory. These are based on the idea that if $\xi > 0$, the tail of an extreme loss distribution follows a power-law times a slowly varying function: $F(x) = k(x)x^{-1/\xi}$ where $k(x)$ varies slowly with x . For example, if we assume for

convenience that $k(x)$ is approximately constant, then [this equation] becomes: $F(x) \approx kx^{-1/\xi}$ ". Notice the assumption that $k(x)$ is approximately constant.

The exponent in Dowd’s power law maintains consistency with the generalized extreme value (GEV) distribution by referring to the exponent as a negative reciprocal, $-1/\xi$, but given this exponent is also a constant along with $k(x)$, we can further simplify the approximation: instead of $F(x) \approx kx^{-1/\xi}$, we can say that $F(x) \approx kx^{-\xi}$ which is sometimes represented as $F(x) \approx kx^{-\alpha}$ where $\alpha = 1/\xi$. For example, Patricia assumes the exponent’s value is -4.0. In the former syntax, $\xi = 0.25$ such that $(-1/\xi) = -4.0$; in the latter version, $\alpha = 4.0$ such that $(-\alpha) = -4.0$, to the same effect. For our purpose here, we will simply refer to $F(x) \approx kx^{-\xi}$.

Consequently, she decides to estimate extremely confident VaRs with a power-law shortcut. Specifically, when Patricia sets xi, $\xi = 4.0$, she observes that the 99.50% VaR is \$6.0 million. With respect to the exponent, she assumes $F(x) \approx kx^{-\xi} = F(x) \approx kx^{-4.0}$. If the loss tail follows the power law, which is **nearest** to the implied 99.90% VaR?

Patricia 被首席风险官（CRO）要求估计高置信水平的价值 at risk（VaR）。最初，她将广义极值（GEV）分布拟合到损失数据。然而，最终受众（董事会）中的一些成员不理解数学。她观察到 Dowd(†) 解释说：“还有其他基于 EV 理论的简便方法来估计 VaR（或 ES）。这些是基于这样的想法：如果 $\xi > 0$ ，极端损失分布的尾部遵循幂律乘以缓慢变化函数： $F(x) = k(x)x^{-1/\xi}$ 其中 $k(x)$ 随 x 缓慢变化。例如，如果我们假设为了方便起见 $k(x)$ 是大约恒定的，那么 [这个方程] 变为： $F(x) \approx kx^{-1/\xi}$ ”。注意假设 $k(x)$ 是大约恒定的。

幂律的指数与广义极值（GEV）分布的指数一致，通过将指数称为负倒数， $-1/\xi$ ，但鉴于这个指数也与 $k(x)$ 一起是常数，我们可以进一步简化近似：代替 $F(x) \approx kx^{-\xi}$ ，我们可以说 $F(x) \approx kx^{-\xi}$ 有时表示为 $F(x) \approx kx^{-\alpha}$ 其中 $\alpha = 1/\xi$ 。例如，Patricia 假设指数的值是 -4.0。在前一种语法中， $\xi = 0.25$ 使得 $(-1/\xi) = -4.0$ ；在后一种版本中， $\alpha = 4.0$ 使得 $(-\alpha) = -4.0$ ，效果相同。为了我们的目的，我们将简单地参考 $F(x) \approx kx^{-\xi}$ 。

因此，她决定使用幂律捷径估计极其自信的 VaR。具体来说，当 Patricia 设置 xi, $\xi = 4.0$ 时，她观察到 99.50% VaR 为 \$6.0 百万。关于指数，她假设 $F(x) \approx kx^{-\xi} = F(x) \approx kx^{-4.0}$ 。如果损失尾部遵循幂律，哪一项最接近 implied 99.90% VaR?

- A. \$5.0 million
- B. \$9.0 million
- C. \$13.0 million
- D. \$21.0 million

答案:B

解析:

B 选项正确。首先要求解 k 。Dowd(†) 使用指数 $(-1/\xi)$ ，但我们将在指数中使用更简单的倒数 $(-\xi)$ 。这样，幂律说: $\text{Prob}(v > x) = F(x) \approx kx^{-\xi}$ 。求解常数 k ，必须有: $k = F(x)x^{\xi}$ 。在这种情况下，当 $x = 6.0$ 时 $F(x) = 0.50\%$ 且 $\xi = 4.0$ ，我们可以找到 $k = 0.50\% \cdot 6.0^4 = 6.480$ 。现在我们知道，因为我们假设常数 (k) 近似，我们的分布是 $F(x) \approx kx^{-\xi} = 6.480x^{-4.0}$ 。我们想要 $F(x) = 0.10\%$ 的 VaR，即 $0.10\% = 6.480x^{-4.0}$ ，其中 $x = (6.480/0.10\%)^{1/4.0} = 8.97$ ，四舍五入到 \$9.0 百万。

考察: Quantitative Analysis: Extreme Value Theory 重要程度: [■■■]

科目: Quantitative Analysis

62. Acme Financial has a position in five derivative contracts with its counterparty. The current mark-to-market values of the positions are the following: +12, -9, +20, -6, and +8. The five positions are all included in a single netting set. Let’ s define the “netting benefit” as the reduction in current exposure due to netting; that is, the difference between the total current exposure with netting and the current exposure “as if” netting did not apply. What is the netting benefit in this set? Acme Financial 在其对手方有五个衍生合约头寸。这些头寸的当前市值如下: +12, -9, +20, -6, 和 +8。这五个头寸都包含在一个单一的净额结算集中。让我们定义“净额收益”为由于净额结算而减少的当前敞口; 即, 净额结算后的总当前敞口与假设没有净额结算时的当前敞口之间的差异。这个集合中的净额收益是多少?

- A. Zero
- B. 3
- C. 15
- D. 25

答案:C

解析:

C 选项正确。正确：15。没有净额结算时，总敞口 = $\max(0, +12) + \max(0, -9) + \max(0, 20) + \max(0, -6) + \max(0, +8) = 40$ 。有净额结算时，总敞口 = $+12 - 9 + 20 - 6 + 8 = 25$ 。注意这与 $-(-9 - 6)$ 相同。

考察: Counterparty Credit Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

63. Patricia is a risk analyst who works for a large international bank. Her team is developing a new risk management framework for the bank and is currently working on risk identification. As part of extreme risk identification, Patricia has been assigned the important task of designing and performing the firm’s scenario analysis. Soon she will facilitate a brainstorming session for which she has prepared the following documents: external loss data, internal loss data (including near misses), risk and control self-assessment (RCSA) results, key risk indicator scores, audit issues, and concentrated exposures. Assuming she continues to follow best (or good) practice with respect to scenario analysis, which of the following is a TRUE statement?

Patricia 是一名风险分析师，在大银行工作。她的团队正在为银行开发新的风险管理框架，目前正在进行风险识别。作为极端风险识别的一部分，Patricia 被分配了重要任务，即设计和执行公司的场景分析。不久她将主持一个头脑风暴会议，她准备了以下文件：外部损失数据、内部损失数据（包括近似事件）、风险和控制自我评估（RCSA）结果、关键风险指标得分、审计问题以及集中敞口。

假设她继续遵循场景分析的最佳（或良好）实践，下列哪一项陈述是正确的？

- A. Because most large operational losses result from external causes, her primary focus should be scenarios driven by external causes
因为大多数大型运营损失来自外部原因，她的主要焦点应该是由外部原因驱动的场景
- B. She should compare her firm’s generated scenarios to industry-specific list(s) published by external sources, but such a comparison should be made after her firm’s own internal generation exercise
她应该比较她公司生成的场景与外部来源发布的行业特定列表，但这样的比较应该在她公司自己的内部生成练习之后进行
- C. The initial output of scenario identification should be limited to approximately 10 or 12 scenarios in order to avoid the fallacy of missing the forest for the trees
场景识别的初始输出应限制在约 10 或 12 个场景，以避免忽视森林的谬误
- D. Because the ultimate goal of scenario analysis is to predict the future with a sufficient degree of certainty, her process should acknowledge and elevate the role of human subjectivity
因为场景分析的最终目标是预测未来，具有足够的确定性，她的过程应该承认并提升人类主观性的作用

答案:B

解析:

B 选项正确。 B. 正确：她应当将自己公司生成的场景与外部来源发布的行业特定场景列表进行比较，但这种比较应在公司自行内部生成场景之后进行。根据 GARP（第 3.2 节），“最后，公司比较自己所生成的场景与行业场景列表是有益的，可以判断是否遗漏了相关场景或风险驱动因素。比如，保险业操作风险协会（ORIC）和 ORX 等行业组织会向其成员提供现成的场景列表。然而，建议企业只在场景生成结束后，再将其自有场景与外部行业列表比较，而非事前进行，以避免外部场景影响公司的生成过程。” (†)

至于 (A)、(C) 和 (D)，它们都是错误的。具体来说：

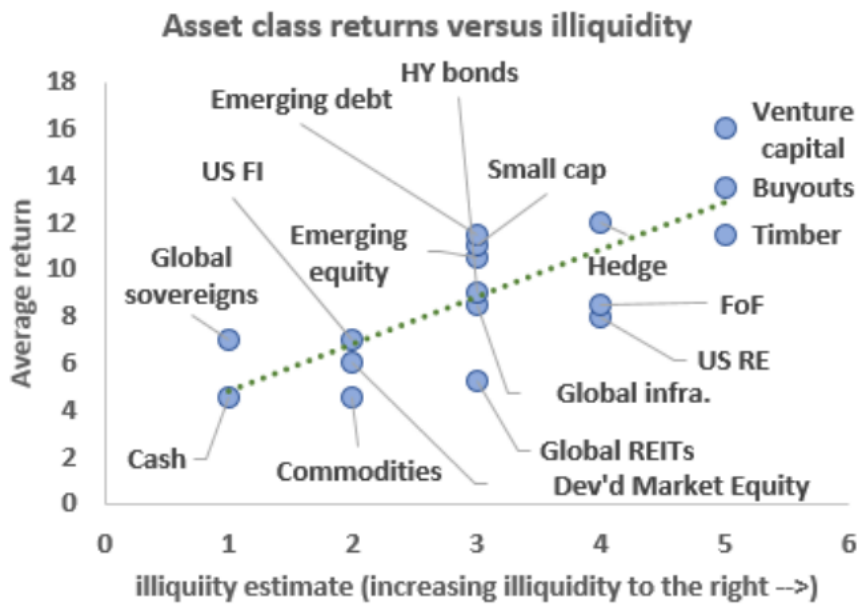
- 大多数大型操作风险损失来源于内部因素
- GARP 建议应有更多场景，例如中等规模公司至少应有 20 30 个场景。对于大型国际银行，~50 个场景也并不少见
- 场景分析不是为了预测未来，而是为了识别多种情景。GARP 还指出：“监管机构要求公司在进行场景识别与评估过程中尽可能减少主观性和偏见。” (†)

(†) 2023 GARP FRM Part II: Operational Risk and Resilience, Pearson Learning Solutions

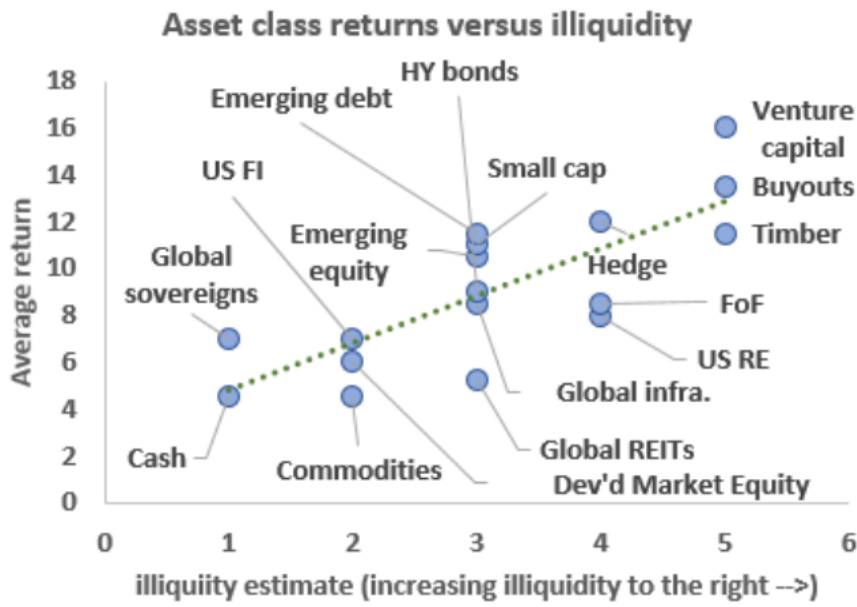
考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

64. In order to identify the presence of illiquidity risk premium(s), Andrew Ang(†) references data presented by Antti Ilmanen in his well-regarded book Expected Returns (An Investor’ s Guide to Harvesting Market Rewards). This data is displayed below as a scatterplot where the y-axis is the long-run average return of the asset class and the x-axis is an index of illiquidity. A higher index (ie, to the right) implies less liquidity. For example, the venture capital as an asset class is assigned to the least liquid (most illiquid) asset class but it also plots the highest long-run average return.



In regard to the illiquidity risk premium, which of the following statements is TRUE?
为了识别流动性风险溢价，Andrew Ang(†) 引用了 Antti Ilmanen 在其广受好评的书籍 Expected Returns (An Investor’ s Guide to Harvesting Market Rewards) 中呈现的数据。这些数据以散点图形式显示，其中 y 轴为资产类别的长期平均回报，x 轴为流动性指数。一个更高的指数（即向右）意味着更少的流动性。例如，风险投资作为一种资产类别被归类为流动性最差（最不流动性）的资产类别，但它也绘制了最高的长期平均回报。



就流动性风险溢价而言，下列哪一项陈述是正确的？

- A. In general illiquid asset classes offer high risk-adjusted returns
一般来说，流动性较差资产类别的风险调整回报率较高
- B. These charts demonstrate that there do exists large illiquidity risk premiums ACROSS asset classes
这些图表表明，确实存在跨资产类别的较大流动性风险溢价
- C. There do exist large illiquidity risk premiums WITHIN many asset classes, but not ACROSS asset classes
许多资产类别内部确实存在较大的流动性风险溢价，但不是跨资产类别

D. Illiquid equities earn the same returns as liquid equities; and illiquid bonds earn the same returns as liquid bonds
流动性较差股票与流动性较好股票获得相同的回报；流动性较差债券与流动性较好债券获得相同的回报

答案:C

解析:

C 选项正确。正确：许多资产类别内部确实存在较大的流动性风险溢价，但不是跨资产类别。

考察: Finance: Portfolio Management 重要程度: [■■■]

科目: Quantitative Analysis

65. Sally represents an investor who likes the strategy and style of a new, entrepreneurial start-up hedge fund. At her client’s request, she is going to conduct due diligence. According to Kevin Mirabile (author of the hedge fund due diligence reading), as Sally conducts her diligence of the hedge fund on behalf of the investor, which of the following is the most important principle or ingredient?

Sally 代表一位喜欢新创业对冲基金策略和风格的投资者。在客户的要求下，她将进行尽职调查。根据 Kevin Mirabile（对冲基金尽职调查阅读的作者），作为 Sally 代表投资者进行对冲基金尽职调查，哪一项是最重要的原则或成分？

- A. She should observe strong evidence that the fund has eliminated business model risk via the SEC’s Zeta Sigma AIML filter
她应该观察到 strong evidence 表明该基金通过 SEC 的 Zeta Sigma AIML 过滤器消除了商业模型风险
- B. She should confirm that the CEO is exceptionally good at all three top-level skills sets: investment, operational, and business
她应该确认 CEO 在所有三个顶级技能集：投资、运营和业务方面都非常出色
- C. She should follow all the steps in the process but with flexibility as due diligence is also an art and each fund is unique with its own strategy
她应该遵循所有步骤，但要有灵活性，因为尽职调查也是一种艺术，每个基金都有自己独特的策略
- D. She should clarify that the fund’s incentives are actual equity ownership grants rather than phantom equity and that they are granted throughout the organization to all full-time employees.
她应该澄清该基金的激励措施是实际的股票所有权授予，而不是幻影股票，并且它们在整个组织中授予所有全职员工。

答案:C

解析:

C 选项正确。她应该遵循所有步骤，但要有灵活性，因为尽职调查也是一种艺术，每个基金都有自己独特的策略。GARP 解释说：“对对冲基金进行尽职调查既像艺术又像科学。投资者必须使用全面的检查清单，确保没有任何遗漏或遗漏，同时保持自由提出开放式问题，以提供对公司哲学或文化的洞察。这里讨论的投资、运营和商业模型风险评估指南主要是通用的。在执行尽职调查时，投资者必须修改他们的方法，使其具有策略针对性。” (Summary, 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.)

关于错误的 (A)、(B) 和 (D)：

- Zeta Sigma AIML 过滤器是虚构的，商业模型风险无法消除
- 作者指出，只有“恐龙”（作为创始人）试图管理所有三个，因为经理们很少在所有三个方面都做得很好（更不用说特别好了）！经理们不应该选择同时担任 CEO 和 CIO。
- 激励结构因组织而异。

考察: Hedge Funds: Due Diligence 重要程度: [■■■]

科目: Quantitative Analysis

66. Bertha is a risk manager who estimates the market risk of one of her firm’ s portfolios. Because geometric returns are time-additive and arithmetic returns are cross-additive, she measures performance with both arithmetic and geometric returns. Her assumptions are gathered below:

- The portfolio value is \$100.0 million and she assumes there are 250 trading days per year.
- Using geometric returns, the average return +12.0% per annum with a standard deviation of 28.0%

- Using arithmetic returns, the average return +14.0% per annum with a standard deviation of 32.0%

She wants to measure the one-day 95.0% value at risk (VaR) under both approaches: assuming arithmetic returns are normal (aka, normal VaR) and assuming geometric returns are normal; aka, lognormal VaR. She assumes the returns are serially independent. Which of the following is **nearest** to the difference between the one-day 95.0% normal VaR and the one-day 95.0% lognormal VaR?

Bertha 是一位风险经理，估计她公司一个投资组合的市场风险。由于几何回报是时间可加的，而算术回报是交叉可加的，她使用算术和几何回报来衡量绩效。她的假设如下：

- 投资组合价值为 \$100.0 百万，她假设每年有 250 个交易日。
- 使用几何回报，平均回报为 +12.0% 每年，标准差为 28.0%
- 使用算术回报，平均回报为 +14.0% 每年，标准差为 32.0%

她想测量两种方法下的一日 95.0% 价值 at risk (VaR)：假设算术回报是正态的（aka，正态 VaR）和假设几何回报是正态的；aka，对数正态 VaR。她假设回报是序列独立的。哪一项最接近一日 95.0% 正态 VaR 与一日 95.0% 对数正态 VaR 之间的差异？

- A. The one-day 95% normal VaR is approximately \$37,700 less than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 少 \$37,700
- B. The one-day 95% normal VaR is approximately \$380,100 less than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 少 \$380,100
- C. The one-day 95% normal VaR is approximately \$29,400 greater than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 多 \$29,400
- D. The one-day 95% normal VaR is approximately \$448,800 greater than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 多 \$448,800

答案:D

解析:

D 选项正确。一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 多 \$448,800。

The one-day 95% normal VaR(%) is given by:

$$-14\%/250 + 32\% \times 1/\sqrt{250} \times 1.645 = 3.273246\%$$

一日 95.0% 正态 VaR (%) 为:

$$-14\%/250 + 32\% \times 1/\sqrt{250} \times 1.645 = 3.273246\%$$

一日 95.0% 对数正态 VaR (%) 为:

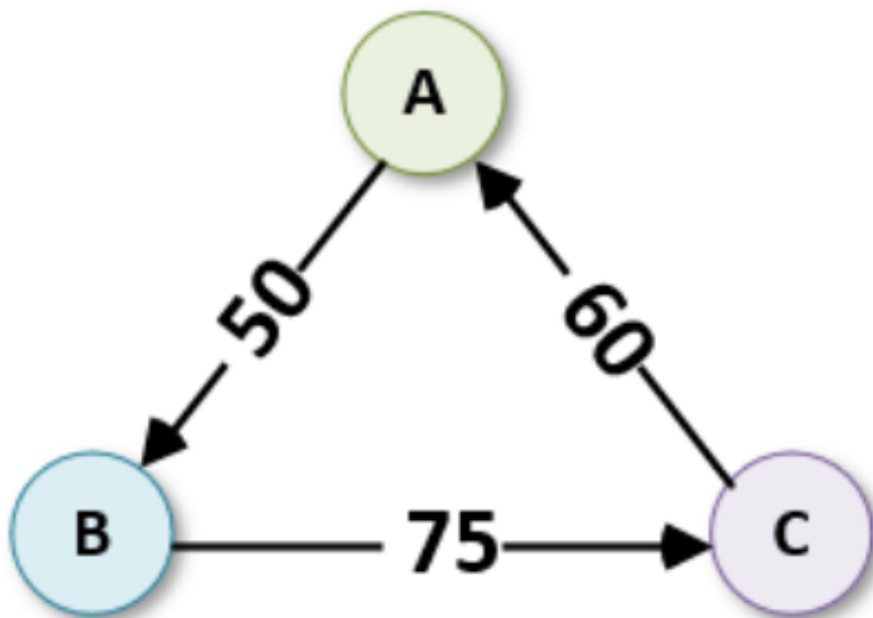
$$1 - \exp(12\%/250 - 28\% \times 1/\sqrt{250} \times 1.645) = 2.824436\%$$

差异为 $3.273246\% - 2.824436\% = 0.448810\%$ ，或大约 $0.448810\% \times \$100.0 \text{ million} = \$448,810$ ；使用精确分位数，差异为 \$448,765.91。

考察: Quantitative Analysis: Value at Risk **重要程度:** [■ ■ □]

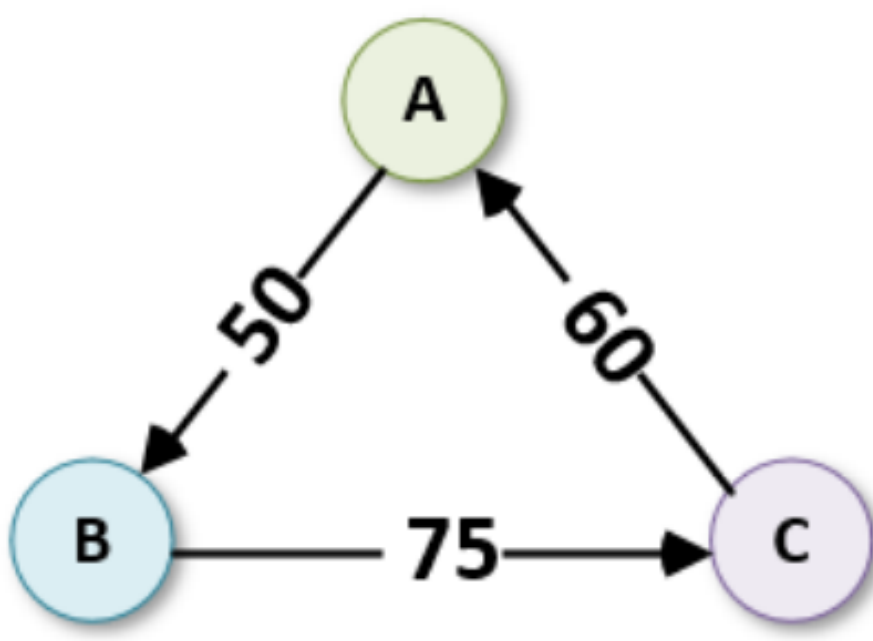
科目: Quantitative Analysis

67. The graph below represents a trilateral scenario between three different counterparties (called A, B, and C) with position sizes in a certain fungible (aka, interchangeable) product. The values represent notional sizes and the direction represents exposures:



In this ring, the current total notional is 370. If trade compression can be achieved, which of the following is an efficient solution?

这个图展示了三个不同对手方（称为 A、B 和 C）在某种可互换产品中的头寸大小。值表示名义金额，方向表示敞口：



在这个环中，当前总名义金额为 370。如果可以实现交易压缩，哪一项是最有效的解决方案？

- A. A -> 25 -> B -> 60 -> C (new total notional = 110)
选项 A：A -> 25 -> B -> 60 -> C (新总名义金额 = 110)
- B. A -> 30 -> B -> 45 -> C (new total notional = 150)
选项 B：A -> 30 -> B -> 45 -> C (新总名义金额 = 150)
- C. B -> 35 -> C -> 30 -> A (new total notional = 65)
选项 C：B -> 35 -> C -> 30 -> A (新总名义金额 = 65)
- D. B -> 25 -> C -> 10 -> A (new total notional = 70)
选项 D：B -> 25 -> C -> 10 -> A (新总名义金额 = 70)

答案:D

解析:

D 选项正确。正确: B -> 25 -> C -> 10 -> A (新总名义金额 = 70)

最简单的方法是识别最低敞口, 在这种情况下为 50, 并从所有三个中减去它。这样:

- 当前 A -> 50 -> B 被消除;
- 当前 B -> 75 -> C 减少为 B -> (75 - 50 = 25) -> C;
- 当前 C -> 60 -> A 减少为 C -> (60 - 50 = 10) -> A

考察: Finance 重要程度: [■ ■ □]

科目: Quantitative Analysis

68. Sally is a risk analyst at an international bank who is leading a project to conduct a causal analysis of the bank’s operational risk. The project will coordinate with, and inform modifications to, the bank’s loss database. One of the project’s goals is to analyze the organization’s controls, control layers, and key risk indicators (KRIs), and key control indicators (KCIIs).

She prefers to use a factor model, and her candidates include the following approaches:

- *Fault tree analysis (FTA)*: this deductive method will build a series of and/or conditions that can happen simultaneously or alternatively
- *Root-cause analysis with bowtie tool*: the visual tool will employ the 5-why analysis to probe deeply into the root causes of events
- *FAIR methodology*: this structured framework will generate scenarios, where each scenario will assume an asset at risk, a threat community, a threat type, and an effect
- *Swiss cheese model*; aka cumulative act effects: by anticipating active failures and identifying latent conditions, this model will promote effective control layering and identifying active failures

Each of these four approaches has advantages and disadvantages. For ANY of these approaches, which of the following paired statements (i.e., a single advantage paired with a single disadvantage) is a TRUE statement?

Sally 是一名国际银行的风险分析师, 领导一个项目, 进行银行运营风险的有因分析。该项目将与银行损失数据库的修改协调, 并告知修改。该项目的目标是分析组织的控制、控制层和关键风险指标 (KRIs) 以及关键控制指标 (KCIIs)。她偏好使用因子模型, 她的候选方法包括以下几种:

- *Fault tree analysis (FTA)*: 这个演绎方法将构建一系列同时或替代发生的事件或条件
- *Root-cause analysis with bowtie tool*: 这个可视化工具将使用 5-why 分析深入探究事件的根源原因
- *FAIR methodology*: 这个结构化框架将生成场景, 每个场景假设一个风险资产、威胁社区、威胁类型和影响
- *Swiss cheese model*; aka cumulative act effects: 通过预测主动失败和识别潜在条件, 这个模型将促进有效的控制层级和识别主动失败

每种方法都有优点和缺点。对于任何一种方法, 下列哪一对陈述 (即单个优点与单个缺点) 是正确的陈述?

- A. Fault tree analysis (FTA) is simple and non-reliant on domain expertise; however, it is unrealistic because it can only assume independent events
故障树分析 (FTA) 简单且不依赖于领域专业知识; 然而, 它是不现实的, 因为它只能假设独立事件
- B. The Bowtie tool is proactive because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes; however, it natively lacks a quantitative framework such that prioritization is harder
Bowtie 工具弓形图工具是前瞻性的, 因为它通过检视控制和间接原因, 在风险尚未发生前评估风险的可能性和/或影响; 但它本身缺少定量框架, 因此优先级排序更困难
- C. The FAIR model is inclusive because it emphasizes subjective identification (aka, findability) of factors; however, it is not a probabilistic approach and may be insufficient when distributions are required
FAIR 模型是包容的, 因为它强调对因素的主观识别 (即可发现性); 然而, 它不是一种概率方法, 在需要分布时可能不足

D. The Swiss cheese model is elegant because clusters and their centroids (aka, holes) are quantitatively retrieved via k-means clustering; however, it struggles when incorporating human factors into a holistic risk perspective he Swiss cheese 模型很优雅，因为聚类及其中心（即“洞”）是通过 k-means 聚类定量获得的；但是，当需要把人因纳入整体风险视角时，它会遇到困难

答案:B

解析:

B 选项正确。正确：Bowtie 工具是前瞻性的，因为它通过检视控制和间接原因，在风险尚未发生前评估风险的可能性和/或影响；然而，它本身缺少定量框架，因此优先级排序更困难

考察: Operational Risk 重要程度: ☒ ☒ ☐

科目: Financial Markets and Products

69. According to Rose and Hudgins, the factors affecting the choice of non-deposit funding sources include the following: the relative costs of each funding source; the risk of each funding source, including the dependability and volatility; the length of time for which funds will be needed; the size of the financial institution needing the funds; and regulations on the use of other sources of funds. In regard to the various sources of non-deposit funding, which of the following statements is TRUE?

根据 Rose 和 Hudgins，影响非存款资金来源选择的因素包括以下几个：每个资金来源的相对成本；每个资金来源的风险，包括可靠性和波动性；资金需求的时间长度；需要资金的金融机构的规模；以及对使用其他资金来源的监管。在各种非存款资金来源方面，下列哪一项陈述是正确的？

- A. The Fed Funds market is the least popular because its effective interest rate is higher than the alternatives
联邦基金市场最不受欢迎，因为它比其他选择的有效利率更高
- B. Repurchase agreements are more popular than Fed Funds because they are simpler, however, they incur greater credit risk
回购协议比联邦基金更受欢迎，因为它们更简单，但它们承担更大的信用风险
- C. The advantages of the commercial paper (CP) market are the lack of credit risk and consistently high (i.e., low volatility) credit availability, however, volume tends to be low and relative cost is high
商业票据（CP）市场的优势是缺乏信用风险和持续高（即低波动性）的信用可用性，但交易量 tends to be low and relative cost is high
- D. Advances from Federal Home Loan Banks (aka, FHLBanks) are highly popular among thrifts and small banks as a stable source of flexible (with respect to maturity) funding at below-market interest rates
联邦住房贷款银行（aka, FHLBanks）的贷款在储蓄机构和中小银行中非常受欢迎，作为稳定来源的灵活（相对于到期日）资金，以低于市场的利率

答案:D

解析:

D 选项正确。联邦住房贷款银行（aka, FHLBanks）的贷款在储蓄机构和中小银行中非常受欢迎，作为稳定来源的灵活（相对于到期日）资金，以低于市场的利率

根据研究笔记：“联邦住房贷款银行（FHLB）系统近年来已进入向住房抵押贷款提供者贷款。由 12 个地区银行组成，FHLB 于 1932 年创建，作为存款机构面对存款人潜在挤兑时的一个稳定资金来源。根据许多学者，FHLB 在“资金来源”领域的作用不仅增加了使用 FHLB 作为资金来源的金融机构（以住房抵押品为担保）的可用资金量，而且还增加了住房贷款的流动性，并提供了激励更多金融机构向住房市场提供信贷。

近年来，FHLB 的影响力迅速扩大，现在已有 6,000 多家商业银行、1,300 多家储蓄机构、700 多家信用合作社和 80 多家保险公司有资格向 FHLB 借款。FHLB 资金的吸引力在于它是一个稳定的融资来源，并能提供低于市场利率的资金。金融机构可以从 FHLB 按灵活的时间期限借款，期限从隔夜到 20 多年不等，利率可以是固定或浮动的。FHLB 能够提供低于市场利率资金的部分原因是其联邦特许章程允许其低成本融资，并将这些节省下来的成本转嫁给会员银行，后者作为股东可获得分红。此外，在机构发生倒闭时，联邦住房贷款银行系统处于有利地位。根据法律规定，FHLB 在追偿资金方面优先于包括 FDIC 在内的其他机构。

关于（A）、（B）和（C），每项都是错误的。具体来说：

- 联邦基金市场是最受欢迎的，其有效利率比其他选择更低，尽管它是有波动的（这是它的缺点）

- 回购协议比联邦基金更受欢迎，因为它们更复杂，但它们承担更少的信用风险，因为它们由抵押品担保（它们通常被视为“抵押的联邦基金交易”）。
- 商业票据（CP）在交易量上很高，成本适中（优势）但可用容量和信用风险（劣势）有波动

考察: Finance: Non-deposit Funding Sources **重要程度:** [■ ■ □]

科目: Quantitative Analysis

70. We can add a momentum factor to the Fama-French so that it becomes a four-factor model. This momentum factor is denoted by WML (i.e., past winners minus past losers) or UMD (i.e., stocks that have gone up minus stocks that have gone down). At least with respect to the historical window analyzed, which is the long period from January 1965 to December 2011, which of the following statements is TRUE about the momentum factor?

我们可以将动量因子添加到 Fama-French 模型中，使其成为四因子模型。这个动量因子由 WML（即过去赢家减去过去输家）或 UMD（即上涨股票减去下跌股票）表示。至少在分析的历史窗口中，从 1965 年 1 月到 2011 年 12 月，哪一项关于动量因子的陈述是正确的？

- A. Momentum is a negative feedback strategy which is inherently stabilizing
动量是一种负反馈策略，本质上具有稳定性
- B. The momentum factor is observed in equities but is NOT observed in bonds, commodities, and real estate
动量因子在股票中被观察到，但在债券、商品和房地产中没有被观察到
- C. Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly negative
动量投资定义上是一种反价值策略；HML 和 WML 之间的相关性 strongly negative
- D. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value
动量策略的累积利润比规模或价值策略的累积利润大一个数量级

答案:D

解析:

D 选项正确。动量策略的累积利润比规模或价值策略的累积利润大一个数量级。

考察: Factor Investing **重要程度:** [■ ■ □]

科目: Quantitative Analysis

71. Peter has collected the historical sample of daily profit and loss (P/L) observations for his firm’s new equity portfolio but his dataset only contains 250 days. Ideally, he would prefer 1,000 observations because the board asked for a presentation that displays value at risk (VaR) at confidence levels of (99.0%, 99.1%, ..., 99.8%, and 99.9%). The board also expressed a preference for a non-parametric approach; apparently, some directors are concerned that parametric approaches might be too difficult to communicate and understand. In addition to granular VaR confidence levels (e.g., 99.1%), if possible, the board would like to see confidence INTERVALS around the VaR estimates in order to gauge their precision; e.g., a 9X% confidence interval around the 99.1% VaR.

Therefore, Peter considers the following four approaches: basic historical simulation (BHS), bootstrapped historical simulation (BOOT), historical simulation using non-parametric density estimation (NPDE), and age-weighted historical simulation (AWHS). Given the realities of his relatively small, empirical dataset, which of the following statements is TRUE?

Peter 收集了他公司新股票组合的每日损益（P/L）历史样本，但他的数据集只有 250 天。理想情况下，他希望有 1,000 个观察值，因为董事会要求一个显示价值 at risk（VaR）的图表，置信水平为 (99.0%, 99.1%, ..., 99.8%, 和 99.9%)。董事会还表达了对非参数方法的偏好；显然，一些董事担心参数方法可能太难沟通和理解。除了细粒度 VaR 置信水平（例如 99.1%），如果可能的话，董事会希望看到 VaR 估计周围的置信区间，以衡量其精度；例如，一个 9X% 置信区间围绕 99.1% VaR。

因此，Peter 考虑了以下四种方法：基本历史模拟（BHS），引导历史模拟（BOOT），使用非参数密度估计的历史模拟（NPDE），和年龄加权历史模拟（AWHS）。鉴于他相对较小的经验数据集的现实情况，哪一项陈述是正确的？

- A. Peter can easily eliminate the BHS approach because basic historical simulation cannot even retrieve a 99.10% (or 99.20% etc.) VaR quantile due to the fact that 1/250 is greater than 0.10%

Peter 可以很容易地消除基本历史模拟（BHS）方法，因为基本历史模拟甚至无法检索到一个 99.10%（或 99.20% 等）VaR 分位数，因为 $1/250$ 大于 0.10%

B. Peter has a dilemma (i.e., none of the four approaches is feasible) as the construction of confidence levels around VaR estimates will require him to employ a parametric (or semi-parametric at least) approach

Peter 有一个困境（即，没有任何一种方法可行），因为构建 VaR 估计周围的置信水平将需要他采用参数（或至少半参数）方法

C. Peter has a dilemma (i.e., no approach is good here) simply because his sample size is too small to meet the board's preferences under any of the four approaches

Peter 有一个困境（即，没有任何一种方法可行），因为他的样本量太小，无法满足董事会在这四种方法下的偏好

D. Non-parametric density estimation (NPDE) can retrieve VaRs at granular confidence levels (e.g., 99.10%) as if the distribution is a continuous probability density function (pdf)

Non-参数密度估计（NPDE）可以在细粒度置信水平（例如 99.10%）下检索 VaR，就像分布是一个连续概率密度函数（pdf）

答案:D

解析:

D 选项正确。非参数密度估计（NPDE）可以在细粒度置信水平（例如 99.10%）下检索 VaR，就像分布是一个连续概率密度函数（pdf）

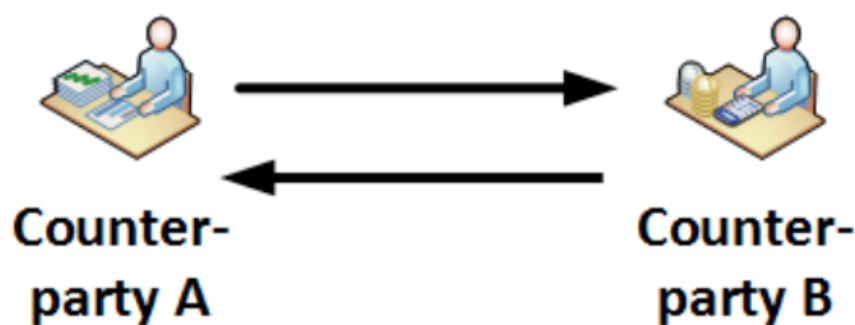
关于（A）、（B）和（C），每项都是错误的。相反，以下是 TRUE 陈述：

- 基本历史模拟肯定返回一个 99.1% VaR，这里它将与 99.0% VaR 相同
- 由于顺序统计量的便利性，可以在所有四种方法下生成置信区间；或者，实际上，可以为任何参数或非参数 VaR 或 ES 估计构建置信区间！
- 小样本量可能有利于引导或非参数密度估计

考察: Market Risk Measurement and Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

72. Two counterparties (A and B, illustrated below) have between them a set of OTC derivative transactions.



Associated with the positions, these counterparties also entered into a bilateral collateral agreement prior to initiation of the underlying transactions. Each of the following is true EXCEPT which is false?

双方交易对手（A 和 B，见下图）之间存在一组场外（OTC）衍生品交易。针对这些头寸，双方在发起基础交易之前还签订了双边抵押协议。以下陈述均为真，除了哪一项为假？

A. If they followed best practice, collateral posting should be continuous for both of them

如果他们遵循最佳实践，抵押品应连续存入双方

B. As a counterparty risk mitigant, collateral posting is generally superior to a reset feature

作为对手方风险缓解措施，抵押品存入通常优于重置功能

- C. Variation margin is a type of collateral that aims to track the change in valuation (aka, change in MTM) of the position over time
变动保证金是一种抵押品，旨在跟踪时间内的头寸估值（aka，MTM 变化）变化
- D. As a bilateral agreement, in the case of a positive MTM, the party will call for collateral; and in the case of negative MTM, the party will be required to themselves post collateral
作为双方协议，在正 MTM 情况下，一方将要求抵押品；在负 MTM 情况下，一方将被迫自己存入抵押品

答案:A

解析:

A 选项正确。错误：连续存入抵押品在操作上太昂贵关于（B）、（C）和（D），每项都是正确的。

考察: Counterparty Credit Risk 重要程度: [■■■□]

科目: Quantitative Analysis

73. MetroScape Bank prides itself on its modern banking solutions and customer-centric approach. During the last quarter, however, a series of events occurred that suggest the bank’s controls should (or at least can) be improved. Specifically, each of the four major types of human error was observed:

- I. Jane, a seasoned cashier at MetroScape, had been handling cash transactions for years. One busy Friday afternoon, while multitasking between customer queries and cash handling, she inadvertently gave out \$500 more to a customer than he had withdrawn. Her muscle memory, developed from years of handling similar transactions, failed her in that moment of distraction.
- II. MetroScape recently redesigned its compensation and incentive programs. The incentive program rewards cross-selling, especially when a valuable customer adds an additional account and/or utilizes a new program. Due to this incentive plan, some branches aggressively cross-sold products to existing customers without sufficient regard for product suitability. Several customers called headquarters to complain about the sales tactics.
- III. A new investment product was introduced at MetroScape, and the bank’s financial advisors were tasked with guiding interested clients. Tom, one of the advisors, faced a unique situation where a client wanted to merge this new product with another existing one. Unsure and without prior experience or training in this scenario, Tom made an educated guess. Unfortunately, his decision led to significant tax implications for the client.
- IV. Lucy, a senior executive at MetroScape, was aware of the bank’s policy against insider trading. However, tempted by the prospect of significant personal gain, she used confidential information about an upcoming merger to buy shares in a company that MetroScape was about to acquire. Her actions, driven by personal greed, were a clear violation of both the bank’s policy and federal regulations.

In regard to these events and their associated control implications, each of the following statements is true EXCEPT which is false?

MetroScape 银行自豪于其现代银行业解决方案和客户中心方法。然而，在最后一个季度，一系列事件表明银行的控制应该（或至少可以）得到改进。具体来说，观察到了四种主要类型的人为错误：

- I. Jane，一位经验丰富的出纳员，多年来一直在处理现金交易。一个繁忙的周五下午，她在处理客户查询和现金处理时，无意中多给了客户 \$500。她的肌肉记忆，来自于多年处理类似交易的经验，在那一刻的分心中失败了。
- II. MetroScape 最近重新设计了其薪酬和激励计划。激励计划奖励交叉销售，特别是在一个重要客户添加另一个账户和/或使用新程序时。由于这个激励计划，一些分支机构对现有客户进行了激进的交叉销售，而没有充分考虑产品的适用性。许多客户打电话到总部抱怨销售策略。
- III. MetroScape 引入了一种新的投资产品，银行的财务顾问被要求引导感兴趣的客户。Tom，一位顾问，面临一个独特的情况，一个客户想要将这个新产品与其他现有产品合并。不确定且没有之前在这个场景中的经验或培训，Tom 做出了一个有根据的猜测。不幸的是，他的决定导致了客户的重要税务影响。
- IV. Lucy，MetroScape 的一位高级执行官，知道银行的政策禁止内部交易。然而，被可能的重大个人收益所诱惑，她使用了关于即将进行的合并的机密信息，购买了 MetroScape 即将收购的公司的股票。她的行为，由个人贪婪驱动，清楚地违反了银行的政策和联邦法规。

针对这些事件及其相关的控制影响，下列陈述均为真，除了哪一项为假？

- A. The fourth (IV.) is a violation which is best mitigated by supervisory controls
第四（IV.）是一种违反，最佳缓解由监管控制
- B. The third (III.) is a type of knowledge-based (KB) mistake which can be mitigated by directive controls
第三（III.）是一种知识基础（KB）错误，可以通过指令控制缓解
- C. The second (II.) is a quality-based (QB) mistake where the best mitigation is typically lean Six Sigma response
第二（II.）是一种质量基础（QB）错误，最佳缓解通常是精益六西格玛响应
- D. The first (I.) is a skill-based (SB) slip, and root-cause analysis may imply process re-engineering, environmental factors, and/or workflow changes
第一（I.）是一种技能基础（SB）滑倒，根因分析可能暗示流程重新工程、环境因素和/或工作流变化

答案:C

解析:

C 选项正确。错误：这是一项知识基础（KB）错误。根据阅读（GARP），知识基础错误的相应响应是指令控制。

考察: Operational Risk: Human Error **重要程度:** ■■□

科目: Quantitative Analysis

74. What is the primary (valid) reason that banks need proper liquidity transfer pricing (LTP)?

银行需要适当流动性转移定价（LTP）的主要（有效）理由是什么？

- A. Because banks engage in maturity transformation
因为银行参与期限转换
- B. Because there is a lack of liquidity transformation products in the marketplace
因为市场上缺乏流动性转换产品
- C. Because the true economic cost of liquidity risk is approximately zero
因为流动性风险的真正经济成本大约为零
- D. Because markets are volatile; e.g., volatility 1-year LIBOR/Swap spread
因为市场是波动的；例如，波动 1 年 LIBOR/Swap 利差

答案:A

解析:

A 选项正确。正确答案：因为银行参与期限转换（maturity transformation 是该章节反复出现的因果主题）。

如 Grant 在 3.1 中所解释，为什么银行需要 LTP：在日常业务中，银行通过用短期存款（负债）为长期贷款（资产）提供资金来赚钱，这一过程通常称为期限转换。正如巴塞尔银行监管委员会（BCBS，2008 年 9 月）所指出的，这“使得银行本质上容易受到流动性风险的影响，无论是机构特定的流动性风险还是影响整个市场的流动性风险”（第 3 页）。但只要银行在产品定价、新产品审批流程以及利润和绩效评估中使用 LTP 来计量流动性的成本、收益和风险，就不应阻止银行参与期限转换。拥有糟糕 LTP 实践的银行更可能积累大量长期非流动性资产、有条件承诺以及短期波动性更大的负债，从而显著增加其资金短缺的脆弱性。”（Joel Grant, 2011, “Liquidity Transfer Pricing: A Guide to Better Practice,” Occasional Paper, Financial Stability Board, Bank for International Settlements.）

关于 (B)、(C) 和 (D)，均为错误陈述。

考察: Liquidity Risk Management **重要程度:** ■■□

科目: Quantitative Analysis

75. In the paper “Finding Bernie Madoff” (by Dimmock and Gerken), the authors produce several findings as they attempt to test the predictability of investment fraud. In regard to their findings specifically, each of the following is TRUE except which is not?

在论文“寻找 Bernie Madoff”（由 Dimmock 和 Gerken 撰写）中，作者们试图测试投资欺诈的可预测性，并产生了几项发现。具体来说，下列陈述均为真，除了哪一项为假？

- A. Disclosed soft dollars do not significantly predict fraud
披露的软美元不能显著预测欺诈
- B. Direct Theft and Self Dealing are the most common types of fraud
直接盗窃和自我交易是最常见的欺诈类型
- C. Current disclosures required by the SEC have significant power for predicting future fraud
当前 SEC 要求的披露对预测未来欺诈具有显著作用
- D. Investors (on average) are compensated for fraud risk through superior performance and/or higher fees 投资者（平均）通过卓越表现和/或更高的费用来补偿欺诈风险

答案:D

解析:

D 选项正确。错误: 相反, 作者发现投资者并没有因为欺诈风险而得到补偿, 无论是通过卓越表现和/或更高的费用关于 (A)、(B) 和 (C), 每项都是正确的。

考察: Operational Due Diligence 重要程度: ■■□

科目: Quantitative Analysis

76. Ralph employed the peaks-over-threshold (POT) approach under extreme value theory (EVT) to the estimation of value at risk (VaR) for his firm’ s portfolio. He has four years of data, $n = 1,000$ days. He would prefer to apply a maximum likelihood estimation method (aka, MLE) to parameter estimation, but he thinks he needs to choose the threshold first. When he selects a threshold of $u = 4.0\%$, he observes $N(u) = 28$ exceedances such that $N(u)/n = 2.80\%$. Fitting the data generates these parameters: scale, $\beta = 0.80\%$; and tail index, $\xi = 0.310$. Based on these assumptions, the 99.90% confident VaR is 8.670%. He subsequently decides to increase the threshold from 4.0% to 5.0% while retaining the tail index at 0.310 and the scale parameter at 0.80%. The number of exceptions declines to $N(u) = 17$ such that $N(u)/n = 17/1,000 = 1.70\%$. His revised 99.90% VaR is similar at 8.630%. In regard to choosing a threshold, which of the following statements is TRUE?

Ralph 在其公司投资组合的价值-at-风险 (VaR) 估计中, 采用了极值理论 (EVT) 下的阈值超越 (POT) 方法。他拥有四年的数据, 即 $n = 1,000$ 个交易日。他更倾向于使用极大似然估计 (MLE) 进行参数估计, 但他认为首先需要选择阈值。当他选择阈值 $u = 4.0\%$ 时, 观察到 $N(u) = 28$ 次超越, 因此 $N(u)/n = 2.80\%$ 。对数据进行拟合得到如下参数: 尺度参数 $\beta = 0.80\%$, 尾指数 $\xi = 0.310$ 。在这些假设下, 99.90% 置信 VaR 为 8.670%。随后, 他决定将阈值从 4.0% 提高到 5.0%, 同时保持尾指数为 0.310、尺度参数为 0.80% 不变。此时超越次数下降到 $N(u) = 17$, 因此 $N(u)/n = 17/1,000 = 1.70\%$ 。调整后得到的 99.90% VaR 与之前相近, 为 8.630%。在阈值选择方面, 下列哪一项陈述是正确的?

- A. The higher threshold offers the benefit of lower bias but the drawback of higher variance
更高的阈值提供了降低偏差但提高方差的优点
- B. The higher threshold offers the benefit of lower variance but the drawback of higher bias
更高的阈值提供了降低方差但提高偏差的优点
- C. Because the desired confidence is 99.90% and $0.10\% = 1/1,000$, the (optimal) threshold should be set equal to the dataset’ s second-worst return
因为所需的置信水平为 99.90% 和 $0.10\% = 1/1,000$, (最优) 阈值应设置为等于数据集的第二差回报
- D. He can avoid a subjective choice of the threshold by applying an analytical maximum log-likelihood estimation (MLE) function to all three parameters simultaneously
他可以通过应用分析极大似然估计 (MLE) 函数同时对所有三个参数进行计算, 避免主观选择阈值

答案:A

解析:

A 选项正确。我们在任何 EVT 方法中都面临偏差-方差权衡。在块方法中, 权衡是块大小 (或数量) 的反映。在 POT 方法中, 它是任意阈值选择, 类似于块大小选择。如果阈值太低, 那么我们的模型将不符合 EVT 的渐近性质, 模型可能不准确 (aka, 偏差)。更高的阈值意味着更低的偏差 (aka, 更高的准确性), 但这必然适合更少的数据点, 因此我们可以预期更高的方差 (aka, 更低的精度)。

Dowd^(†) 在半参数估计方法（例如，Hill 估计器）的上下文中参考了偏差-方差权衡：“... 基于 k 的选择意味着偏差和方差之间的权衡。如果我们增加 k ，我们获得更多数据，因此移动到分布的中心。这增加了我们的估计器的精度（因此降低了它的方差），但也增加了尾部估计器的偏差，通过在分布中心附近放置相对更多的权重来观察。或者，如果我们减少 k 并进一步沿着尾部移动，我们减少偏差，但有更少的数据来使用，并且得到更高的方差。”

考察: Quantitative Analysis: Extreme Value Theory **重要程度:** [■ ■ □]

科目: Quantitative Analysis

77. Consider two counterparties who are about to enter into bilateral framework under an ISDA Master Agreement. They are currently negotiating the credit support annex (CSA) which will govern the terms of collateral posting. Each of the following is true EXCEPT which is inaccurate?

考虑两个交易对手，他们即将根据 ISDA 主协议进入双边框架。他们目前正在谈判信用支持附件（CSA），该附件将规定抵押品存入的条款。下列陈述均为真，除了哪一项为假？

- A. If the goal is to minimize funding costs, both parties prefer to avoid rehypothecation rights
如果目标是降低融资成本，双方都倾向于避免再抵押权
- B. With respect to VARIATION margin, both parties probably prefer rehypothecation rights; and probably will NOT require segregation
在变动保证金方面，双方可能更喜欢再抵押权；可能不会要求分离
- C. With respect to INITIAL margin, both parties probably are likely to insist on non-rehypothecation; and probably will require segregation
在初始保证金方面，双方可能倾向于坚持非再抵押；可能需要分离
- D. Whoever is the collateral poster (aka, collateral giver) prefers the method of Security Interest rather than the method of Title Transfer (which is preferred by the collateral receiver)
谁是抵押品存入者（aka，抵押品提供者）更喜欢安全利益方法而不是标题转移方法（由抵押品接收者更喜欢）

答案:A

解析:

A 选项正确。错误：再抵押最小化融资成本。权衡是再抵押最小化融资成本但增加交易对手风险。

考察: Counterparty Risk **重要程度:** [■ ■ □]

科目: Quantitative Analysis

78. Reporting on operational risks presents a unique set of challenges. As GARP explains, “Unlike other risk factors, assessing the volatility of operational risk is especially challenging given this is a relatively new discipline. Credit and market risk metrics often have over a hundred years of historical trend information. Operational risk measurement is relatively new, requiring more attention to the qualitative and quantitative tracking measures used to evaluate changes in risk profiles.”^(†)

Let’ s assume the organization’ s central operational risk function (ORF) has primary responsibility for operational risk reporting. In regard to the challenges to reporting operational risks, each of the following statements is true EXCEPT which is false?

报告操作风险面临一系列独特挑战。正如 GARP 所解释的那样：“与其他风险因素不同，评估操作风险的波动性尤为困难，因为这是一个相对较新的领域。信用风险和市场风险度量往往拥有一百多年历史趋势信息。操作风险计量则相对较新，因此在用以评估风险状况变化的定性和定量跟踪指标方面需要投入更多关注。”^(†) 假设机构的中央操作风险职能部门（ORF）对操作风险报告负有首要责任。关于操作风险报告所面临的挑战，下列陈述均为真，除了哪一项为假？

- A. The ORF should focus scarce resources on preventing and addressing major incidents rather than getting distracted by daily volatility
中央操作风险职能部门应将稀缺资源集中在预防和解决重大事件，而不是被日常波动分散注意力
- B. When the purpose is benchmarking across business units, losses should be reported in dollar terms (i.e., rather than percentage or basis points) because low percentages could mask high dollar losses
当目的是跨业务单位进行基准测试时，损失应以美元（即，而不是百分比或基点）报告，因为低百分比可能掩盖高美元损失

- C. The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are consistent and random, consider including their average cost in the service price
中央操作风险职能部门应定期审查小、频繁的损失，以发现潜在的控制违规或流程缺陷；如果这些损失是连续且随机的，应考虑将它们的平均成本包含在服务价格中
- D. The ORF can address the challenge of compiling qualitative risk data through either (i) conversion and summation, (ii) categorization, and (iii) worst-case reporting.
中央操作风险职能部门可以通过（i）转换和汇总，（ii）分类，和（iii）最坏情况报告来应对编译定性风险数据的挑战。

答案:B

解析:

B 选项正确。错误：相反，百分比比美元更受欢迎。

GARP 说：“报告运营损失作为毛收入、总成本（用于成本中心）、总预算（用于项目）或资本基点，有助于跨业务单位比较类似实体。报告以百分比而不是绝对值也使报告可扩展并适用于所有规模的实体。理想情况下，运营风险管理人员也将更好地理解实施适当控制和运营限制的成本效益评估和权衡。” (†)

关于（A）、（C）和（D），每项都是正确的。

- 中央操作风险职能部门应将稀缺资源集中在预防和解决重大事件，而不是被日常波动分散注意力。GARP 说：“适当分配风险管理资源是专注于预防和解决大型事件，而不是被日常波动分散和分散注意力。” (†)
- 中央操作风险职能部门应定期审查小、频繁的损失，以发现潜在的控制违规或流程缺陷；如果这些损失是连续且随机的，应考虑将它们的平均成本包含在服务价格中。GARP 说：“小而频繁的损失应定期识别和分析，以检测任何可能表明控制违规或流程缺陷的模式，这需要采取行动计划。如果这些损失随机发生，没有特定的结构，并且结构有限、稳定和重复，它们的平均成本可以作为服务成本的一部分传递给客户。” (†)
- 中央操作风险职能部门可以通过（i）转换和汇总，（ii）分类，和（iii）最坏情况报告来应对编译定性风险数据的挑战。

(†) GARP Chapter 6, Risk Reporting, 2023 FRM Part 2, Topic 7.

考察: Operational Risk Reporting **重要程度:** [■ ■ □]

科目: Quantitative Analysis

79. According to Patrick McGuire, which of the following best summarizes the proximate cause of the US dollar shortage during the global financial crisis (GFC)?

根据 Patrick McGuire，哪一项最准确地总结了全球金融危机（GFC）期间美元短缺的直接原因？

- A. Depositors initiated a classic run on the banks and converted their domestic deposits to foreign currency
存款人发起了一场经典的银行挤兑，并将其国内存款转换为外国货币
- B. The effective maturity of banks’ dollar funding lengthened while their dollar assets shortened, and this decreased their maturity mismatch.
银行美元资金的有效期限延长，而美元资产期限缩短，这减少了它们的期限错配。
- C. The effective maturity of banks’ dollar funding shortened while their dollar assets lengthened, and this increased their maturity mismatch.
银行美元资金的有效期限缩短，而美元资产期限延长，这增加了它们的期限错配。
- D. Banks shifted their portfolios to non-bank assets that included banks’ retail and corporate lending, lending to hedge funds, and holdings of securities ranging from US Treasury and agency securities to structured finance products
银行将其投资组合转移到非银行资产，包括银行零售和公司贷款、对冲基金贷款以及从美国国债和机构证券到结构化金融产品的证券持有。

答案:C

解析:

C 选项正确。银行美元资金的有效期限缩短，而美元资产期限延长，这增加了它们的期限错配。

McGuire (emphasis ours): “危机期间的事件导致了银行短期资金来源的严重中断。银行间市场陷入瘫痪，外汇掉期市场的混乱使得通过掉期获得美元的成本更高。银行的资金压力也因非银行资金来源的不稳定性而加剧，特别是美元货币市场基金和持有

美元的中央银行。市场压力意味着银行美元资金的有效期限缩短，而美元资产期限延长，因为许多资产在流动性不足的市场上难以出售。这种内生性的期限错配，难以事前对冲，导致了全球美元短缺。我们对银行美元资金缺口在危机开始时的估计揭示了为什么国际政策响应是必要的，以及为什么它采取了全球中央银行掉期线的形式。” (Patrick McGuire, Gotz von Peter, 2009. “The US Dollar Shortage in Global Banking and the International Policy Response,” BIS Working Papers, Bank for International Settlements.)

考察: Finance: Global Banking **重要程度:** [■■■□]

科目: Quantitative Analysis

80. Janice is preparing to conduct factor regressions in order to discover the factor loadings in her firm’s equity portfolio. She will begin by regressing three explanatory (aka, independent) variables: the market factor (MKT) plus two additional factors. Each of the following pairs of factors are good candidates for these additional factors EXCEPT which of the following pair is NOT a good candidate?

Janice 准备进行因子回归，以发现其公司股票投资组合中的因子载荷。她将首先回归三个解释变量：市场因子（MKT）加上两个额外因子。下列每一对因子中，哪一对不是好的候选因子？

- A. Quality and momentum (style factors)
质量与动量（风格因子）
- B. Strategy and reputation (style factors)
- C. Economic growth and inflation (macro factors)
经济增长与通胀（宏观因子）
- D. Real interest rates and liquidity (macro factors)
实际利率与流动性（宏观因子）

答案:B

解析:

B 选项正确。错误：在现实中，直接投资于声誉是困难的，几乎不可能想象如何将策略操作化为可投资的因子。根据 Andrew Ang（见 <https://www.blackrock.com/us/individual/investment-ideas/what-is-factor-investing>），有两种主要类型的因子：宏观经济因子和风格因子。宏观经济因子包括经济增长、信用（违约风险）、实际利率、新兴市场（政治和主权风险）、通胀和流动性。风格因子包括价值、质量、最小波动、规模、动量和 carry。在这方面，（A）、（C）和（D）每项都是正确的：

- 质量和动量是真正的风格因子
- 经济增长和通胀是真正的宏观因子

考察: Factor Investing **重要程度:** [■■■□]

科目: Quantitative Analysis